

Notes on Gauge Theory

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Preface

本次讨论班的内容主要是规范理论，我们将会介绍规范理论所需的几何基础、用几何语言描述规范理论，以及用规范理论研究低维拓扑。

首先我们会介绍规范理论的几何基础，如联络、主丛、特征类等，这些内容在物理学中是非常重要的工具，在数学中也有着广泛的应用。我们将会从黎曼几何中的 Levi-Civita 联络开始，逐渐引入更一般的仿射联络和主丛上的联络，并介绍它们的曲率和特征类等相关内容。

物理学家用联络描述规范场论中的规范场，曲率则对应场强，我们将会用几何语言描述电磁理论和 Yang-Mills 理论等规范场论的基本内容。

最后，数学家用规范理论构造了很多重要的拓扑不变量，如 Donaldson 不变量、Seiberg-Witten 不变量等，我们将会介绍这些不变量的定义，以及它们在低维拓扑中的应用。

以下是本次讨论班的参考资料：

- 《Differential Geometry - Connections, Curvature and Characteristic Classes》(Loring W. Tu)
- 《黎曼几何引论 (上下两册)》(陈维桓, 李兴校)
- 《Lectures on Seiberg-Witten Invariants》(John Douglas Moore)
- 何思奇老师在陈所暑期学校的讲义 <http://www.cim.nankai.edu.cn/2020/0709/c11453a284191/page.htm>

5 Prologue: Electromagnetism and $U(1)$ Gauge Theory

5.1 Maxwell's equations

We begin the story with Maxwell's equations. Recall that Maxwell's equations in vacuum are

$$\begin{cases} \operatorname{div} \mathbf{E} = 0, \\ \operatorname{div} \mathbf{B} = 0, \\ \operatorname{curl} \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \\ \operatorname{curl} \mathbf{B} = \frac{\partial \mathbf{E}}{\partial t}. \end{cases}$$

Here

$$\begin{aligned} \mathbf{E} &= (E_x, E_y, E_z) \quad \text{is the electric field,} \\ \mathbf{B} &= (B_x, B_y, B_z) \quad \text{is the magnetic field.} \end{aligned}$$

Each component is a function of the spacetime coordinates (t, x, y, z) .

Below we will list some concepts and notations commonly used in physics:

- 4-vector potential A_μ ¹

$$A_\mu = (\phi, A_x, A_y, A_z),$$

where ϕ is the *scalar potential* and $\mathbf{A} = (A_x, A_y, A_z)$ is the *vector potential*.

- Electromagnetic field strength tensor $F_{\mu\nu}$

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{pmatrix}.$$

- Relation between A_μ and $F_{\mu\nu}$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \implies \begin{cases} \mathbf{E} = \operatorname{grad} \phi - \frac{\partial \mathbf{A}}{\partial t}, \\ \mathbf{B} = \operatorname{curl} \mathbf{A}. \end{cases}$$

- Gauge transformation of the 4-vector potential

$$A_\mu \mapsto A_\mu + \partial_\mu \chi \implies \begin{cases} \phi \mapsto \phi + \frac{\partial \chi}{\partial t}, \\ \mathbf{A} \mapsto \mathbf{A} + \operatorname{grad} \chi. \end{cases}$$

where χ is an arbitrary smooth function. One can check that this transformation leaves the electromagnetic field strength tensor $F_{\mu\nu}$ invariant.

¹ $\mu = 0, 1, 2, 3$ are the spacetime indices, where 0 corresponds to the time coordinate t , and 1, 2, 3 correspond to the spatial coordinates x, y, z .

²Here we use the notation $\partial_0 := \frac{\partial}{\partial t}$, $\partial_1 := \frac{\partial}{\partial x}$, $\partial_2 := \frac{\partial}{\partial y}$, $\partial_3 := \frac{\partial}{\partial z}$.

Although Maxwell's equations are already quite concise, we can still transform them into an even more elegant expression using the language of differential forms.

Let $\mathbb{R}^{3,1}$ be the spacetime with coordinates (t, x, y, z) , equipped with the Minkowski metric

$$ds^2 = dt^2 - dx^2 - dy^2 - dz^2.$$

For simplicity, we set the speed of light $c = 1$.

The Hodge star operator $*$ on $\Omega^*(\mathbb{R}^{3,1})$ is defined by

$$\alpha \wedge *\beta = \langle \alpha, \beta \rangle dt \wedge dx \wedge dy \wedge dz, \quad \forall \alpha, \beta \in \Omega^*(\mathbb{R}^{3,1}).$$

Explicitly, we have

$$\begin{aligned} *1 &= dt \wedge dx \wedge dy \wedge dz, \\ *dt &= dx \wedge dy \wedge dz, \\ *dx &= dt \wedge dy \wedge dz, \\ *dy &= dt \wedge dz \wedge dx, \\ *dz &= dt \wedge dx \wedge dy, \\ *(dt \wedge dx) &= -dy \wedge dz, \\ *(dt \wedge dy) &= -dz \wedge dx, \\ *(dt \wedge dz) &= -dx \wedge dy, \\ *(dx \wedge dy) &= dt \wedge dz, \\ *(dy \wedge dz) &= dt \wedge dx, \\ *(dz \wedge dx) &= dt \wedge dy, \\ *(dt \wedge dx \wedge dy) &= dz, \\ *(dt \wedge dy \wedge dz) &= dx, \\ *(dt \wedge dz \wedge dx) &= dy, \\ *(dx \wedge dy \wedge dz) &= dt, \\ *(dt \wedge dx \wedge dy \wedge dz) &= -1. \end{aligned}$$

It is direct to check that

$$*^2 = (-1)^{p+1} \quad \text{on } \Omega^p(\mathbb{R}^{3,1}).$$

Now we can introduce the electromagnetic field strength 2-form

$$F = -dt \wedge (E_x dx + E_y dy + E_z dz) + B_x dy \wedge dz + B_y dz \wedge dx + B_z dx \wedge dy.$$

Then

$$\begin{aligned} dF &= dt \wedge \left[(\partial_x E_y - \partial_y E_x) dx \wedge dy + (\partial_y E_z - \partial_z E_y) dy \wedge dz + (\partial_z E_x - \partial_x E_z) dz \wedge dx \right] \\ &\quad + dt \wedge \left(\partial_t B_x dy \wedge dz + \partial_t B_y dz \wedge dx + \partial_t B_z dx \wedge dy \right) \\ &\quad + \left(\partial_x B_x + \partial_y B_y + \partial_z B_z \right) dx \wedge dy \wedge dz, \end{aligned}$$

Thus

$$\begin{cases} \operatorname{div} \mathbf{B} = 0, \\ \operatorname{curl} \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \end{cases} \iff dF = 0.$$

On the other hand,

$$*F = E_x dy \wedge dz + E_y dz \wedge dx + E_z dx \wedge dy + dt \wedge (B_x dx + B_y dy + B_z dz).$$

So

$$\begin{aligned} d * F &= -dt \wedge \left[(\partial_x B_y - \partial_y B_x) dx \wedge dy + (\partial_z B_x - \partial_x B_z) dz \wedge dx + (\partial_y B_z - \partial_z B_y) dy \wedge dz \right] \\ &\quad + dt \wedge \left(\partial_t E_x dy \wedge dz + \partial_t E_y dz \wedge dx + \partial_t E_z dx \wedge dy \right) \\ &\quad + \left(\partial_x E_x + \partial_y E_y + \partial_z E_z \right) dx \wedge dy \wedge dz. \end{aligned}$$

Thus

$$\begin{cases} \operatorname{div} \mathbf{E} = 0, \\ \operatorname{curl} \mathbf{B} = \frac{\partial \mathbf{E}}{\partial t}. \end{cases} \iff d * F = 0.$$

Since $*$: $\Omega^p(\mathbb{R}^{3,1}) \rightarrow \Omega^{4-p}(\mathbb{R}^{3,1})$ is an isomorphism,

$$d * F = 0 \iff *d * F = 0.$$

Therefore,

$$\begin{cases} \operatorname{div} \mathbf{E} = 0, \\ \operatorname{div} \mathbf{B} = 0, \\ \operatorname{curl} \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \\ \operatorname{curl} \mathbf{B} = \frac{\partial \mathbf{E}}{\partial t}. \end{cases} \iff \begin{cases} dF = 0, \\ d * F = 0. \end{cases}$$

Here $d^* = *d*$ is the dual of d .

In Riemannian geometry, a p -form α is called *harmonic* if $d\alpha = 0$ and $d^*\alpha = 0$. Thus Maxwell's equations in vacuum corresponds to a harmonic 2-form F .

Poincaré lemma says all closed forms on $\mathbb{R}^{3,1}$ are exact. So if $dF = 0$, there exists a 1-form A such that

$$F = dA.$$

If we write

$$A = \phi dt + A_x dx + A_y dy + A_z dz,$$

Then

$$F = dA \implies \begin{cases} E_x = \partial_x \phi - \partial_t A_x, \\ E_y = \partial_y \phi - \partial_t A_y, \\ E_z = \partial_z \phi - \partial_t A_z, \\ B_x = \partial_y A_z - \partial_z A_y, \\ B_y = \partial_z A_x - \partial_x A_z, \\ B_z = \partial_x A_y - \partial_y A_x. \end{cases} \implies \begin{cases} \mathbf{E} = \operatorname{grad} \phi - \frac{\partial \mathbf{A}}{\partial t}, \\ \mathbf{B} = \operatorname{curl} \mathbf{A}. \end{cases}$$

It is easy to see that A is not unique. In fact, if we transform $A \mapsto A + d\chi$ for some smooth function χ , then

$$F \mapsto dA + d^2\chi = dA = F.$$

Explicitly,

$$A \mapsto A + d\chi \implies \begin{cases} \phi \mapsto \phi + \partial_t \chi, \\ A_x \mapsto A_x + \partial_x \chi, \\ A_y \mapsto A_y + \partial_y \chi, \\ A_z \mapsto A_z + \partial_z \chi. \end{cases} \implies \begin{cases} \phi \mapsto \phi + \frac{\partial \chi}{\partial t}, \\ \mathbf{A} \mapsto \mathbf{A} + \nabla \chi. \end{cases}$$

We can summarize the above discussion in a table:

Physics	Differential Geometry
4-vector potential A_μ	1-form A
Electromagnetic field strength tensor $F_{\mu\nu}$	2-form F
$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$	$F = dA$
Gauge transformation $A_\mu \mapsto A_\mu + \partial_\mu \chi$	$A \mapsto A + d\chi$
Maxwell's equations	$dF = 0, d^*F = 0$

5.2 $U(1)$ gauge theory

Let $\psi(x)$ be a complex-valued function on $\mathbb{R}^{3,1}$, which represents the wave function of a matter field. Here x denotes the spacetime coordinates. In physics, $|\psi(x)|^2$ represents the probability amplitude of finding the particle at a given point in spacetime. If we transform

$$\psi(x) \mapsto \psi'(x) = e^{i\chi} \psi(x)$$

for some real constant χ . This transformation is called a *global $U(1)$ gauge transformation*. Since the probability amplitude $|\psi'(x)|^2 = |\psi(x)|^2$ remains the same, the physics is invariant under global $U(1)$ gauge transformations.

Now we want to promote the global $U(1)$ gauge symmetry to a local one. That is, we want the physics to be invariant under the transformation

$$\psi(x) \mapsto \psi'(x) = e^{i\chi(x)} \psi(x)$$

for some smooth function $\chi \in C^\infty(\mathbb{R}^{3,1})$.

For some physical reason, we want the derivative of ψ to be covariant under transformations. This holds in the case of global $U(1)$ gauge transformations, since

$$d\psi \mapsto d\psi' = e^{i\chi} d\psi.$$

However, in the case of local $U(1)$ gauge transformations,

$$d\psi \mapsto d\psi' = e^{i\chi(x)} d\psi + (de^{i\chi(x)})\psi.$$

The extra term $(de^{i\chi(x)})\psi$ breaks the covariance of the derivative.

To fix this problem, we introduce a 1-form A and define the *covariant derivative* by

$$\nabla\psi = d\psi + iA\psi.$$

Under the local $U(1)$ gauge transformation,

$$\nabla \mapsto \nabla' = d + iA',$$

we want $\nabla\psi$ to be covariant, i.e.

$$\nabla\psi \mapsto \nabla'\psi' = e^{i\chi(x)}\nabla\psi.$$

Then by

$$\begin{aligned}\nabla'\psi' &= d\psi' + iA'\psi' = e^{i\chi(x)}d\psi + ie^{i\chi(x)}(A' + d\chi)\psi, \\ e^{i\chi(x)}\nabla\psi &= e^{i\chi(x)}(d\psi + iA\psi) = e^{i\chi(x)}d\psi + ie^{i\chi(x)}A\psi,\end{aligned}$$

we must have

$$A' = A - d\chi.$$

Ignoring the sign, this is exactly the gauge transformation of the electromagnetic potential A in electromagnetism. Thus, the electromagnetic potential A can be interpreted as a gauge field that ensures the covariance of the derivative under local $U(1)$ gauge transformations.

6 Affine Connections

6.1 Levi-Civita Connections

When it comes to connections, most of us probably think first of the Levi-Civita connection in Riemannian geometry. So let us first review its most basic definition, and then try to rewrite everything we are familiar with in the language of differential forms.

6.1.1 Covariant Derivative

Definition 6.1. A **connection** ∇ on a manifold M is a map

$$\begin{aligned} \nabla : C^\infty(M, TM) \times C^\infty(M, TM) &\rightarrow C^\infty(M, TM) \\ (Y, X) &\mapsto \nabla_Y X \end{aligned}$$

satisfying the following properties:

$$\begin{aligned} \text{(C1)} \quad & \nabla_{f_1 Y_1 + f_2 Y_2} X = f_1 \nabla_{Y_1} X + f_2 \nabla_{Y_2} X, \\ \text{(C2)} \quad & \nabla_Y (\lambda_1 X_1 + \lambda_2 X_2) = \lambda_1 \nabla_Y X_1 + \lambda_2 \nabla_Y X_2, \\ \text{(C3)} \quad & \nabla_Y fX = df(Y) \cdot X + f \nabla_Y X. \end{aligned}$$

where $X, X_1, X_2, Y, Y_1, Y_2 \in \Gamma(TM)$, $f, f_1, f_2 \in C^\infty(M)$, and $\lambda_1, \lambda_2 \in \mathbb{R}$.

Definition 6.2. The **torsion** of a connection ∇ is defined as the tensor

$$T(X, Y) := \nabla_X Y - \nabla_Y X - [X, Y].$$

The **curvature** of a connection ∇ is defined as the tensor

$$R(X, Y)Z := \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z.$$

Remark. The torsion T can only be defined for connections on the tangent bundle, since it involves both $\nabla_X Y$ and $\nabla_Y X$.

Definition 6.3. A connection ∇ of a Riemannian manifold (M, g) is called a **Levi-Civita connection** if it is torsion-free and compatible with the metric g , i.e.

$$\begin{aligned} \text{(LC1)} \quad & \nabla_X Y - \nabla_Y X = [X, Y], \\ \text{(LC2)} \quad & X(g(Y, Z)) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z). \end{aligned}$$

Exterior covariant derivative

A connection ∇ can also be viewed as a map

$$\begin{aligned} \nabla : C^\infty(M, TM) &\rightarrow C^\infty(M, T^*M \otimes TM) \\ X &\mapsto \nabla X \end{aligned}$$

where ∇X is a 1-form taken value in TM^3 and $(\nabla X)(Y) := \nabla_Y X$. ∇ is not a $C^\infty(M)$ -linear map, instead,

$$\nabla(fX) = df \otimes X + f \nabla X.$$

³One can simply view ∇X as a $(1, 1)$ -tensor field.

We can extend ∇ to the exterior covariant derivative

$$\begin{aligned}\nabla : \Omega^*(M, TM) &\rightarrow \Omega^{*+1}(M, TM)^4 \\ \eta &\mapsto \nabla\eta,\end{aligned}$$

satisfying the Leibniz rule:

$$\nabla(\alpha \wedge \eta) = d\alpha \wedge \eta + (-1)^q \alpha \wedge \nabla\eta, \quad \forall \alpha \in \Omega^q(M), \eta \in \Omega^*(M, TM),$$

For example, let $\eta = \alpha \otimes X$ be a section of $\Omega^r(M, TM)$, then

$$\nabla\eta = \nabla(\alpha \otimes X) = d\alpha \otimes X + (-1)^r \alpha \wedge \nabla X.$$

Under the above setup, we can define $\nabla^2 = \nabla \circ \nabla, \nabla^3, \dots, \nabla^k$, and we can prove the following property:

Proposition 6.4. $\nabla^2 : \Omega^*(M, TM) \rightarrow \Omega^{*+2}(M, TM)$ is an $\Omega^*(M)$ -linear map, i.e. for any $\alpha \in \Omega^*(M), \eta \in \Omega^*(M, TM)$,

$$\nabla^2(\alpha \wedge \eta) = \alpha \wedge \nabla^2\eta.$$

Thus $R := \nabla^2$ defines a section of $\Omega^2(M, \text{End}(TM))$, which is called the **curvature** of the connection ∇ .

Proof.

$$\begin{aligned}\nabla^2(\alpha \wedge \eta) &= \nabla(d\alpha \wedge \eta + (-1)^q \alpha \wedge d\nabla\eta) \\ &= d^2\alpha \wedge \eta + (-1)^{q+1} d\alpha \wedge d\nabla\eta + (-1)^q d\alpha \wedge \nabla\eta + \alpha \wedge \nabla^2\eta \\ &= \alpha \wedge \nabla^2\eta.\end{aligned} \quad \square$$

Unlike the usual exterior derivative d , we have $\nabla^2 \neq 0$ in general. The following proposition explains why ∇^2 is called the curvature of ∇ :

Proposition 6.5. For $X, Y, Z \in C^\infty(M, TM)$, we have

$$(\nabla^2 Z)(X, Y) = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z.$$

Proof. We will prove it after introducing the connection 1-forms. □

6.1.2 Connection 1-form

Locally, we always have a frame. Let $\{e_i\}$ be a local frame of TM on an open set $U \subset M$, and $\{\theta^i\}$ be the dual frame. then we can write

$$\nabla_{e_i} e_j = \Gamma_{ij}^k e_k,$$

where Γ_{ij}^k are the Christoffel symbols of the connection ∇ w.r.t. $\{e_i\}$. Define

$$\omega_j^k := \Gamma_{ij}^k \theta^i,$$

⁴ $\Omega^r(M, TM) := C^\infty(M, \wedge^r T^*M \otimes TM).$

we call ω_j^k the **connection 1-form** of ∇ w.r.t. $\{e_i\}$. So

$$\nabla e_j = \omega_j^k \otimes e_k.$$

Let the matrix of 1-forms $\omega = (\omega_j^k)$ ⁵, then we can write the above equation in a more compact way:

$$\nabla(e_1, \dots, e_n) = (e_1, \dots, e_n) \omega = (e_1, \dots, e_n) \begin{pmatrix} \omega_1^1 & \dots & \omega_1^n \\ \vdots & & \vdots \\ \omega_n^1 & \dots & \omega_n^n \end{pmatrix}^6$$

For any vector field $X = X^i e_i$, we have

$$\begin{aligned} \nabla X &= \nabla \left((e_1, \dots, e_n) \begin{pmatrix} X^1 \\ \vdots \\ X^n \end{pmatrix} \right) \\ &= \left(\nabla(e_1, \dots, e_n) \right) \begin{pmatrix} X^1 \\ \vdots \\ X^n \end{pmatrix} + (e_1, \dots, e_n) \begin{pmatrix} dX^1 \\ \vdots \\ dX^n \end{pmatrix} \\ &= (e_1, \dots, e_n) \left(d \begin{pmatrix} X^1 \\ \vdots \\ X^n \end{pmatrix} + \omega \begin{pmatrix} X^1 \\ \vdots \\ X^n \end{pmatrix} \right). \end{aligned}$$

Thus under a local frame, ∇ can be expressed as

$$\nabla = d + \omega.$$

We can compute the curvature in terms of the connection 1-form ω :

$$\begin{aligned} \nabla^2(e_1, \dots, e_n) &= \nabla(\nabla(e_1, \dots, e_n)) = \nabla((e_1, \dots, e_n) \omega) \\ &= (\nabla(e_1, \dots, e_n)) \omega + (e_1, \dots, e_n) d\omega \\ &= (e_1, \dots, e_n) \omega \wedge \omega + (e_1, \dots, e_n) d\omega \\ &= (e_1, \dots, e_n) (d\omega + \omega \wedge \omega). \end{aligned}$$

Define

$$\Omega := d\omega + \omega \wedge \omega, \tag{6.1}$$

then Ω is a matrix of 2-forms, and we call Ω the **curvature 2-form** of the connection ∇ w.r.t. $\{e_i\}$.

Proposition 6.6. *Let $R(e_i, e_j)e_k = R_{ijk}^l e_l$, then*

$$\Omega_k^l(e_i, e_j) = R_{ijk}^l.$$

⁵The upper index k is the row index, and the lower index j is the column index.

⁶This is an awkward notation. Since we arrange the frame as row vectors so matrix ω should multiply on the right. For a vector field X and a differential forms $\alpha \in \Omega^q(M)$, we have $X \otimes \alpha = \alpha \otimes X$. For a general $\eta \in \Omega^p(M, TM)$, we have $\eta \otimes \alpha = (-1)^{pq} \alpha \otimes \eta$.

Proof. Since $\Omega_k^l = d\omega_k^l + \omega_m^l \wedge \omega_k^m$,

$$\begin{aligned} (d\omega_k^l)(e_i, e_j)e_l &= e_i(\omega_k^l(e_j))e_l - e_j(\omega_k^l(e_i))e_l - \omega_k^l([e_i, e_j])e_l \\ &= \nabla_{e_i}(\omega_k^l(e_j)e_l) - \omega_k^l(e_j)\nabla_{e_i}e_l - \nabla_{e_j}(\omega_k^l(e_i)e_l) + \omega_k^l(e_i)\nabla_{e_j}e_l - \omega_k^l([e_i, e_j])e_l \\ &= \nabla_{e_i}\nabla_{e_j}e_k - \omega_k^l(e_j)\nabla_{e_i}e_l - \nabla_{e_j}\nabla_{e_i}e_k + \omega_k^l(e_i)\nabla_{e_j}e_l - \omega_k^l([e_i, e_j])e_l, \end{aligned}$$

and

$$\begin{aligned} (\omega_m^l \wedge \omega_k^m)(e_i, e_j)e_l &= \omega_m^l(e_i)\omega_k^m(e_j)e_l - \omega_m^l(e_j)\omega_k^m(e_i)e_l \\ &= \omega_k^m(e_j)\nabla_{e_i}e_m - \omega_k^m(e_i)\nabla_{e_j}e_m. \end{aligned}$$

Thus

$$\Omega_k^l(e_i, e_j)e_l = \nabla_{e_i}\nabla_{e_j}e_k - \nabla_{e_j}\nabla_{e_i}e_k - \nabla_{[e_i, e_j]}e_k = R(e_i, e_j)e_k. \quad \square$$

Corollary 6.7. *This proposition immediately implies Prop ?? since*

$$(\nabla^2 e_k)(e_i, e_j) = \Omega_k^l(e_i, e_j)e_l = R(e_i, e_j)e_k.$$

Let $\{\tilde{e}_j\}$ be another local frame, and the transition matrix from $\{e_i\}$ to $\{\tilde{e}_j\}$ is $a = (a_i^j)$, i.e.

$$(\tilde{e}_1, \dots, \tilde{e}_n) = (e_1, \dots, e_n) a,$$

then

$$\begin{aligned} (\tilde{e}_1, \dots, \tilde{e}_n)\tilde{\omega} &= \nabla(\tilde{e}_1, \dots, \tilde{e}_n) = \nabla((e_1, \dots, e_n) a) \\ &= (\nabla(e_1, \dots, e_n))a + (e_1, \dots, e_n)da \\ &= (e_1, \dots, e_n)\omega a + (e_1, \dots, e_n)da \\ &= (\tilde{e}_1, \dots, \tilde{e}_n)(a^{-1}\omega a + da). \end{aligned}$$

Thus

$$\tilde{\omega} = a^{-1}da + a^{-1}\omega a. \quad (6.2)$$

Applying the transformation formula (??) to (??), we get

$$\begin{aligned} \tilde{\Omega} &= d\tilde{\omega} + \tilde{\omega} \wedge \tilde{\omega} \\ &= d(a^{-1}da + a^{-1}\omega a) + (a^{-1}da + a^{-1}\omega a) \wedge (a^{-1}da + a^{-1}\omega a) \\ &= da^{-1} \wedge da + da^{-1} \wedge \omega a + a^{-1}d\omega a - a^{-1}\omega \wedge da \\ &\quad + a^{-1}da \wedge a^{-1}da + a^{-1}da \wedge a^{-1}\omega a + a^{-1}\omega \wedge da + a^{-1}\omega \wedge \omega a, \end{aligned}$$

since

$$a^{-1}a = I \implies da^{-1}a + a^{-1}da = 0 \implies da^{-1} = -a^{-1}da a^{-1},$$

we have

$$\begin{aligned} \tilde{\Omega} &= -a^{-1}da a^{-1} \wedge da - a^{-1}da a^{-1} \wedge \omega a + a^{-1}d\omega a - a^{-1}\omega \wedge da \\ &\quad + a^{-1}da \wedge a^{-1}da + a^{-1}da \wedge a^{-1}\omega a + a^{-1}\omega \wedge da + a^{-1}\omega \wedge \omega a \\ &= a^{-1}(d\omega + \omega \wedge \omega)a = a^{-1}\Omega a. \end{aligned}$$

Thus transformation formula for curvature between $\{e_i\}$ and $\{\tilde{e}_j\}$ is given by

$$\tilde{\Omega} = a^{-1}\Omega a. \quad (6.3)$$

The above formula shows that although Ω isn't a globally defined form, two different curvature forms only differ by a conjugation. So if we apply a quantity invariant under conjugation, we obtain a globally defined form. This is the idea of Chern-Weil theory, which we will discuss in later section.

6.2 General Affine Connections

Let $\pi : E \rightarrow M$ be a rank m vector bundle, we can also define affine connections on E in a similar way as we did for the tangent bundle.

6.2.1 Covariant Derivative

The content of this section is almost a verbatim copy of the previous content.

Definition 6.8. An (affine) connection ∇ on a vector bundle E is a map

$$\begin{aligned} \nabla : C^\infty(M, TM) \times C^\infty(M, E) &\rightarrow C^\infty(M, E) \\ (X, s) &\mapsto \nabla_X s \end{aligned}$$

satisfying

$$(C1) \quad \nabla_{f_1 Y_1 + f_2 Y_2} X = f_1 \nabla_{Y_1} X + f_2 \nabla_{Y_2} X,$$

$$(C2) \quad \nabla_Y (\lambda_1 X_1 + \lambda_2 X_2) = \lambda_1 \nabla_Y X_1 + \lambda_2 \nabla_Y X_2,$$

$$(C3) \quad \nabla_Y fX = df(Y) \cdot X + f \nabla_Y X.$$

Let $(\nabla s)(X) := \nabla_X s$, we can extend ∇ to the exterior covariant derivative

$$\begin{aligned} \nabla : \Omega^*(M, E) &\rightarrow \Omega^{*+1}(M, E) \\ \eta &\mapsto \nabla \eta, \end{aligned}$$

satisfying the Leibniz rule: for any $\alpha \in \Omega^q(M), \eta \in \Omega^*(M, E)$,

$$\nabla(\alpha \wedge \eta) = d\alpha \wedge \eta + (-1)^q \alpha \wedge \nabla \eta.$$

Remark. The reason why we call ∇ an **affine** connection is that given two connections ∇_1 and ∇_2 on E , $(1-t)\nabla_1 + t\nabla_2$ is also a connection on E for any $t \in \mathbb{R}$.

Remark. The existence of connections on E can be proved by using partitions of unity, which we left as an exercise.

Proposition 6.9. Let ∇_1 and ∇_2 be two connections on E , then $\nabla_1 - \nabla_2$ is $C^\infty(M)$ -linear in s , thus defines a section $A \in \Omega^1(M, \text{End}(E))$. Conversely, for any $A \in \Omega^1(M, \text{End}(E))$ and a connection ∇ on E , $\nabla + A$ is also a connection on E . *Consequently, the space of connections on E is an affine space modeled on $\Omega^1(M, \text{End}(E))$.*

Proof. For $s \in C^\infty(M, E), f \in C^\infty(M)$,

$$(\nabla_1 - \nabla_2)(fs) = df \otimes s + f \nabla_1 s - df \otimes s - f \nabla_2 s = f(\nabla_1 - \nabla_2)(s),$$

thus $\nabla_1 - \nabla_2$ is $C^\infty(M)$ -linear in s . Conversely, for $A \in \Omega^1(M, \text{End}(E))$ and a connection ∇ on E ,

$$(\nabla + A)(fs) = \nabla(fs) + A(fs) = df \otimes s + f \nabla s + fAs = df \otimes s + f(\nabla + A)(s). \quad \square$$

Curvature of connections

Proposition 6.10. $\nabla^2 : \Omega^*(M, E) \rightarrow \Omega^{*+2}(M, E)$ is an $\Omega^*(M)$ -linear map, i.e., for any $\alpha \in \Omega^*(M), \eta \in \Omega^*(M, E)$,

$$\nabla^2(\alpha \wedge \eta) = \alpha \wedge \nabla^2 \eta.$$

Proof. For $\alpha \in \Omega^q(M), \eta \in \Omega^*(M, E)$,

$$\begin{aligned} \nabla^2(\alpha \wedge \eta) &= \nabla(d\alpha \wedge \eta + (-1)^q \alpha \wedge \nabla \eta) \\ &= d^2 \alpha \wedge \eta + (-1)^{q+1} d\alpha \wedge \nabla \eta + (-1)^q d\alpha \wedge \nabla \eta + \alpha \wedge \nabla^2 \eta \\ &= \alpha \wedge \nabla^2 \eta. \end{aligned} \quad \square$$

Definition 6.11. The curvature of ∇ is defined as $R := \nabla^2 \in \Omega^2(M, \text{End}(E))$.

Metrics and metric compatibility

Definition 6.12. Let E be a (complex) vector bundle over M . A **(Hermitian) metric** on E is a smooth section g of $E^* \otimes E^*$ such that for any $x \in M$, g_x is a (Hermitian) inner product $\langle \cdot, \cdot \rangle_{E_x}$ on the fibre E_x .

Definition 6.13. A connection ∇ on E is called **compatible with the metric g** if for any $X \in C^\infty(M, TM)$ and $s_1, s_2 \in C^\infty(M, E)$,

$$X \langle s_1, s_2 \rangle = \langle \nabla_X s_1, s_2 \rangle + \langle s_1, \nabla_X s_2 \rangle.$$

Remark. For a (complex) vector bundle E over M , we can always construct a (Hermitian) metric on E by using partitions of unity.

Remark. Given a (Hermitian) metric g on E , we can always construct a connection ∇ compatible with g by using partitions of unity. Note that the connection compatible with g is not unique.

Proposition 6.14. Let ∇_1 and ∇_2 be two connections on E compatible with metric g , then $\nabla_1 - \nabla_2$ is skew-symmetric w.r.t. g , i.e., $\nabla_1 - \nabla_2 \in \Omega^1(M, \mathfrak{so}(E, g))$. Conversely, for any $A \in \Omega^1(M, \mathfrak{so}(E, g))$ and a connection ∇ compatible with g , $\nabla + A$ is also compatible with g . *Consequently, the space of connections on E compatible with g is an affine space modeled on $\Omega^1(M, \mathfrak{so}(E, g))$.*

Proof. For $s_1, s_2 \in C^\infty(M, E), X \in C^\infty(M, TM)$,

$$\begin{aligned} &\begin{cases} X \langle s_1, s_2 \rangle = \langle \nabla_{1,X} s_1, s_2 \rangle + \langle s_1, \nabla_{1,X} s_2 \rangle \\ X \langle s_1, s_2 \rangle = \langle \nabla_{2,X} s_1, s_2 \rangle + \langle s_1, \nabla_{2,X} s_2 \rangle \end{cases} \\ &\implies \langle (\nabla_1 - \nabla_2)_X s_1, s_2 \rangle + \langle s_1, (\nabla_1 - \nabla_2)_X s_2 \rangle = 0. \end{aligned}$$

Conversely, for $A \in \Omega^1(M, \mathfrak{so}(E, g))$ and a connection ∇ compatible with g ,

$$\begin{aligned} &\langle (\nabla + A)_X s_1, s_2 \rangle + \langle s_1, (\nabla + A)_X s_2 \rangle \\ &= \langle \nabla_X s_1, s_2 \rangle + \langle s_1, \nabla_X s_2 \rangle + \langle A(X) s_1, s_2 \rangle + \langle s_1, A(X) s_2 \rangle \\ &= X \langle s_1, s_2 \rangle. \end{aligned} \quad \square$$

6.2.2 Connection 1-form

Let E be a rank m vector bundle over M , and $\{e_i\}$ be a local frame of E on an open set $U \subset M$. Define ω_j^k by

$$\nabla_X e_j = \omega_j^k(X) e_k,$$

this 1-form matrix $\omega = (\omega_j^k)$ is called **the connection 1-form** of ∇ w.r.t. $\{e_i\}$. Then we can write the above equation in a more compact way:

$$\nabla(e_1, \dots, e_m) = (e_1, \dots, e_m) \omega = (e_1, \dots, e_m) \begin{pmatrix} \omega_1^1 & \cdots & \omega_1^m \\ \vdots & & \vdots \\ \omega_m^1 & \cdots & \omega_m^m \end{pmatrix}.$$

Thus for any section $s = s^i e_i$, we have

$$\begin{aligned} \nabla s &= \nabla \left((e_1, \dots, e_m) \begin{pmatrix} s^1 \\ \vdots \\ s^m \end{pmatrix} \right) = \left(\nabla(e_1, \dots, e_m) \right) \begin{pmatrix} s^1 \\ \vdots \\ s^m \end{pmatrix} + (e_1, \dots, e_m) \begin{pmatrix} ds^1 \\ \vdots \\ ds^m \end{pmatrix} \\ &= (e_1, \dots, e_m) \left(d \begin{pmatrix} s^1 \\ \vdots \\ s^m \end{pmatrix} + \omega \begin{pmatrix} s^1 \\ \vdots \\ s^m \end{pmatrix} \right). \end{aligned}$$

So under a local frame, ∇ can be expressed as

$$\nabla = d + \omega.$$

Proposition 6.15. Suppose $\{\tilde{e}_i\}$ are another local frame of E on U , and the transition matrix from $\{e_i\}$ to $\{\tilde{e}_i\}$ is $a = (a_i^j)$, i.e.

$$(\tilde{e}_1, \dots, \tilde{e}_m) = (e_1, \dots, e_m) a,$$

Let $\tilde{\omega}$ be the connection 1-form w.r.t. $\{\tilde{e}_i\}$, then the connection 1-forms transform as

$$\tilde{\omega} = a^{-1} da + a^{-1} \omega a.$$

A local frame of E is the same as a local trivialization. Let $\{(U_\alpha, \psi_\alpha)\}$ be a local trivialization of E , $g_{\alpha\beta}$ be the transition functions. A section $s \in C^\infty(M, E)$ possesses a local representative s_α , which is an m -tuple of smooth functions on U_α . Also, ∇s possesses a local representative $(\nabla s)_\alpha$, which is an m -tuple of 1-forms on U_α . Just as in the preceding paragraph, we can write

$$(\nabla s)_\alpha = ds_\alpha + \omega_\alpha s_\alpha, \tag{6.4}$$

where ω_α is an $m \times m$ matrix of 1-forms on U_α .

Since the connection is globally defined, the local representatives of ∇s must satisfy

$$ds_\beta + \omega_\beta s_\beta = g_{\alpha\beta}^{-1} (ds_\alpha + \omega_\alpha s_\alpha) \quad \text{on } U_\alpha \cap U_\beta.$$

Apply $s_\alpha = g_{\alpha\beta} s_\beta$ to the above equation,

$$ds_\beta + \omega_\beta s_\beta = g_{\alpha\beta}^{-1} (d(g_{\alpha\beta} s_\beta) + \omega_\alpha g_{\alpha\beta} s_\beta)$$

$$\begin{aligned}
&= g_{\alpha\beta}^{-1}((dg_{\alpha\beta})s_\beta + g_{\alpha\beta}ds_\beta + \omega_\alpha g_{\alpha\beta}s_\beta) \\
&= ds_\beta + \left(g_{\alpha\beta}^{-1}dg_{\alpha\beta} + g_{\alpha\beta}^{-1}\omega_\alpha g_{\alpha\beta}\right)s_\beta,
\end{aligned}$$

Thus

$$\omega_\beta = g_{\alpha\beta}^{-1}dg_{\alpha\beta} + g_{\alpha\beta}^{-1}\omega_\alpha g_{\alpha\beta}. \quad (6.5)$$

The above discussion can be summarized as the following theorem:

Theorem 6.16. *Let $\{(U_\alpha, \psi_\alpha)\}$ be a local trivialization of E , with transition functions $g_{\alpha\beta}$, and ω_α be an $m \times m$ matrix of 1-forms on U_α satisfying the transformation formula (??). This information uniquely determines a connection ∇ on E .*

Let (E, g) be a real vector bundle equipped with a metric, we can choose the local frame $\{e_i\}$ on U to be orthonormal, then for any $X \in C^\infty(U, TU)$,

$$0 = X\langle e_i, e_j \rangle = \langle \omega_i^k(X)e_k, e_j \rangle + \langle e_i, \omega_j^k(X)e_k \rangle = \omega_j^i(X) + \omega_i^j(X),$$

so the 1-form matrix ω is skew-symmetric, i.e., $\omega \in \Omega^1(U, \mathfrak{so}(m))$.

Let $\{(U_\alpha, \psi_\alpha)\}$ be a local trivialization of (E, g) . We can assume

$$\psi_\alpha : \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times \mathbb{R}^m$$

is isometric on each fibre, where the metric on $U_\alpha \times \mathbb{R}^m$ is Euclidean metric on each fibre. Then the transition functions $g_{\alpha\beta}$ take values in $O(m)$, and each ω_α is in $\Omega^1(U_\alpha, \mathfrak{so}(m))$.

If E is a complex vector bundle and g is a Hermitian metric. Then all the discussion above still holds, except that ω_α is in $\Omega^1(U_\alpha, \mathfrak{u}(m))$ and $g_{\alpha\beta}$ takes values in $U(m)$.

Example 6.17. Let $\pi : L \rightarrow M$ be a complex line bundle, g be a Hermitian metric on L and ∇ be a connection on L compatible with g . Choose a local trivialization $\{(U_\alpha, \psi_\alpha)\}$ of L such that $\psi_\alpha : \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times \mathbb{C}$ is an isometry. Then $g_{\alpha\beta}$ takes values in $U(1)$, so we can write $g_{\alpha\beta} = e^{i\theta_{\alpha\beta}}$ for some real-valued function $\theta_{\alpha\beta}$ on $U_\alpha \cap U_\beta$. Meanwhile, ω_α takes values in $\mathfrak{u}(1)$, so we can write $\omega_\alpha = iA_\alpha$ for some real-valued 1-form A_α on U_α . For any section s of L , on U_α , we have

$$(\nabla s)_\alpha = ds_\alpha + \omega_\alpha s_\alpha, \quad \omega_\alpha = iA_\alpha,$$

On $U_\alpha \cap U_\beta$,

$$\omega_\beta = g_{\alpha\beta}^{-1}dg_{\alpha\beta} + g_{\alpha\beta}^{-1}\omega_\alpha g_{\alpha\beta} = e^{-i\theta_{\alpha\beta}}de^{i\theta_{\alpha\beta}} + e^{-i\theta_{\alpha\beta}}\omega_\alpha e^{i\theta_{\alpha\beta}} = \omega_\alpha + id\theta_{\alpha\beta} \quad (= \omega_\alpha - id\theta_{\beta\alpha})$$

Recall $U(1)$ -gauge theory in physics, the following table shows the correspondence between concepts in physics and differential geometry:

Physics	Differential Geometry
wave function ψ	section s of complex line bundle L
$\psi \mapsto \psi' = e^{i\chi}\psi$	$s_\beta = g_{\beta\alpha}s_\alpha$
covariant derivative	connection ∇ on L
$D'\psi' = e^{i\chi}D\psi$	$(\nabla s)_\beta = g_{\beta\alpha}(\nabla s)_\alpha$ ⁷
$D = d + iA$	$(\nabla s)_\alpha = ds_\alpha + \omega_\alpha s_\alpha$
$A' = A - d\chi$	$\omega_\beta = \omega_\alpha - id\theta_{\beta\alpha}$

⁷That is to say, ∇s is globally defined.

6.2.3 Horizontal Distribution

In this subsection, we give another interpretation of a connection on a vector bundle, namely a horizontal distribution, and prove that it is equivalent to the definition given in terms of covariant derivatives.

Let $\pi : E \rightarrow M$ be a rank m vector bundle, and $p \in E_x$ be a point in the fibre over $x \in M$. The projection π induces a linear map $\pi_{*p} : T_p E \rightarrow T_x M$. We define $\mathcal{V}_p := \ker \pi_{*p}$, which is called the **vertical tangent space** at p . Thus we have a short exact sequence

$$0 \longrightarrow \mathcal{V}_p \hookrightarrow T_p E \xrightarrow{\pi_*} T_{\pi(p)} M \longrightarrow 0$$

and

$$\dim \mathcal{V}_p = \dim T_p E - \dim T_{\pi(p)} M = m.$$

The kernel of π_* form a subbundle of TE . We call it the **vertical subbundle** of TE and denote it by $\mathcal{V}$.

Proposition 6.18. *For each $p \in E$, there is an isomorphism $E_{\pi(p)} \cong \mathcal{V}_p$, which induces an isomorphism $\pi^* E \cong \mathcal{V}$.*

Proof. A point $(p, p') \in \pi^* E$ must satisfy $\pi(p) = \pi(p')$, so we can define a curve $\gamma(t) = p + tp'$ in E . Then $\gamma'(0) \in T_p E$ gives a tangent vector at p , this gives an injective morphism

$$\pi^* E \hookrightarrow TE, \quad (p, p') \mapsto (p, \gamma'(0)).$$

So we can view $\pi^* E$ as a subbundle of TE . Moreover, since $\gamma(t) \in E_{\pi(p)}$, we have $\pi(\gamma(t)) = \pi(p)$, thus $\pi_{*p}(\gamma'(0)) = 0$. Therefore, $\pi^* E \subset \ker \pi_*$.

Since π_{*p} is a surjective linear map, we have

$$\dim \ker \pi_{*p} = \dim T_p E - \dim T_{\pi(p)} M = m = (\pi^* E)_p.$$

Thus $\pi^* E = \ker \pi_*$. □

For any $p \in E$, we can define a map

$$\iota_p : E_{\pi(p)} \rightarrow \mathcal{V}_p \subset T_p E, \quad p' \mapsto \gamma'(0),$$

where $\gamma(t) = p + tp'$ is a curve in E . Then ι_p is an isomorphism between $E_{\pi(p)}$ and $\mathcal{V}_p$.

Let $\pi^* TM$ be the pull-back of TM by π , then $\pi_* : TE \rightarrow TM$ can be viewed as a morphism $\tilde{\pi}_* : TE \rightarrow \pi^* TM$. The above proposition implies the following short exact sequence of vector bundles over E :

$$0 \longrightarrow \mathcal{V} \hookrightarrow TE \xrightarrow{\tilde{\pi}_*} \pi^* TM \longrightarrow 0$$

In smooth category, all short exact sequences of vector bundles split, so we can find a subbundle $\mathcal{H} \hookrightarrow TE$ such that

$$TE = \mathcal{H} \oplus \mathcal{V}, \quad \tilde{\pi}_* : \mathcal{H} \rightarrow \pi^* TM \text{ is an isomorphism.}$$

On the vector bundle E , we have two natural operations: scalar multiplication and addition.

- The scalar multiplication:

$$m : \mathbb{R} \times E \rightarrow E, \quad m(\lambda, p) = \lambda p.$$

When fixing λ , we get a map $m_\lambda : E \rightarrow E$, $m_\lambda(p) = \lambda p$, which is the scalar multiplication by λ .

- The addition:

$$a : E \times_M E \rightarrow E, \quad a(p, q) = p + q,$$

where $E \times_M E = \{(p, q) \in E \times E : \pi(p) = \pi(q)\}$.

Definition 6.19. A **horizontal distribution** of E is a smooth subbundle $\mathcal{H} \subset TE$ satisfying the following properties

1. There exists a splitting map $k : \pi^*TM \rightarrow TE$ such that $\tilde{\pi}_* \circ k = \text{id}$.

$$0 \longrightarrow \mathcal{V} \hookrightarrow TE \begin{array}{c} \xrightarrow{\tilde{\pi}_*} \\ \xleftarrow{k} \end{array} \pi^*TM \longrightarrow 0$$

That is to say,

$$(H1) \quad TE = \mathcal{H} \oplus \mathcal{V}.$$

2. $\mathcal{H}$ is invariant under the scalar multiplication on E , i.e. for any $\lambda \in \mathbb{R}$, $p \in E$,

$$(H2) \quad m_{\lambda*p} \mathcal{H}_p = \mathcal{H}_{\lambda p}.$$

3. $\mathcal{H}$ is invariant under the addition on E , i.e. for any $(p, q) \in E \times_M E$,

$$(H3) \quad a_{*(p,q)}(\mathcal{H}_p, \mathcal{H}_q) = \mathcal{H}_{p+q}.$$

The last two conditions are saying that $\mathcal{H}$ is compatible with the linear structure of E .

There is no canonical way to choose a horizontal distribution, so a horizontal distribution is an extra structure on E .

In what follows, we reveal that covariant derivatives and horizontal distributions can actually determine each other.

Covariant Derivative determines Horizontal Distribution

Let ∇ be a connection on E . Given any smooth curve $\gamma : (-\varepsilon, \varepsilon) \rightarrow M$ with $\gamma(0) = x$, we say $\tilde{\gamma} : (-\varepsilon, \varepsilon) \rightarrow E$ is a **lift** of γ if $\pi(\tilde{\gamma}(t)) = \gamma(t)$ for all t .

Definition 6.20. A lift $\tilde{\gamma}$ of γ is called a **horizontal lift** if

$$\nabla_{\gamma'(t)} \tilde{\gamma}(t) = 0, \quad \forall t \in (-\varepsilon, \varepsilon).$$

Example 6.21. In the case of a Riemannian manifold (M, g) , the horizontal lift is exactly the parallel transport.

Proposition 6.22. *For any (picewise) smooth curve $\gamma : (-\varepsilon, \varepsilon) \rightarrow M$ and any point $p \in E_{\gamma(0)}$, there exists a unique horizontal lift $\tilde{\gamma}$ of γ starting at p .*

Sketch of proof. The proof is exactly the same as the existence and uniqueness of geodesics. The outline of the proof is to write the equations as a system of ODEs in a local frame; the conclusion then follows from the existence and uniqueness theory of linear ODEs. $\square$

For any vector $u \in T_x M$ and any point $p \in E_x$, we can find a curve γ such that $\gamma(0) = x$ and $\gamma'(0) = u$, the horizontal lift $\tilde{\gamma}$ gives a tangent vector $\tilde{\gamma}'(0) \in T_p E$.

Fact 1. $\tilde{\gamma}'(0) \in T_p E$ is independent of the choice of γ .

Sketch of proof. By solving the ODEs, we know that $\tilde{\gamma}'(0)$ depends only on the derivative of γ at 0, which is $\gamma'(0) = u$. $\square$

So there is a well-defined map

$$T_{\pi(p)} M \rightarrow T_p E, \quad (u^h)_p := \tilde{\gamma}'(0).$$

We call $(u^h)_p$ the **horizontal lift** of u at p . By definition,

$$\pi(\tilde{\gamma}(t)) = \gamma(t) \implies \pi_{*p}(u^h)_p = \pi_{*p}(\tilde{\gamma}'(0)) = \gamma'(0) = u.$$

So we obtain a linear injection from $T_{\pi(p)} M$ to $T_p E$, whose image forms a distribution on E denoted by $\mathcal{H}$, i.e.,

$$\mathcal{H} := \{(p, (u^h)_p) \mid p \in E, u \in T_{\pi(p)} M\}.$$

Fact 2. $\mathcal{H}$ is a smooth subbundle of TE .

Sketch of proof. This fact relies on the smooth dependence of solutions of linear ODEs on initial conditions and parameters. $\square$

Fact 3. $\mathcal{H}$ is invariant under the scalar multiplication and addition on E .

Proof. Let $p, q \in E$ such that $\pi(p) = \pi(q)$, $\lambda \in \mathbb{R}$, and γ be a curve such that $\gamma(0) = \pi(p)$. Let $\tilde{\gamma}_p, \tilde{\gamma}_q$ be the horizontal lift of γ starting at p, q respectively, then by the uniqueness of horizontal lift, $\lambda \tilde{\gamma}_p$ is the horizontal lift of γ starting at λp , and $\tilde{\gamma}_p + \tilde{\gamma}_q$ is the horizontal lift of γ starting at $p + q$. Thus

$$\begin{aligned} m_{\lambda * p} \tilde{\gamma}'(0) &= (\lambda \tilde{\gamma}_p)'(0) \in \mathcal{H}_{\lambda p}, \\ a_{*(p,q)}(\tilde{\gamma}'_p(0), \tilde{\gamma}'_q(0)) &= (\tilde{\gamma}_p + \tilde{\gamma}_q)'(0) \in \mathcal{H}_{p+q}. \end{aligned} \quad \square$$

The above discussion is summarized in the following theorem.

Theorem 6.23. $\mathcal{H}$ is a horizontal distribution of E .

Horizontal Distribution determines Covariant Derivative

Conversaly, given a horizontal distribution $\mathcal{H}$, we can derive a connection ∇ on E as follows.

Definition 6.24. For any $p \in E$ and $u \in TE$, we can write $u = h(u) + v(u)$ where $h(u) \in \mathcal{H}$ and $v(u) \in \mathcal{V}$. We call $h(u)$ and $v(u)$ the horizontal and vertical component of u respectively.

For any $X \in C^\infty(M, TM)$ and $s \in C^\infty(M, E)$, the tangent map of s gives a map $s_* : TM \rightarrow TE$, then at any point $x \in M$ we can define $(\nabla_X s)(x)$ by

$$(\nabla_X s)(x) := \iota_{s(x)}^{-1} (v(s_{*x} X_x)).$$

Note that $\iota_{s(x)} : E_x \rightarrow \mathcal{V}_{s(x)}$ is the isomorphism defined above. The idea is taking the vertical component of $s_{*x}(X_x)$.

Theorem 6.25. ∇ defined above is a connection on E .

To prove this theorem, we need the following lemma.

Lemma 6.26. For $\lambda \in \mathbb{R}$, $(p, q) \in E \times_M E$ and $u, w \in E_{\pi(p)}$, we have

$$\begin{aligned} m_{\lambda*p} \iota_p(u) &= \iota_{\lambda p}(\lambda u), \\ a_{*(p,q)}(\iota_p(u), \iota_q(w)) &= \iota_{p+q}(u + w). \end{aligned}$$

Proof. By the definition of ι_p ,

$$\begin{aligned} m_{\lambda*p} \iota_p(u) &= \left. \frac{d}{dt} \right|_{t=0} m_\lambda(p + tu) = \left. \frac{d}{dt} \right|_{t=0} (\lambda p + t\lambda u) = \iota_{\lambda p}(\lambda u), \\ a_{*(p,q)}(\iota_p(u), \iota_q(w)) &= \left. \frac{d}{dt} \right|_{t=0} (p + tu) + (q + tw) = \iota_{p+q}(u + w). \end{aligned} \quad \square$$

Lemma 6.27. For any $\lambda \in \mathbb{R}$, $(p, q) \in E \times_M E$ and $u \in TE_p$, $w \in TE_q$,

$$\begin{aligned} h(m_{\lambda*p} u) &= m_{\lambda*p}(h(u)), & v(m_{\lambda*p} u) &= m_{\lambda*p}(v(u)), \\ h(a_{*(p,q)}(u, w)) &= a_{*(p,q)}(h(u), h(w)), & v(a_{*(p,q)}(u, w)) &= a_{*(p,q)}(v(u), v(w)). \end{aligned}$$

Proof. Decompose u and $m_{\lambda*p} u$ into their horizontal and vertical components, we get

$$m_{\lambda*p}(h(u)) + m_{\lambda*p}(v(u)) = m_{\lambda*p}(u) = h(m_{\lambda*p} u) + v(m_{\lambda*p} u).$$

Since $\mathcal{H}$ is compatible with the linear structure of the vector bundle, we have $m_{\lambda*p}(h(u)) \in \mathcal{H}_{\lambda p}$. Besides, Lemma ?? implies $m_{\lambda*p}(v(u)) \in \mathcal{V}_{\lambda p}$, thus

$$m_{\lambda*p}(h(u)) = h(m_{\lambda*p} u), \quad m_{\lambda*p}(v(u)) = v(m_{\lambda*p} u).$$

The proof for the addition is exactly the same. $\square$

Lemma 6.28. For $\nu \in T_\lambda \mathbb{R} \cong \mathbb{R}$ and $u \in T_p E$, we have

$$m_{*(\lambda,p)}(\nu, u) = \nu \cdot \iota_p(p) + m_{\lambda*p} u.$$

Proof. let $\gamma(t)$ be a curve in E such that $\gamma(0) = p$ and $\gamma'(0) = u$, then

$$\begin{aligned} m_{*(\lambda,p)}(\nu, u) &= \left. \frac{d}{dt} \right|_{t=0} (\lambda + t\nu)\gamma(t) \\ &= \left. \frac{d}{dt} \right|_{t=0} (\lambda p + t\nu p) + \left. \frac{d}{dt} \right|_{t=0} \lambda \gamma(t) \\ &= \nu \cdot \iota_p(p) + m_{\lambda * p} u. \end{aligned}$$

□

Proof of Theorem ??. We need to verify that ∇ satisfies the properties of a connection.

- $\nabla_{f_1 X_1 + f_2 X_2} s = f_1 \nabla_{X_1} s + f_2 \nabla_{X_2} s$:

$$\begin{aligned} \nabla_{f_1 X_1 + f_2 X_2} s(x) &= \iota_{s(x)}^{-1} \left(v(s_{*x}(f_1 X_1 + f_2 X_2)) \right) \\ &= \iota_{s(x)}^{-1} \left(v(s_{*x}(f_1(x) X_{1x})) + v(s_{*x}(f_2(x) X_{2x})) \right) \\ &= \iota_{s(x)}^{-1} \left(f_1(x) v(s_{*x} X_{1x}) + f_2(x) v(s_{*x} X_{2x}) \right) \\ &= f_1 \nabla_{X_1} s(x) + f_2 \nabla_{X_2} s(x). \end{aligned}$$

- $\nabla_X(\lambda_1 s_1 + \lambda_2 s_2) = \lambda_1 \nabla_X s_1 + \lambda_2 \nabla_X s_2$:

For $\mathbb{R}$ -linearity,

$$\begin{aligned} \nabla_X(\lambda s)(x) &= \iota_{(\lambda s)(x)}^{-1} \left(v((\lambda s)_{*x}(X_x)) \right) \\ &= \iota_{(\lambda s)(x)}^{-1} \left(v(m_{\lambda * s(x)} s_{*x}(X_x)) \right) \\ &= \iota_{(\lambda s)(x)}^{-1} \left(m_{\lambda * s(x)} v(s_{*x}(X_x)) \right) \quad (\text{By Lemma ??}) \\ &= \iota_{(\lambda s)(x)}^{-1} \left(\iota_{(\lambda s)(x)} (\lambda \nabla_X s(x)) \right) \quad (\text{By Lemma ??}) \\ &= \lambda \nabla_X s(x), \end{aligned}$$

For additivity,

$$\begin{aligned} \nabla_X(s_1 + s_2)(x) &= \iota_{(s_1 + s_2)(x)}^{-1} \left(v((s_1 + s_2)_{*x} X_x) \right) \\ &= \iota_{(s_1 + s_2)(x)}^{-1} \left(v(a_{*(s_1(x), s_2(x))}(s_{1*x} X_x, s_{2*x} X_x)) \right) \\ &= \iota_{(s_1 + s_2)(x)}^{-1} \left(a_{*(s_1(x), s_2(x))}(v(s_{1*x} X_x), v(s_{2*x} X_x)) \right) \quad (\text{By Lemma ??}) \\ &= \iota_{(s_1 + s_2)(x)}^{-1} \left(\iota_{(s_1 + s_2)(x)} (\nabla_X s_1(x) + \nabla_X s_2(x)) \right) \quad (\text{By Lemma ??}) \\ &= \nabla_X s_1(x) + \nabla_X s_2(x). \end{aligned}$$

- $\nabla_X(fs) = df(X) \cdot s + f \nabla_X s$:

$$\begin{aligned} \nabla_X(fs)(x) &= \iota_{(fs)(x)}^{-1} \left(v((fs)_{*x} X_x) \right) \\ &= \iota_{(fs)(x)}^{-1} \left(v(m_{f(x)*s(x)}(f_{*x} X_x, s_{*x} X_x)) \right) \end{aligned}$$

$$\begin{aligned}
&= \iota_{(fs)(x)}^{-1} \left(df(X)_x \cdot \iota_{(fs)(x)}(s(x)) + v(m_{f(x)*s(x)} s_{*x} X_x) \right) \quad (\text{By Lemma ??}) \\
&= \iota_{(fs)(x)}^{-1} \left(\iota_{(fs)(x)}(df(X)_x \cdot s(x) + f(x)\nabla_X s(x)) \right) \quad (\text{By Lemma ??}) \\
&= df(X)_x \cdot s(x) + f(x)\nabla_X s(x). \quad \square
\end{aligned}$$

6.2.4 Parallel Transport

Let ∇ be a connection on E . For any (piecewise) smooth curve $\gamma : [a, b] \rightarrow M$ and any point $v \in E_{\gamma(a)}$, There exists a unique horizontal lift $\tilde{\gamma}_v$ of γ starting at v . Varying $v \in E_{\gamma(a)}$, we get different point $w = \tilde{\gamma}_v(b) \in E_{\gamma(b)}$. This gives a map

$$P^\gamma : E_{\gamma(a)} \rightarrow E_{\gamma(b)}, \quad P^\gamma(v) := \tilde{\gamma}_v(b).$$

Proposition 6.29. P^γ is a linear isomorphism.

Proof. For any $v, w \in E_{\gamma(a)}$ and $\lambda \in \mathbb{R}$, the uniqueness of horizontal lift implies that

$$\lambda \tilde{\gamma}_v = \tilde{\gamma}_{\lambda v}, \quad \tilde{\gamma}_v + \tilde{\gamma}_w = \tilde{\gamma}_{v+w}.$$

Thus

$$\begin{aligned}
P^\gamma(\lambda v) &= \tilde{\gamma}_{\lambda v}(b) = (\lambda \tilde{\gamma}_v)(b) = \lambda P^\gamma(v), \\
P^\gamma(v + w) &= \tilde{\gamma}_{v+w}(b) = (\tilde{\gamma}_v + \tilde{\gamma}_w)(b) = P^\gamma(v) + P^\gamma(w).
\end{aligned}$$

So P^γ is a linear map. Let $\bar{\gamma}(t) := \gamma(a + b - t)$ be the reversed curve of γ , then one can check that $P^{\bar{\gamma}} : E_{\gamma(b)} \rightarrow E_{\gamma(a)}$ is the inverse map of P^γ . $\square$

Definition 6.30. P^γ is called **the parallel transport** along γ .

Remark. If E possesses a metric g and ∇ is compatible with g , then $P^\gamma : E_{\gamma(a)} \rightarrow E_{\gamma(b)}$ is an isometry. The converse is also true: if P^γ is an isometry for any curve γ , then ∇ is compatible with g .

Remark. P^γ highly depends on the choice of γ , not only on the endpoints $\gamma(a)$ and $\gamma(b)$. In general, if γ_1 and γ_2 are two curves with the same endpoints, P^{γ_1} and P^{γ_2} can be very different. In other words, if γ is a loop, $P^\gamma : E_{\gamma(a)} \rightarrow E_{\gamma(a)}$ may not be the identity map.

Example 6.31. Consider S^2 with the standard metric and the Levi-Civita connection ∇ . Let γ be a loop on S^2 as shown in the figure below, After parallel transport along γ , the tangent vector $\gamma'(0)$ is rotated by 90° .

Given a connection ∇ on E , for any two points $x, y \in M$, we can link x and y by a curve γ , then ∇ gives a linear isomorphism $P^\gamma : E_x \rightarrow E_y$. In this sense, ∇ **connects** any two fibers of E . This is the reason why ∇ is called a **connection**.

The key point is that if any curve γ linking x and y gives an isomorphism $\Phi^\gamma : E_x \rightarrow E_y$ and Φ^γ satisfies some natural properties, then we can define a connection ∇ on E such that $P^\gamma = \Phi^\gamma$. So intuitively, **a connection can be thought of as a family of isomorphisms between fibers at endpoints of curves.**

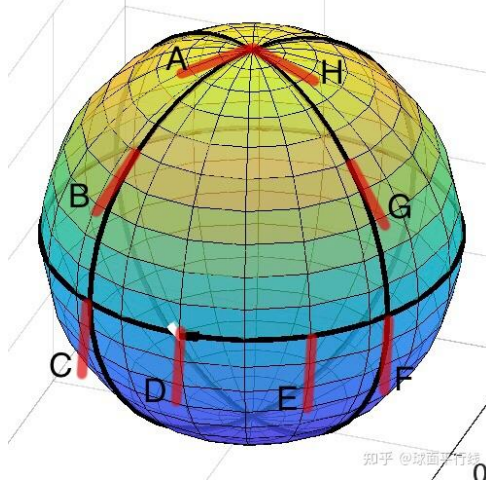


Figure 1: The parallel transport along γ in S^2

Proposition 6.32. Let ∇ be a connection on E , $X_x \in T_x M$ be a tangent vector at x and $s : M \rightarrow E$ be a section of E . Then there exists a curve $\gamma : (-\varepsilon, \varepsilon) \rightarrow M$ such that $\gamma(0) = x$ and $\gamma'(0) = X_x$. If we denote $P_t : E_x \rightarrow E_{\gamma(t)}$ as the parallel transport along $\gamma|_{[0,t]}$, then

$$\nabla_{X_x} s = \left. \frac{d}{dt} \right|_{t=0} P_t^{-1}(s(\gamma(t))). \quad (6.6)$$

Proof. Let $e_1, \dots, e_m$ be a base of E_x and set $e_i(t) := P_t(e_i)$, then $e_1(t), \dots, e_m(t)$ is a base of $E_{\gamma(t)}$. We can write

$$s(\gamma(t)) = \sum_{i=1}^m s^i(t) e_i(t),$$

Then

$$\nabla_X s = \nabla_{\gamma'(0)} s = \sum_{i=1}^m \frac{ds^i}{dt}(0) e_i.$$

On the other hand,

$$P_t^{-1}(s(\gamma(t))) = \sum_{i=1}^m s^i(t) e_i,$$

so

$$\left. \frac{d}{dt} \right|_{t=0} P_t^{-1}(s(\gamma(t))) = \sum_{i=1}^m \frac{ds^i}{dt}(0) e_i = \nabla_X s. \quad \square$$

Proposition 6.33. Let $P^\gamma : E_{\gamma(a)} \rightarrow E_{\gamma(b)}$ be the parallel transport along γ , then P^γ satisfies the following properties:

- If the end point of γ_1 is the starting point of γ_2 , we can concatenate γ_1 and γ_2 to get $\gamma_1 * \gamma_2$, then

$$(P1) \quad P^{\gamma_1 * \gamma_2} = P^{\gamma_2} \circ P^{\gamma_1}.$$

- If $\bar{\gamma}$ is the reversed curve of γ , then

$$(P2) \quad P^{\bar{\gamma}} = (P^{\gamma})^{-1}.$$

- Let $\mathcal{U}$ be an open subset in $\mathbb{R}^k$ and $\gamma : \mathcal{U} \times [0, 1] \rightarrow M$ is a smooth map. For any $u \in \mathcal{U}$, we denote $\gamma_u(t) := \gamma(u, t)$, then $\{\gamma_u\}$ is a family of curves smoothly depending on u . Let $s : \mathcal{U} \rightarrow E$ be a smooth map such that $s(u) \in E_{\gamma_u(0)}$. Then the map $\mathcal{U} \rightarrow E$, $u \mapsto P^{\gamma_u}(s(u))$ is also smooth. For simplicity, we state this conclusion as

$$(P3) \quad P^{\gamma} \text{ depends smoothly on } \gamma.$$

- The last property is a bit technical. Let $\gamma_1, \gamma_2 : [0, 1] \rightarrow M$ be two curves such that $\gamma_1(0) = \gamma_2(0)$, and $\gamma_1'(0) = \gamma_2'(0)$, Let $P_t^{\gamma_i}$ be the parallel transport along $\gamma_i|_{[0, t]}$, then

$$\left. \frac{d}{dt} \right|_{t=0} P_t^{\gamma_1} = \left. \frac{d}{dt} \right|_{t=0} P_t^{\gamma_2}.$$

For simplicity, we state this conclusion as

$$(P4) \quad \text{If } \gamma_1, \gamma_2 \text{ have contact of order 1, then } P^{\gamma_1}, P^{\gamma_2} \text{ have contact of order 1.}$$

Proof. (P1) and (P2) are straightforward from the uniqueness of solution of ODEs.

(P3) relies on the smooth dependence of solutions of ODEs on initial conditions and parameters.

For (P4), given any $v \in E_{\gamma_1(0)}$, $\tilde{\gamma}_{i,v}(t) := P_t^{\gamma_i}(v)$ is the horizontal lift of γ_i starting at v . The proposition is equivalent to saying that $\tilde{\gamma}'_{1,v}(0) = \tilde{\gamma}'_{2,v}(0)$, which can be verified by calculating under a local trivialization. $\square$

Parallel Transport determines Covariant Derivative

Theorem 6.34. Let E be a vector bundle over M . Suppose there is a family of isomorphisms

$$\left\{ \Phi^{\gamma} : E_{\gamma(0)} \rightarrow E_{\gamma(1)} \mid \gamma : [0, 1] \rightarrow M \right\}$$

satisfying (P1)-(P4), then there exists a unique connection ∇ on E such that $P^{\gamma} = \Phi^{\gamma}$ for any curve γ .

Proof. For any vector field $X \in C^{\infty}(M, TM)$ and section $s \in C^{\infty}(M, E)$, at $x \in M$, we can find a curve $\gamma : (-\varepsilon, \varepsilon) \rightarrow M$ such that $\gamma(0) = x$ and $\gamma'(0) = X_x$. We define $(\nabla_X s)(x)$ by the formula (??), i.e.,

$$(\nabla_X s)(x) := \left. \frac{d}{dt} \right|_{t=0} (\Phi_t^{\gamma})^{-1}(s(\gamma(t))).$$

By (P4), this definition does not depend on the choice of γ . By (P3), $\nabla_X s$ is a smooth section of E .

We can verify that ∇ satisfies (C2) and (C3) by direct calculation. For (C1), we need to verify that $\nabla_{X_x}s$ is $\mathbb{R}$ -linear in X_x . Let $\varphi(X_x) = \nabla_{X_x}s$, then φ is a map from T_xM to E_x . For any $\lambda \in \mathbb{R}$, set $\gamma_\lambda(t) = \gamma(\lambda t)$, then $\gamma_\lambda(0) = x$ and $\gamma'_\lambda(0) = \lambda X_x$, so

$$\begin{aligned}\varphi(\lambda X_x) &= \nabla_{\gamma'_\lambda(0)}s = \left. \frac{d}{dt} \right|_{t=0} (\Phi_t^{\gamma_\lambda})^{-1}(s(\gamma_\lambda(t))) \\ &= \left. \frac{d}{dt} \right|_{t=0} (\Phi_{\lambda t}^\gamma)^{-1}(s(\gamma(\lambda t))) \\ &= \lambda \left. \frac{d}{dt} \right|_{t=0} (\Phi_t^\gamma)^{-1}(s(\gamma(t))) \\ &= \lambda \nabla_{\gamma'(0)}s = \lambda \varphi(X_x).\end{aligned}$$

It is easy to verify that φ is additive, thus φ is linear.

Finally, we can verify that $P^\gamma = \Phi^\gamma$ for any curve γ by direct calculation. $\square$

Above theorem actually says that the covariant derivative ∇ is the **infinitesimal** version of the parallel transport P^γ .

Parallel Transport and Horizontal Distribution

Let $\pi : E \rightarrow M$ be a vector bundle. Given a family of parallel transports $\{P^\gamma\}$, we can define a horizontal distribution $\mathcal{H}$ by

$$\mathcal{H}_p := \left\{ \left. \frac{d}{dt} \right|_{t=0} P_t^\gamma(p) \mid \gamma : [0, 1] \rightarrow M \text{ with } \gamma(0) = p \right\} \subset T_p E.$$

Varying $p \in E$, we get a distribution $\mathcal{H}$ of E . Note that this definition is consistent with the previous one. Thus we have

Theorem 6.35. *If $\{P^\gamma\}$ is a family of parallel transports satisfying (P1)-(P4), then $\mathcal{H}$ satisfies (H1)-(H3), thus $\mathcal{H}$ is a horizontal distribution of E .*

One can view $\mathcal{H}$ as **the collection of all directions one can possibly move via parallel transports**.

Conversely, given a horizontal distribution $\mathcal{H}$, we can define the horizontal lift, then the parallel transport. Explicitly, for any curve $\gamma : [a, b] \rightarrow M$ and any point $v \in E_{\gamma(a)}$, there exists a unique curve $\tilde{\gamma}_v^\mathcal{H} : [a, b] \rightarrow E$ such that

$$\pi(\tilde{\gamma}_v^\mathcal{H}(t)) = \gamma(t), \quad \tilde{\gamma}_v^\mathcal{H}(a) = v, \quad \tilde{\gamma}_v^{\mathcal{H}'}(t) \in \mathcal{H}_{\tilde{\gamma}_v^\mathcal{H}(t)}.$$

The last condition can be translated into an ODE. Since we have a decomposition $TE = \mathcal{H} \oplus \mathcal{V}$, we can denote the projection to $\mathcal{V}$ by v . Then

$$\tilde{\gamma}_v^{\mathcal{H}'}(t) \in \mathcal{H}_{\tilde{\gamma}_v^\mathcal{H}(t)} \iff v(\tilde{\gamma}_v^{\mathcal{H}'}(t)) = 0,$$

which can be written as a linear ODEs under a local trivialization. Thus we can solve $\tilde{\gamma}_v^\mathcal{H}$ by the theory of ODEs. Now we can define $P^\gamma(v) := \tilde{\gamma}_v^\mathcal{H}(b)$.

Proposition 6.36. Fix a curve $\gamma : [0, 1] \rightarrow M$ connecting x and y , for any $v, w \in E_x$, $\lambda \in \mathbb{R}$,

$$\begin{aligned}\tilde{\gamma}_{v+w}^{\mathcal{H}}(t) &= \tilde{\gamma}_v^{\mathcal{H}}(t) + \tilde{\gamma}_w^{\mathcal{H}}(t), \\ \tilde{\gamma}_{\lambda v}^{\mathcal{H}}(t) &= \lambda \tilde{\gamma}_v^{\mathcal{H}}(t).\end{aligned}$$

Thus $P^\gamma : E_x \rightarrow E_{\gamma(1)}$ is a linear map. Moreover, P^γ is an isomorphism with inverse $P^{\bar{\gamma}}$ where $\bar{\gamma}$ is the reversed curve of γ .

Proof. The first two equalities are implied by the linearity of $\mathcal{H}$, and the last one can be verified by direct calculation. $\square$

Theorem 6.37. If $\mathcal{H}$ is a horizontal distribution of E , then the parallel transport defined above satisfies (P1)-(P4).

Proof. The proof is similar to the previous one. $\square$

In summary, three different concepts, i.e., covariant derivative ∇ , parallel transport $\{P^\gamma\}$ and horizontal distribution $\mathcal{H}$, are equivalent to each other.

$$\text{covariant derivative} \iff \text{parallel transport} \iff \text{horizontal distribution}.$$

In fact, both ∇ and $\mathcal{H}$ can be viewed as the **infinitesimal** version of $\{P^\gamma\}$.

Exercises

Exercise 6.1. Given a connection ∇ , we can define the covariant derivative of a 1-form ω by

$$(\nabla_X \omega)(Y) := X(\omega(Y)) - \omega(\nabla_X Y).$$

For an (r, s) -tensor field T , the covariant derivative $\nabla_Z T$ is

$$\begin{aligned} (\nabla_Z T)(X_1, \dots, X_r, \omega^1, \dots, \omega^s) &:= Z(T(X_1, \dots, X_r, \omega^1, \dots, \omega^s)) \\ &\quad - \sum_{i=1}^r T(X_1, \dots, \nabla_Z X_i, \dots, X_r, \omega^1, \dots, \omega^s) \\ &\quad - \sum_{j=1}^s T(X_1, \dots, X_r, \omega^1, \dots, \nabla_Z \omega^j, \dots, \omega^s). \end{aligned}$$

If we rearrange the terms, we see that this is precisely the Leibniz rule.

The curvature operator can also act on general tensors

$$R(X, Y)T := \nabla_X \nabla_Y T - \nabla_Y \nabla_X T - \nabla_{[X, Y]} T.$$

We define the covariant differential $\nabla_{\text{tensor}} T$ as the $(r, s+1)$ -tensor field

$$(\nabla_{\text{tensor}} T)(Z, \dots, X_i, \dots, \dots, \omega^j, \dots) := (\nabla_Z T)(\dots, X_i, \dots, \dots, \omega^j, \dots)$$

This is another way to extend the covariant derivative to tensor fields, and we denote it by ∇_{tensor} to distinguish it from the exterior covariant derivative.

[Prove that](#) for a torsion-free connection, ∇_{tensor} satisfies **the Ricci identity**

$$(\nabla_{\text{tensor}}^2 T)(X, Y, \dots, \dots) - (\nabla_{\text{tensor}}^2 T)(Y, X, \dots, \dots) = (R(X, Y)T)(\dots, \dots)$$

Exercise 6.2. For a vector field $Z \in C^\infty(M, TM)$, suppose ∇ is torsion-free. Both $\nabla^2 Z$ and $\nabla_{\text{tensor}}^2 Z$ are $(1, 2)$ -tensor fields, but by Ricci identity, we have

$$(\nabla_{\text{tensor}}^2 Z)(X, Y) - (\nabla_{\text{tensor}}^2 Z)(Y, X) = R(X, Y)Z = (\nabla^2 Z)(X, Y).$$

Can you explain this phenomenon?

Exercise 6.3 (Torsion 2-form). Let ∇ be a connection on TM , $\{e_i\}$ be a local frame of TM , $\{\theta^i\}$ be the corresponding dual coframe and ω be the connection 1-form of ∇ w.r.t. $\{e_i\}$. Define

$$\tau^i := d\theta^i + \omega_j^i \wedge \theta^j.$$

If we arrange $\theta = (\theta^1, \dots, \theta^n)^T$ as a column vector, then the above formula can be written as

$$\tau = d\theta + \omega \wedge \theta.$$

then τ is called the **torsion 2-form** of ∇ w.r.t. $\{e_i\}$.

Let $T(e_i, e_j) = T_{ij}^k e_k$, where T is the torsion of ∇ . [Prove that](#)

$$\tau^k(e_i, e_j) = T_{ij}^k.$$

Thus

$$\tau^k = \frac{1}{2} T_{ij}^k \theta^i \wedge \theta^j.$$

Exercise 6.4 (Levi-Civita connection). Let (M, g) be a Riemannian manifold, ∇ be a connection on TM . Show that

1. ∇ is torsion-free if and only if $\tau = 0$, i.e.,

$$d\theta^i + \omega_j^i \wedge \theta^j = 0.$$

2. If $\{e_i\}$ is a local orthonormal frame, then ∇ is compatible with g if and only if ω is skew-symmetric, i.e.,

$$\omega_j^i + \omega_i^j = 0.$$

3. Under a local orthonormal frame, there exists a unique connection 1-form ω satisfying

$$d\theta^i + \omega_j^i \wedge \theta^j = 0, \quad \omega_j^i + \omega_i^j = 0.$$

This is the existence and uniqueness of the Levi-Civita connection.

Exercise 6.5. Let ∇ be a connection on TM , $\{e_i\}$ be a local frame of TM , $\{\theta^i\}$ be the corresponding dual coframe and ω be the connection 1-form of ∇ w.r.t. $\{e_i\}$. The following two equations

$$d\theta = \tau - \omega \wedge \theta,$$

$$d\omega = \Omega - \omega \wedge \omega,$$

are called the **structure equations**. Show that τ and Ω satisfy

$$d\tau = \Omega \wedge \theta - \omega \wedge \tau,$$

$$d\Omega = \Omega \wedge \omega - \omega \wedge \Omega.$$

These two equations are called the **Bianchi identities**. Furthermore, show that

$$d(\Omega^k) = \Omega^k \wedge \omega - \omega \wedge \Omega^k.$$

Exercise 6.6. Show that if ∇ is torsion-free, then $d\tau = \Omega \wedge \theta - \omega \wedge \tau$ is equivalent to the **first Bianchi identity**

$$R(X, Y)Z + R(Y, Z)X + R(Z, X)Y = 0.$$

Exercise 6.7. Show that if ∇ is torsion-free, then $d\Omega = \Omega \wedge \omega - \omega \wedge \Omega$ is equivalent to the **second Bianchi identity**

$$(\nabla_X R)(Y, Z) + (\nabla_Y R)(Z, X) + (\nabla_Z R)(X, Y) = 0.$$

Exercise 6.8. For $\{e_i\} = \{\frac{\partial}{\partial x^i}\}$, $\{\tilde{e}_j\} = \{\frac{\partial}{\partial \tilde{x}^j}\}$, $\omega_j^k = \Gamma_{ij}^k dx^i$, show that the transformation formula

$$\tilde{\omega} = a^{-1} da + a^{-1} \omega a$$

reduces to the well-known transformation formula of Christoffel symbols:

$$\tilde{\Gamma}_{ij}^k = \frac{\partial \tilde{x}^k}{\partial x^l} \frac{\partial x^m}{\partial \tilde{x}^i} \frac{\partial x^n}{\partial \tilde{x}^j} \Gamma_{mn}^l + \frac{\partial^2 x^l}{\partial \tilde{x}^i \partial \tilde{x}^j} \frac{\partial \tilde{x}^k}{\partial x^l}.$$

7 Principal Bundles and Connections

7.1 Principal Bundles

Principal bundles are both a special type of fiber bundle and the most important one. In the following, we will introduce the definition of principal bundles and some basic structures on them.

Definition 7.1 (G -action). A **smooth right action** of a Lie group G on a manifold P is a smooth map

$$\mu : M \times G \rightarrow M,$$

denoted by $\mu(x, g) := x \cdot g$, such that for all $x \in M$ and $g, h \in G$,

$$x \cdot e = x, \quad (x \cdot g) \cdot h = x \cdot (gh).$$

where e is the identity element of G . A manifold M with a smooth right G -action is called a **(right) G -manifold**.

We say the action is **free** if for all $x \in M$, $x \cdot g = x$ implies $g = e$.

Definition 7.2 (G -equivariant map). Let M and N be two G -manifolds. A smooth map $f : M \rightarrow N$ is called **G -equivariant** if for all $x \in M$ and $g \in G$,

$$f(x \cdot g) = f(x) \cdot g.$$

Definition 7.3 (Principal G -bundle). A **principal G -bundle** is a fiber bundle $\pi : P \rightarrow M$ with fiber G together with a free right G -action on P such that for each $x \in M$, the local trivialization

$$\phi_U : \pi^{-1}(U) \rightarrow U \times G,$$

is G -equivariant, where the right G -action on $U \times G$ is given by

$$(x, h) \cdot g = (x, hg), \quad \forall x \in U, g, h \in G.$$

By a **local trivialization** we mean a open neighborhood U of x and a diffeomorphism $\phi_U : \pi^{-1}(U) \rightarrow U \times G$ satisfying the following commutative diagram:

$$\begin{array}{ccc} \pi^{-1}(U) & \xrightarrow{\phi_U} & U \times G \\ & \searrow \pi & \swarrow \text{pr}_1 \\ & U & \end{array}$$

where $\text{pr}_1 : U \times G \rightarrow U$ is the projection map.

Example 7.4. Let G be a Lie group and M be a smooth manifold, then the trivial bundle $p_1 : M \times G \rightarrow M$ is a principal G -bundle over M .

Example 7.5. If G is a Lie group and M is a smooth manifold, G acts on M freely and properly, then the quotient space M/G is a smooth manifold and the projection map $\pi : M \rightarrow M/G$ is a principal G -bundle over M/G . Especially, if G is a finite group equipped with the discrete topology, then a principal G -bundle is exactly a covering space.

Example 7.6. Let G be a Lie group and H be a closed subgroup of G . Then the quotient space G/H is a smooth manifold and the projection map $\pi : G \rightarrow G/H$ is a principal H -bundle over G/H . For example, the Hopf fibration $S^1 \rightarrow S^3 \rightarrow S^2$.

Example 7.7. Let $\pi : E \rightarrow M$ be a rank m real (complex) vector bundle over M . Then the frame bundle $\text{Fr}(E)$ is a principal $\text{GL}(m, \mathbb{R})$ -bundle ($\text{GL}(m, \mathbb{C})$ -bundle) over M .

Remark. In general, $\text{Aut}(E)$ is not a principal bundle.

Cocycle Definition of Principal Bundles (Čech Cohomology Perspective)

Proposition 7.8. Let $\pi : P \rightarrow M$ be a principal G -bundle. Then the right G -action on P is transitive on each fiber.

Proof. Since G acts transitively on $\{x\} \times G$ and the fiber diffeomorphism $\phi_{U,x} : P_x \rightarrow G$ is G -equivariant, thus G acts transitively on P_x . $\square$

Lemma 7.9. For any group G , all right G -equivariant maps are exactly the left translations.

Proof. Let $f : G \rightarrow G$ be a right G -equivariant map, then for all $g \in G$,

$$f(g) = f(e \cdot g) = f(e) \cdot g = l_{f(e)}(g),$$

where $l_{f(e)} : G \rightarrow G$ is the left translation by $f(e)$. $\square$

Definition 7.10 (Cocycle definition). A **principal G -bundle** is a fiber bundle $\pi : P \rightarrow M$ with fiber G together with a local trivialization $\{(U_\alpha, \phi_\alpha)\}$ such that for each $U_{\alpha\beta} := U_\alpha \cap U_\beta \neq \emptyset$, there exists a smooth map $g_{\alpha\beta} : U_{\alpha\beta} \rightarrow G$ such that

$$\phi_\alpha \circ \phi_\beta^{-1}(x, h) = (x, g_{\alpha\beta}(x)h), \quad \forall (x, h) \in U_{\alpha\beta} \times G.$$

In addition, the transition functions $\{g_{\alpha\beta}\}$ satisfy the following cocycle condition:

$$g_{\alpha\beta}(x)g_{\beta\gamma}(x) = g_{\alpha\gamma}(x), \quad \forall x \in U_\alpha \cap U_\beta \cap U_\gamma \neq \emptyset.$$

Remark. By Lemma ??, one can easily check that the two definitions of principal bundles are equivalent. These two definitions characterize G -principal bundles from different perspectives.

Example 7.11. Suppose E is a rank m real (complex) vector bundle over M . Let $\{(U_\alpha, \psi_\alpha)\}$ be a local trivialization

$$\begin{aligned} \psi_\alpha : \pi^{-1}(U_\alpha) &\rightarrow U_\alpha \times \mathbb{R}^m \\ (x, v) &\mapsto (x, \psi_{\alpha,x}(v)), \end{aligned}$$

with transition functions $\{g_{\alpha\beta}\}$:

$$\begin{aligned} \psi_\alpha \circ \psi_\beta^{-1} : U_{\alpha\beta} \times \mathbb{R}^m &\rightarrow U_{\alpha\beta} \times \mathbb{R}^m \\ (x, v_\beta) &\mapsto (x, \psi_{\alpha,x} \circ \psi_{\beta,x}^{-1}(v_\beta)) = (x, g_{\alpha\beta}(x)v_\beta). \end{aligned}$$

We define the local trivialization $\{(U_\alpha, \tilde{\psi}_\alpha)\}$ of $\text{Fr}(E)$ by

$$\begin{aligned}\tilde{\psi}_\alpha &: \pi_{\text{Fr}}^{-1}(U_\alpha) \rightarrow U_\alpha \times \text{GL}(m, \mathbb{R}) \\ (x, (e_1, \dots, e_m)) &\mapsto (x, \psi_{\alpha,x}(e_1, \dots, e_m)).\end{aligned}$$

here $(e_1, \dots, e_m)$ is a frame of E_x , and $\psi_{\alpha,x}(e_1, \dots, e_m) = (\psi_{\alpha,x}(e_1), \dots, \psi_{\alpha,x}(e_m))$ is the corresponding frame of $\mathbb{R}^m$. Since

$$\begin{aligned}\tilde{\psi}_\alpha \circ \tilde{\psi}_\beta^{-1} &: U_{\alpha\beta} \times \text{GL}(m, \mathbb{R}) \rightarrow U_{\alpha\beta} \times \text{GL}(m, \mathbb{R}) \\ (x, (e_{1,\beta}, \dots, e_{m,\beta})) &\mapsto (x, \psi_{\alpha,x} \circ \psi_{\beta,x}^{-1}(e_{1,\beta}, \dots, e_{m,\beta})) = (x, g_{\alpha\beta}(x)(e_{1,\beta}, \dots, e_{m,\beta})).\end{aligned}$$

The transition functions of $\text{Fr}(E)$ are exactly the same as those of E .

Fundamental Vector Fields

Suppose P is a G -manifold, then for each $A \in \mathfrak{g}$, we can define a vector field $\underline{A}$ on P by

$$\underline{A}_p := \left. \frac{d}{dt} \right|_{t=0} p \cdot \exp(tA), \quad \forall p \in P.$$

The vector field $\underline{A}$ is called the **fundamental vector field** associated to A .

Fact 4. For each $A \in \mathfrak{g}$, $\underline{A}$ is a C^∞ vector field on P .

The fundamental vector field construction gives rise to a map

$$\sigma : \mathfrak{g} \rightarrow \mathfrak{X}(P),^8 \quad \sigma(A) := \underline{A}.$$

For $p \in P$, define $j_p : G \rightarrow P$ by $j_p(g) := p \cdot g$, then $j_{p*} := j_{p*e}$ is

$$j_{p*}(A) = \left. \frac{d}{dt} \right|_{t=0} j_p(\exp(tA)) = \left. \frac{d}{dt} \right|_{t=0} p \cdot \exp(tA) = \underline{A}_p.$$

This gives another discription of the fundamental vector field: $\underline{A}_p = j_{p*}(A)$. It also implies that the map $\sigma : \mathfrak{g} \rightarrow \mathfrak{X}(P)$ is $\mathbb{R}$ -linear. In fact, we have

Fact 5. The map $\sigma : \mathfrak{g} \rightarrow \mathfrak{X}(P)$ is a Lie algebra homomorphism, i.e.,

$$[\underline{A}, \underline{B}] = \underline{[A, B]}, \quad \forall A, B \in \mathfrak{g}.$$

Fact 6. Let P be a G -manifold, then the fundamental vector fields $\underline{A}$ vanishes at p if and only if $A \in \text{Lie}(\text{Stab}(p))$. Thus for any $p \in P$, the kernel of $j_{p*} : \mathfrak{g} \rightarrow T_p P$ is exactly $\text{Lie}(\text{Stab}(p))$. In particular, if the G -action on P is free, then j_{p*} is injective for all $p \in P$.

Example 7.12. Let G acts on itself by **right translation**, then the fundamental vector field $\underline{A}$ is exactly the **left-invariant** vector field generated by A .

For $g \in G$, let $c_g : G \rightarrow G$ be the conjugation map $c_g(h) := ghg^{-1}$. The differential of c_g at the identity element e is the adjoint representation $\text{Ad}_g := (c_g)_* : \mathfrak{g} \rightarrow \mathfrak{g}$.

⁸ $\mathfrak{X}(P) := C^\infty(P, TP)$ is the space of smooth vector fields on P .

Proposition 7.13. *For each $A \in \mathfrak{g}$ and $g \in G$, we have*

$$(r_g)_* \underline{A} = \underline{\text{Ad}_{g^{-1}}(A)}.$$

where $r_g : P \rightarrow P$ is the right translation by g , i.e., $r_g(p) := p \cdot g$.

Proof. For each $p \in P$ and $h \in G$, we have

$$(r_g \circ j_p)(h) = r_g(p \cdot h) = (p \cdot h) \cdot g = pg \cdot (g^{-1}hg) = (j_{pg} \circ c_{g^{-1}})(h).$$

By the chain rule,

$$(r_g)_*(\underline{A}_p) = (r_g)_*j_{p*}(A) = j_{pg*}(c_{g^{-1}})_*(A) = j_{pg*}(\text{Ad}_{g^{-1}}(A)) = \underline{\text{Ad}_{g^{-1}}(A)}_{pg}. \quad \square$$

7.1.1 Vertical Subbundle of TP

Let $\pi : P \rightarrow M$ be a principal G -bundle, at each $p \in P$, the differential $\pi_{*p} : T_p P \rightarrow T_{\pi(p)} M$ is surjective. We define $\mathcal{V}_p := \ker \pi_{*p} \subset T_p P$ to be the **vertical tangent space** at p . Then $\mathcal{V} := \bigsqcup_{p \in P} \mathcal{V}_p$ is a subbundle of TP , called the **vertical subbundle** of TP . An element of $\mathcal{V}_p$ is called a **vertical tangent vector** at p .

We have a short exact sequence of vector spaces

$$0 \longrightarrow \mathcal{V}_p \hookrightarrow T_p P \xrightarrow{\pi_{*p}} T_{\pi(p)} M \longrightarrow 0$$

and

$$\dim \mathcal{V}_p = \dim T_p P - \dim T_{\pi(p)} M = \dim G.$$

Proposition 7.14. *For any $A \in \mathfrak{g}$, the fundamental vector field $\underline{A}$ is vertical at every point $p \in P$.*

Proof. Since $(\pi \circ j_p)(g) = \pi(p \cdot g) = \pi(p)$ is a constant map for all $g \in G$, we have $\pi_{*p}(\underline{A}_p) = \pi_{*p}j_{p*}(A) = (\pi \circ j_p)_*(A) = 0$. $\square$

This proposition implies that $j_{p*}(\mathfrak{g}) \subset \mathcal{V}_p$ for all $p \in P$. In fact, we have the following stronger result.

Proposition 7.15. *For each $p \in P$, the map $j_{p*} : \mathfrak{g} \rightarrow \mathcal{V}_p$ is an isomorphism.*

Proof. Since G acts freely on P , the map j_{p*} is injective. On the other hand, $\dim \mathcal{V}_p = \dim G = \dim \mathfrak{g}$, thus j_{p*} is an isomorphism. $\square$

Corollary 7.16. *The vertical vectors at $p \in P$ are exactly the fundamental vector at p . Moreover, the vertical subbundle $\mathcal{V}$ of TP is isomorphic to the trivial bundle $P \times \mathfrak{g}$.*

Proof. The first statement follows from the previous proposition. For the second statement, let $A_1, \dots, A_l$ be a basis of $\mathfrak{g}$, where $l = \dim G$, then $\underline{A}_1, \dots, \underline{A}_l$ form a global frame of $\mathcal{V}$, thus $\mathcal{V}$ is a trivial bundle. $\square$

7.2 Connections on Principal Bundles

7.2.1 Horizontal Distributions

The map $\pi_* : TP \rightarrow TM$ induces a bundle map $\tilde{\pi}_* : TP \rightarrow \pi^*TM$ over P . Then we have a short exact sequence of vector bundles over P :

$$0 \longrightarrow \mathcal{V} \hookrightarrow TP \xrightarrow{\tilde{\pi}_*} \pi^*TM \longrightarrow 0$$

In the category of smooth vector bundles, all short exact sequence of vector bundles splits. Thus there exists a smooth subbundle $\mathcal{H}$ of TP such that $TP = \mathcal{V} \oplus \mathcal{H}$.

Definition 7.17. Let $\pi : P \rightarrow M$ be a principal G -bundle, a **horizontal distribution** of TP is a smooth subbundle $\mathcal{H}$ of TP such that

- $TP = \mathcal{V} \oplus \mathcal{H}$,
- $\mathcal{H}$ is **right-invariant**, i.e., $r_{g*}\mathcal{H} = \mathcal{H}$, $\forall g \in G$, where $r_g : P \rightarrow P$, $r_g(p) = p \cdot g$.

Definition 7.18. Given a horizontal distribution $\mathcal{H}$ of TP , For each $X \in C^\infty(P, TP)$, we can write

$$X = hX + vX, \quad hX \in C^\infty(P, \mathcal{H}), vX \in C^\infty(P, \mathcal{V}).$$

We call hX the **horizontal component** of X and vX the **vertical component** of X , both of them are smooth vector fields on P .

Proposition 7.19. For each $g \in G$, the right translation r_{g*} commutes with the horizontal and vertical projections h and v , i.e.,

$$r_{g*}(hX) = h(r_{g*}X), \quad r_{g*}(vX) = v(r_{g*}X).$$

Proof. Since $\mathcal{H}$ is right-invariant, $r_{g*}(hX)$ is still a horizontal vector field and $r_{g*}(vX)$ is still a vertical vector field, thus

$$r_{g*}(X) = r_{g*}(hX) + r_{g*}(vX) = h(r_{g*}X) + v(r_{g*}X) \implies \begin{cases} r_{g*}(hX) = h(r_{g*}X), \\ r_{g*}(vX) = v(r_{g*}X). \end{cases} \quad \square$$

Remark. In general, there is no canonical choice of $\mathcal{H}$. A choice of $\mathcal{H}$ is a splitting $k : \pi^*TM \rightarrow TP$ such that $\tilde{\pi}_* \circ k = \text{id}_{\pi^*TM}$, together with a G -invariance condition.

$$0 \longrightarrow \mathcal{V} \hookrightarrow TP \xrightleftharpoons[k]{\tilde{\pi}_*} \pi^*TM \longrightarrow 0$$

7.2.2 Ehresmann Connections

Let $\pi : P \rightarrow M$ be a principal G -bundle, $\mathcal{H}$ be a horizontal distribution of TP . We define the projection map⁹

$$\text{pr}_{\mathcal{V}} : TP = \mathcal{V} \oplus \mathcal{H} \rightarrow \mathcal{V},$$

⁹Although $\mathcal{V}$ is intrinsically defined, the vertical component of a tangent vector depends on the choice of $\mathcal{H}$.

Now we have a splitting:

$$0 \longrightarrow \mathcal{V} \xrightleftharpoons[\text{pr}_{\mathcal{V}}]{\iota} TP \xrightarrow{\tilde{\pi}_*} \pi^*TM \longrightarrow 0$$

where $\iota : \mathcal{V} \hookrightarrow TP$ is the inclusion map. We define a $\mathfrak{g}$ -valued 1-form ω on P by

$$\omega_p := j_{p*}^{-1} \circ \text{pr}_{\mathcal{V}} : T_pP \xrightarrow{\text{pr}_{\mathcal{V}}} \mathcal{V}_p \xrightarrow{j_{p*}^{-1}} \mathfrak{g}.$$

In terms of ω , the vertical component of $Y_p \in T_pP$ is given by

$$\text{pr}_{\mathcal{V}}(Y_p) = j_{p*}(\omega_p(Y_p)).$$

In summary, given a projection $\text{pr}_{\mathcal{V}} : TP \rightarrow \mathcal{V}$, we induce the following composition of maps:

$$\omega : TP \xrightarrow{\text{pr}_{\mathcal{V}}} \mathcal{V} \xrightarrow{\cong} P \times \mathfrak{g} \xrightarrow{\text{pr}_2} \mathfrak{g} \rightsquigarrow \omega \in \Omega^1(P, \mathfrak{g}).$$

So the information of the horizontal distribution $\mathcal{H}$ is encoded in ω .

Theorem 7.20. *If $\mathcal{H}$ is a smooth, right-invariant horizontal distribution, then the $\mathfrak{g}$ -valued 1-form ω defined above satisfies the following properties:*

- $\omega_p(\underline{A}_p) = A, \forall p \in P, A \in \mathfrak{g}.$
- (G -equivariance) $(r_g)^*\omega = (\text{Ad}_{g^{-1}})\omega, \forall g \in G.$
- ω is smooth.

We say a $\mathfrak{g}$ -valued 1-form ω on P satisfying the above properties is an **Ehresmann connection** or simply a **connection** on P .

Proof. 1. Since $\underline{A}$ is already vertical,

$$\omega_p(\underline{A}_p) = j_{p*}^{-1}(\underline{A}_p) = A.$$

2. For each $p \in P$ and $Y_p \in T_pP$, we need to show

$$\omega_{pg}((r_g)_*Y_p) = (\text{Ad}_{g^{-1}})\omega_p(Y_p).$$

Since both side are linear in Y_p , we only need to check the above equality for $Y_p \in \mathcal{V}_p$ and $Y_p \in \mathcal{H}_p$ respectively.

If $Y_p \in \mathcal{V}_p$, then $Y_p = \underline{A}_p$ for some $A \in \mathfrak{g}$, thus

$$\omega_{pg}(r_{g*}\underline{A}_p) = \omega_{pg}(\underline{\text{Ad}_{g^{-1}}(A)})_{pg} = \text{Ad}_{g^{-1}}(A) = (\text{Ad}_{g^{-1}})\omega_p(\underline{A}_p).$$

If $Y_p \in \mathcal{H}_p$, by the right-invariance of $\mathcal{H}$, we have $r_{g*}(Y_p) \in \mathcal{H}_{pg}$, thus

$$\omega_{pg}(r_{g*}Y_p) = 0 = (\text{Ad}_{g^{-1}})\omega_p(Y_p).$$

3. Omit. □

Viewing ω as a map from TP to $\mathfrak{g}$. Both TP and $\mathfrak{g}$ are G -manifolds, where

$$g \cdot Y_p := (r_g)_* Y_p, \quad g \cdot A := \text{Ad}_{g^{-1}}(A).$$

Then the G -equivariance condition of ω

$$\omega_{pg}((r_g)_* Y_p) = (\text{Ad}_{g^{-1}}) \omega_p(Y_p),$$

can be rephrased as $\omega : TP \rightarrow \mathfrak{g}$ is a G -equivariant map.

Example 7.21. The Maurer-Cartan form ω_{MC} on a Lie group G is the unique $\mathfrak{g}$ -valued 1-form on G such that

- $\omega_{\text{MC}}(A) = A$ for all $A \in \mathfrak{g} = T_e G$,
- ω_{MC} is left-invariant, i.e., $l_g^* \omega_{\text{MC}} = \omega_{\text{MC}}$ for all $g \in G$.

Let ω_{MC} be the Maurer-Cartan form on G , then $\pi^* \omega_{\text{MC}}$ is an Ehresmann connection on the trivial bundle $\pi : M \times G \rightarrow M$.

Remark. Let ω be an Ehresmann connection on P , for any $p \in P$, $j_p : G \rightarrow P$ pulls back ω to ω_{MC} on G , i.e., $j_p^* \omega = \omega_{\text{MC}}$.

The Horizontal Distribution of an Ehresmann Connection

In the previous section we have shown that a smooth, right-invariant horizontal distribution $\mathcal{H}$ on P induces an Ehresmann connection ω . We now prove the converse.

Theorem 7.22. *If ω is an Ehresmann connection on the principal G -bundle $\pi : P \rightarrow M$, then $\mathcal{H}_p := \ker \omega_p$, $p \in P$ defines a smooth, right-invariant horizontal distribution on P .*

Proof. We need to verify three properties:

1. For each $p \in P$, there is an exact sequence

$$0 \longrightarrow \mathcal{H}_p \hookrightarrow T_p P \xrightarrow{\omega_p} \mathfrak{g} \longrightarrow 0$$

The map $j_{p*} : \mathfrak{g} \rightarrow \mathcal{V}_p \subset T_p P$ provides a splitting of the above sequence, thus

$$T_p P \cong \mathfrak{g} \oplus \mathcal{H}_p \cong \mathcal{V}_p \oplus \mathcal{H}_p.$$

2. Suppose $Y_p \in \mathcal{H}_p$, then $\omega_p(Y_p) = 0$. By the G -equivalence of ω , we have

$$\omega_{pg}(r_{g*} Y_p) = (r_g^* \omega)_p(Y_p) = (\text{Ad}_{g^{-1}}) \omega_p(Y_p) = 0.$$

Hence $r_{g*} Y_p \in \mathcal{H}_{pg}$.

3. $\mathcal{H}$ is a smooth distribution. We omit the details here. □

7.2.3 Horizontal Lift and Parallel Transport

Let $\pi : P \rightarrow M$ be a principal G -bundle and ω be an Ehresmann connection on P . For each (piecewise) smooth curve $\gamma : [0, 1] \rightarrow M$ and $p \in P$ such that $\pi(p) = \gamma(0)$, there exists a unique (piecewise) smooth curve $\tilde{\gamma} : [0, 1] \rightarrow P$ such that

$$\tilde{\gamma}(0) = p, \quad \pi \circ \tilde{\gamma} = \gamma, \quad \text{and} \quad \tilde{\gamma}'(t) \in \mathcal{H}_{\tilde{\gamma}(t)}, \quad \forall t \in [0, 1].$$

Definition 7.23. We call $\tilde{\gamma}$ the **horizontal lift** of γ starting at p . If we denote $\tilde{\gamma}_p$ to be the horizontal lift of γ starting at p , then the map

$$P^\gamma : \text{Fiber}_{\gamma(0)} \rightarrow \text{Fiber}_{\gamma(1)}, \quad P^\gamma(p) := \tilde{\gamma}_p(1),$$

is called the **parallel transport** along γ .

Proposition 7.24. Let $\tilde{\gamma}_p$ be the horizontal lift of γ starting at p , then $\tilde{\gamma}_{pg} = \tilde{\gamma}_p \cdot g$ for any $g \in G$.

Proof. By the right-invariance of $\mathcal{H}$, we have

$$(\tilde{\gamma}_p \cdot g)(0) = pg, \quad \pi \circ (\tilde{\gamma}_p \cdot g) = \pi \circ \tilde{\gamma}_p = \gamma \quad \text{and} \quad (\tilde{\gamma}_p \cdot g)'(t) = r_{g*} \tilde{\gamma}'_p(t) \in \mathcal{H}_{(\tilde{\gamma}_p \cdot g)(t)}, \quad \forall t \in [0, 1].$$

Thus $\tilde{\gamma}_p \cdot g$ is the horizontal lift of γ starting at pg , i.e., $\tilde{\gamma}_{pg} = \tilde{\gamma}_p \cdot g$. $\square$

Corollary 7.25. The parallel transport P^γ is a diffeomorphism between the fibers $\text{Fiber}_{\gamma(0)}$ and $\text{Fiber}_{\gamma(1)}$.

For any two points $x, y \in M$, we can choose a curve γ connecting x and y , then the parallel transport P^γ gives a diffeomorphism between the fibers Fiber_x and Fiber_y . In this sense, a connection provides a way to identify fibers over different points of the base manifold.

Like the case of affine connections on vector bundles, a family of parallel transports along all curves in M also determines a unique Ehresmann connection on P .

7.2.4 Connections on the frame bundle $\text{Fr}(E)$

In the previous sections, we have defined affine connections on vector bundles E . Since the frame bundle $\text{Fr}(E)$ is a natural principal bundle corresponding to E , we can study the relation between connections on E and connections on $\text{Fr}(E)$.

Let ∇ be an affine connection on E . Then for any curve $\gamma : [0, 1] \rightarrow M$, we have a parallel transport $P_t^\gamma : E_{\gamma(0)} \rightarrow E_{\gamma(t)}$ defined by ∇ . Since P_t^γ is a linear isomorphism, it induces a diffeomorphism $\tilde{P}_t^\gamma : \text{Fr}(E_{\gamma(0)}) \rightarrow \text{Fr}(E_{\gamma(t)})$ by

$$\tilde{P}_t^\gamma(e_1, \dots, e_m) := (P_t^\gamma(e_1), \dots, P_t^\gamma(e_m)),$$

where $(e_1, \dots, e_m) \in \text{Fr}(E_{\gamma(0)})$ is a frame of $E_{\gamma(0)}$. Then $\tilde{\gamma}_e(t) = \tilde{P}_t^\gamma(e_1, \dots, e_m)$ is a lift of γ starting at $e = (e_1, \dots, e_m)$.

Definition 7.26. The **horizontal distribution** of $\text{Fr}(E)$ induced by ∇ is defined by

$$\mathcal{H}_e := \{\tilde{\gamma}'_e(0) \mid \gamma : [0, 1] \rightarrow M, \gamma(0) = \pi(e)\},$$

and $\mathcal{H} := \bigsqcup_{e \in \text{Fr}(E)} \mathcal{H}_e$

Proposition 7.27. $\mathcal{H}$ is a smooth, right-invariant horizontal distribution on $\text{Fr}(E)$.

Proof. 1. Smoothness: omit. 2. $T_e \text{Fr}(E) = \mathcal{V}_e \oplus \mathcal{H}_e$: omit.

3. Right-invariance: Let $e = (e_1, \dots, e_m) \in \text{Fr}(E)$ and $a \in \text{GL}(m, \mathbb{R})$, since P_t^γ is a linear isomorphism, we have

$$\tilde{\gamma}_{e \cdot a}(t) = \tilde{P}_t^\gamma((e_1, \dots, e_m)a) = (P_t^\gamma(e_1), \dots, P_t^\gamma(e_m))a = \tilde{P}_t^\gamma(e_1, \dots, e_m) \cdot a = \tilde{\gamma}_e(t) \cdot a,$$

Thus $r_{a*} \tilde{\gamma}'_e(0) = \tilde{\gamma}'_{e \cdot a}(0)$, which implies $r_{a*} \mathcal{H}_e = \mathcal{H}_{e \cdot a}$. $\square$

Since now we got a horizontal distribution $\mathcal{H}$ on $\text{Fr}(E)$, we can define an Ehresmann connection on $\text{Fr}(E)$ — the $\mathfrak{gl}(n, \mathbb{R})$ -valued 1-form ω . Recall that given a local trivialization $\{(U_\alpha, \psi_\alpha)\}$ of E , we have a local connection 1-form ω_α on U_α defined by ∇ . There is no doubt that $\omega_\alpha \in \Omega^1(U_\alpha, \mathfrak{gl}(m, \mathbb{R}))$ is related to $\omega \in \Omega^1(\text{Fr}(E), \mathfrak{gl}(m, \mathbb{R}))$ in some way.

Theorem 7.28. In the notation above, let ω be the Ehresmann connection on $\text{Fr}(E)$ induced by ∇ , then we have

$$\omega_\alpha = s_\alpha^* \omega.$$

Note that $s_\alpha : U_\alpha \rightarrow \text{Fr}(E)$ can be viewed as a local section of $\text{Fr}(E)$. *So the local connection 1-forms ω_α can be viewed as the pullback of an intrinsic object ω on $\text{Fr}(E)$.*

To prove the above theorem, we need the following lemmas.

Lemma 7.29. Let $\pi : E \rightarrow M$ be a rank m bundle with a connection ∇ , $x \in M$ and $e_x \in \text{Fr}(E_x)$. For any smooth curve $\gamma : [0, 1] \rightarrow M$ starting at x , there is a unique horizontal lift $\tilde{\gamma} : [0, 1] \rightarrow \text{Fr}(E)$ with the initial point $e_x = (e_1, \dots, e_m)$. Let $s : U \rightarrow \text{Fr}(E)$ be a local frame of E around x , with $s(x) = e_x$, then $s(\gamma(t))$ is also a lift of γ starting at e_x . Both $\tilde{\gamma}(t)$ and $s(\gamma(t))$ are frames of $E_{\gamma(t)}$, thus there exists a unique $a(t) \in \text{GL}(m, \mathbb{R})$ such that

$$s(\gamma(t)) = \tilde{\gamma}(t) \cdot a(t).$$

Then we have the following formula

$$s_*(\gamma'(0)) = \tilde{\gamma}'(0) + \underline{a'(0)}_{e_x}.$$

Proof. Let $\mu : \text{Fr}(E) \times \text{GL}(m, \mathbb{R}) \rightarrow \text{Fr}(E)$ be the right action of $\text{GL}(m, \mathbb{R})$ on $\text{Fr}(E)$. Then

$$s(\gamma(t)) = \mu(\tilde{\gamma}(t), a(t)),$$

with $\gamma(0) = x$, $\tilde{\gamma}(0) = e_x$ and $a(0) = I$.

$$\begin{aligned} s_*(\gamma'(0)) &= \mu_{*(\tilde{\gamma}(0), a(0))}(\tilde{\gamma}'(0), a'(0)) \\ &= \mu_{*(e_x, I)}(\tilde{\gamma}'(0), 0) + \mu_{*(e_x, I)}(0, a'(0)) \\ &= \tilde{\gamma}'(0) + \underline{a'(0)}_{e_x}. \end{aligned} \quad \square$$

Lemma 7.30. Since $s = (s_1, \dots, s_m)$ is a local frame of E , we can define a local connection 1-form ω_s of ∇ w.r.t s . Then we have $\omega_s(\gamma'(0)) = a'(0)$.

Proof. Let $\tilde{\gamma}(t) = (\tilde{\gamma}_1(t), \dots, \tilde{\gamma}_m(t))$, then $\tilde{\gamma}_i(t)$ is the parallel transport of e_i along γ . Now we have

$$s(\gamma(t))_i = \tilde{\gamma}_j(t)a_i^j(t),$$

take the covariant derivative of both side at $t = 0$, we get

$$\omega_i^j(\gamma'(0))s_j(\gamma(0)) = \nabla_{\gamma'(0)}s_i = \nabla_{\gamma'(0)}(\tilde{\gamma}_j(t)a_i^j(t)) = a_i^{j'}(0)\tilde{\gamma}_j(0).$$

Both $s(\gamma(0))$ and $\tilde{\gamma}(0)$ are $e_x = (e_1, \dots, e_m)$, thus we have $\omega_s(\gamma'(0)) = a'(0)$. $\square$

Corollary 7.31. *In the notation above, we have*

$$\tilde{\gamma}'(0) = s_*(\gamma'(0)) - \underline{\omega_s(\gamma'(0))}_{e_x}.$$

Corollary 7.32. *Let ∇ be an affine connection on $\pi : E \rightarrow M$. Suppose $\{(U_\alpha, s_\alpha)\}$ is a local trivialization of E , then we have a family of connection 1-forms ω_α of ∇ w.r.t s_α . For $x \in U_\alpha$, $p = s_\alpha(x) \in \text{Fr}(E_x)$ and $X_x \in T_x M$, let $\tilde{X}_p \in T_p \text{Fr}(E)$ be the horizontal lift of X_x in $\text{Fr}(E)$, then we have*

$$\tilde{X}_p = (s_\alpha)_*(X_x) - \underline{\omega_\alpha(X_x)}_p.$$

Now we are ready to prove the previous theorem.

Proof of Theorem ??. Let $x \in U_\alpha$, $p = s_\alpha(x) \in \text{Fr}(E_x)$ and $X_x \in T_x M$. Let $\tilde{X}_p \in T_p \text{Fr}(E)$ be the horizontal lift of X_x in $\text{Fr}(E)$, then we have

$$\tilde{X}_p = (s_\alpha)_*(X_x) - \underline{\omega_\alpha(X_x)}_p.$$

Applying ω_p to both side, we get

$$0 = \omega_p(\tilde{X}_p) = (s_\alpha^*\omega)_x(X_x) - \omega_p(\underline{\omega_\alpha(X_x)}_p) = (s_\alpha^*\omega)_x(X_x) - \omega_\alpha(X_x).$$

Since X_x is arbitrary, we have $\omega_\alpha = s_\alpha^*\omega$. $\square$

7.2.5 Curvature of Connections

Recall that for an affine connection ∇ on a vector bundle E , we have defined the curvature $R = \nabla^2 \in \Omega^2(M, \text{End}(E))$. Under a local trivialization $\{(U_\alpha, s_\alpha)\}$ of E , we have a family of local connection 1-forms ω_α , and the curvature 2-form Ω_α is given by

$$\Omega_\alpha = d\omega_\alpha + \omega_\alpha \wedge \omega_\alpha.$$

If we rewrite it in terms of Lie bracket, we have

$$\Omega_\alpha = d\omega_\alpha + \frac{1}{2}[\omega_\alpha, \omega_\alpha].$$

This formula can generalize to the case of Ehresmann connections on principal bundles.

Definition 7.33. Let $\pi : P \rightarrow M$ be a principal G -bundle and ω be an Ehresmann connection on P . The **curvature** of ω is the $\mathfrak{g}$ -valued 2-form $\Omega \in \Omega^2(P, \mathfrak{g})$ defined by

$$\Omega = d\omega + \frac{1}{2}[\omega, \omega].$$

Proposition 7.34. Let E be a vector bundle over M and $\text{Fr}(E)$ be the frame bundle of E . The affine connection ∇ on E induces an Ehresmann connection ω on $\text{Fr}(E)$. Let $\{(U_\alpha, s_\alpha)\}$ be a local trivialization of E , then the curvature Ω of ω satisfies

$$\Omega_\alpha = s_\alpha^* \Omega,$$

where Ω_α is the curvature 2-form of ∇ w.r.t s_α . *Thus the curvature form Ω pulls back by sections to the local curvature forms Ω_α .*

Proof. By Theorem ??, we have $\omega_\alpha = s_\alpha^* \omega$, thus

$$s_\alpha^* \Omega = s_\alpha^* d\omega + \frac{1}{2} s_\alpha^* [\omega, \omega] = d(s_\alpha^* \omega) + \frac{1}{2} [s_\alpha^* \omega, s_\alpha^* \omega] = d\omega_\alpha + \frac{1}{2} [\omega_\alpha, \omega_\alpha] = \Omega_\alpha. \quad \square$$

Theorem 7.35. Let ω be a connection on a principal G -bundle $\pi : P \rightarrow M$, then the curvature Ω of ω satisfies the following properties:

1. (Horizontal) $\Omega_p(X_p, Y_p) = (d\omega)_p(hX_p, hY_p)$ for all $p \in P$ and $X_p, Y_p \in T_p P$.
2. (G -equivariance) $(r_g)^* \Omega = (\text{Ad}_{g^{-1}}) \Omega$ for all $g \in G$.
3. (Second Bianchi identity) $d\Omega = [\Omega, \omega]$.

To prove the above theorem, we need some lemmas.

Lemma 7.36. Let $\pi : P \rightarrow M$ be a principal G -bundle and ω be an Ehresmann connection on P . Fix a point $p \in P$.

1. Every vertical vector $X_p \in T_p P$ can be extended to a fundamental vector field $\underline{A}$ for some $A \in \mathfrak{g}$.
2. Every horizontal vector $Y_p \in T_p P$ can be extended to a horizontal lift $\tilde{B}$ of some vector field B on M .

Proof. 1. By the isomorphism $j_{p*} : \mathfrak{g} \rightarrow \mathcal{V}_p$, there exists a unique $A \in \mathfrak{g}$ such that $X_p = j_{p*}(A) = \underline{A}_p$. Then the fundamental vector field $\underline{A}$ extends X_p .

2. Let $x = \pi(p)$ and $B_x = \pi_*(Y_p) \in T_x M$. We can extend B_x to a vector field B on M . Then the horizontal lift $\tilde{B}$ of B extends Y_p . $\square$

Lemma 7.37. Suppose P is a principal G -bundle with a connection. Let $\underline{A}$ be a fundamental vector field.

1. If Y is a horizontal vector field on P , then $[\underline{A}, Y]$ is also horizontal.
2. If Y is a right-invariant vector field on P , then $[\underline{A}, Y] = 0$.

Proof. The fundamental vector field $\underline{A}$ generates a flow $\phi_t(p) = p \cdot e(tA) = r_{e^{tA}}(p)$ on P . By the property of Lie bracket, we have

$$[\underline{A}, Y]_p = \mathcal{L}_{\underline{A}} Y_p = \frac{d}{dt} \Big|_{t=0} (\phi_t)_* Y_p = \lim_{t \rightarrow 0} \frac{(r_{e^{-tA}})_* Y_{pe^{tA}} - Y_p}{t}.$$

1. If Y is horizontal, then by the right-invariance of $\mathcal{H}$, $(r_{e^{-tA}})_* Y_{pe^{tA}}$ is also horizontal for all t . Thus $[\underline{A}, Y]_p$ is horizontal.

2. If Y is right-invariant, then $(r_{e^{-tA}})_* Y_{pe^{tA}} = Y_p$ for all t . Thus $[\underline{A}, Y]_p = 0$. $\square$

Proof of the theorem. 1. Since both sides of the equation are bilinear in X_p and Y_p , we can decompose X_p and Y_p into their vertical and horizontal components, and check the equality for each case.

Case 1: Both X_p and Y_p are horizontal, then

$$\begin{aligned}\Omega_p(X_p, Y_p) &= (d\omega)_p(X_p, Y_p) + \frac{1}{2}[\omega_p, \omega_p](X_p, Y_p) \\ &= (d\omega)_p(X_p, Y_p) + \frac{1}{2}([\omega_p(X_p), \omega_p(Y_p)] - [\omega_p(Y_p), \omega_p(X_p)]) \\ &= (d\omega)_p(X_p, Y_p) = (d\omega)_p(hX_p, hY_p).\end{aligned}$$

Case 2: X_p is vertical and Y_p is horizontal. For the same reason, $\frac{1}{2}[\omega_p, \omega_p](X_p, Y_p) = 0$. We can extend X_p to a fundamental vector field $\underline{A}$ and Y_p to a horizontal lift $\tilde{B}$. Then

$$d\omega(\underline{A}, \tilde{B}) = \underline{A}(\omega(\tilde{B})) - \tilde{B}(\omega(\underline{A})) - \omega([\underline{A}, \tilde{B}]) = 0 - \tilde{B}(A) - 0 = 0.$$

Here $(\omega(\underline{A})) = A$ is a constant function on P . So the left-hand side is

$$\Omega_p(X_p, Y_p) = (d\omega)_p(\underline{A}_p, \tilde{B}_p) + \frac{1}{2}[\omega_p, \omega_p](\underline{A}_p, \tilde{B}_p) = 0.$$

The right-hand side is also zero since $hX_p = 0$.

Case 3: Both X_p and Y_p are vertical. We can extend X_p and Y_p to fundamental vector fields $\underline{A}$ and $\underline{B}$ respectively. Then

$$d\omega(\underline{A}, \underline{B}) = \underline{A}(\omega(\underline{B})) - \underline{B}(\omega(\underline{A})) - \omega([\underline{A}, \underline{B}]) = 0 - 0 - \omega([\underline{A}, \underline{B}]) = -[A, B].$$

Here $(\omega(\underline{A})) = A$ is a constant function on P . And

$$\frac{1}{2}[\omega, \omega](\underline{A}, \underline{B}) = \frac{1}{2}([\omega(\underline{A}), \omega(\underline{B})] - [\omega(\underline{B}), \omega(\underline{A})]) = [A, B].$$

Thus the left-hand side of the equation is

$$\Omega_p(X_p, Y_p) = -[A, B] + [A, B] = 0.$$

The right-hand side is also zero since $hX_p = hY_p = 0$.

2. By the G -equivariance of ω , i.e., $r_g^*\omega = \text{Ad}_{g^{-1}}\omega$, we have

$$\begin{aligned}(r_g)^*\Omega &= r_g^*d\omega + \frac{1}{2}r_g^*[\omega, \omega] = d(r_g^*\omega) + \frac{1}{2}[r_g^*\omega, r_g^*\omega] \\ &= d((\text{Ad}_{g^{-1}})\omega) + \frac{1}{2}[(\text{Ad}_{g^{-1}})\omega, (\text{Ad}_{g^{-1}})\omega] \\ &= (\text{Ad}_{g^{-1}})d\omega + \frac{1}{2}(\text{Ad}_{g^{-1}})[\omega, \omega] = (\text{Ad}_{g^{-1}})\Omega.\end{aligned}$$

Here we use the fact that $\text{Ad}_{g^{-1}}$ is a Lie algebra homomorphism.

3. By strictly calculation, we have

$$\begin{aligned}d\Omega &= d^2\omega + \frac{1}{2}d[\omega, \omega] = 0 + \frac{1}{2}([d\omega, \omega] - [\omega, d\omega]) = [d\omega, \omega] \\ &= [\Omega - \frac{1}{2}[\omega, \omega], \omega] = [\Omega, \omega] - \frac{1}{2}[[\omega, \omega], \omega] = [\Omega, \omega].\end{aligned}$$

□

Example 7.38. In case $P = \text{Fr}(E)$ is the frame bundle of a rank m vector bundle E , the second Bianchi identity can be rewritten as

$$d\Omega = [\Omega, \omega] = \Omega \wedge \omega - \omega \wedge \Omega.$$

We denote $\Omega^k = \Omega \wedge \cdots \wedge \Omega$ (k times), then we have **generalized Bianchi identities**

$$d\Omega^k = [\Omega^k, \omega] = \Omega^k \wedge \omega - \omega \wedge \Omega^k.$$

This identity is easy to verify by direct calculation. It is important in the theory of characteristic classes.

Example 7.39. If $P = M \times G$ is a trivial bundle, with Ehresmann connection $\omega = \pi^* \omega_{\text{MC}}$, then

$$\Omega = d\omega + \frac{1}{2}[\omega, \omega] = \pi^* \left(d\omega_{\text{MC}} + \frac{1}{2}[\omega_{\text{MC}}, \omega_{\text{MC}}] \right) = 0.$$

Example 7.40. If P is a principal $U(1)$ -bundle over M , then the Ehresmann connection $\omega = iA \in \Omega^1(P, \mathfrak{u}(1))$ and the curvature $\Omega = iF \in \Omega^2(P, \mathfrak{u}(1))$.¹⁰ Since the Lie bracket on $\mathfrak{u}(1)$ is trivial, we have

$$\Omega = d\omega + \frac{1}{2}[\omega, \omega] = d\omega \implies F = dA.$$

¹⁰ $\mathfrak{u}(1)$ is the Lie algebra of $U(1)$, which can be identified with $i\mathbb{R}$.

8 Associated Bundles and Covariant Derivative

8.1 Associated Bundles

Let $\pi : P \rightarrow M$ be a principal G -bundle, F be a smooth manifold with a left G -action $\rho : G \rightarrow \text{Diff}(F)$.

Definition 8.1. The **associated bundle** of P with fiber F is the quotient space

$$E := P \times_{\rho} F := (P \times F) / \sim,$$

where the equivalence relation $\sim$ is defined by

$$(p, f) \sim (pg, g^{-1}f), \quad \forall p \in P, f \in F, g \in G.$$

Denote the equivalence class of (p, f) by $[p, f]$. We can define a projection map $\pi_E : E \rightarrow M$ by $\pi_E([p, f]) = \pi(p)$. Since

$$\pi_E([pg, g^{-1}f]) = \pi(pg) = \pi(p) = \pi_E([p, f]),$$

π_E is well-defined.

Example 8.2. If $P = U \times G$ is a trivial principal G -bundle over U , then $(U \times G) \times_{\rho} F$ is isomorphic to $U \times F$. Since

$$(U \times G) \times_{\rho} F \rightarrow U \times F, \quad [(x, g), f] = [(x, e) \cdot g, f] = [(x, e), g \cdot f] \mapsto (x, g \cdot f),$$

is a well-defined diffeomorphism.

Fact 7. The associated bundle $E := P \times_{\rho} F$ with the projection π_E gives a fiber bundle over M with fiber F . Let $\{(U_{\alpha}, \phi_{\alpha})\}$ be a local trivialization of P

$$\phi_{\alpha} : \pi^{-1}(U_{\alpha}) \rightarrow U_{\alpha} \times G, \quad p \mapsto (\pi(p), g_{\alpha}(p)),$$

The definition of $g_{\alpha}(p)$ is as follows: Let $s_{\alpha} : U_{\alpha} \rightarrow P$ be a local section defined by $s_{\alpha}(x) = \phi_{\alpha}^{-1}(x, e)$. Since the action of G on P is free and transitive on each fiber, for any $p \in \pi^{-1}(U_{\alpha})$, there exists a unique $g_{\alpha}(p) \in G$ such that $p = s_{\alpha}(\pi(p))g_{\alpha}(p)$. The transition functions of this local trivialization are given by

$$g_{\alpha\beta}(x) = g_{\alpha}(s_{\beta}(x)).$$

Then we can construct a local trivialization $\{(U_{\alpha}, \psi_{\alpha})\}$ of E by

$$\psi_{\alpha} : \pi_E^{-1}(U_{\alpha}) \rightarrow U_{\alpha} \times F, \quad [p, f] \mapsto (\pi(p), \rho(g_{\alpha}(p))f).$$

One can check that ψ_{α} is well-defined and is a diffeomorphism. The transition functions of this local trivialization are

$$\rho \circ g_{\alpha\beta} : U_{\alpha} \cap U_{\beta} \rightarrow \text{Diff}(F).$$

Example 8.3. If $P = M \times G$ is a trivial, then $P \times_\rho F$ is trivial. If the action ρ is trivial, then $P \times_\rho F$ is also trivial.

Example 8.4. If $F = V$ is a finite-dimensional vector space and $\rho : G \rightarrow \text{GL}(V)$ is a linear representation of G , then the associated bundle $E = P \times_\rho V$ is a vector bundle over M with fiber V . The vector bundle structure is given by

$$[p, v] + [p, w] := [p, v + w], \quad \lambda[p, v] := [p, \lambda v], \quad \forall p \in P, v, w \in V, \lambda \in \mathbb{R}.$$

One can check that the vector bundle structure is well-defined.

Example 8.5. Let $P = \text{Fr}(E)$ be the frame bundle of a rank m vector bundle E over M .

- $V = \mathbb{R}^m$, $\text{GL}(m, \mathbb{R}) \curvearrowright \mathbb{R}^m : g \cdot v = gv$, we get $\text{Fr}(E) \times_{\rho_{\text{can}}} \mathbb{R}^m \cong E$.
- $V = \mathbb{R}^{m \times m}$, $\text{GL}(m, \mathbb{R}) \curvearrowright \mathbb{R}^{m \times m} : g \cdot A = gAg^{-1}$, we get $\text{End}(E)$.
- $V = \bigwedge^k (\mathbb{R}^m)^*$, $\text{GL}(m, \mathbb{R}) \curvearrowright \bigwedge^k (\mathbb{R}^m)^* : g \cdot \alpha(v_1, \dots, v_k) = \alpha(g^{-1}v_1, \dots, g^{-1}v_k)$, we get $\bigwedge^k E^*$.
- $V = \text{Sym}^k (\mathbb{R}^m)^*$, $\text{GL}(m, \mathbb{R}) \curvearrowright \text{Sym}^k (\mathbb{R}^m)^* : g \cdot \alpha(v_1, \dots, v_k) = \alpha(g^{-1}v_1, \dots, g^{-1}v_k)$, we get $\text{Sym}^k E^*$.
- $V = \mathbb{R}$, $\text{GL}(m, \mathbb{R}) \curvearrowright \mathbb{R} : g \cdot r = (\det g)r$, we get $\det E$.

Example 8.6. Take $F = G$ and $\rho : G \rightarrow \text{Aut}(G) : \rho(g)(h) = ghg^{-1}$ the conjugation action, then the associated bundle $P \times_{\text{conj}} G$ is denoted by $\text{Ad}P$. Take $V = \mathfrak{g}$ and $\rho = \text{Ad} : G \rightarrow \text{GL}(\mathfrak{g})$ the adjoint representation. The associated bundle $\text{ad}P := P \times_{\text{Ad}} \mathfrak{g}$ is called the **adjoint bundle** of P .

Over this section, we mainly focus on the case where $F = V$ and $\rho : G \rightarrow \text{GL}(V)$ is a linear representation of G .

Fix $x \in M$, let P_x be the fiber of P over x and E_x be the fiber of E over x . Then for every $p \in P_x$, there is a canonical way to identify E_x with V by

$$f_p : V \rightarrow E_x, \quad f_p(v) = [p, v].$$

Lemma 8.7. *The map f_p is a linear isomorphism.*

Proof. Injectivity: If $[p, v] = [p, w]$, then there exists $g \in G$ such that $pg = p$ and $g^{-1}v = w$. Since the action of G on P is free, we have $g = e$ and thus $v = w$.

Surjectivity: For any $[q, w] \in E_x$, $q \in P_x$, so there exists $g \in G$ such that $q = pg$. Then we have $[q, w] = [pg, w] = [p, gw] = f_p(gw)$. $\square$

Lemma 8.8. *Let $f_p : V \rightarrow E_x$ be the map defined above. For any $g \in G$, $f_{pg} = f_p \circ \rho(g)$.*

Proof. For any $v \in V$, we have

$$f_{pg}(v) = [pg, v] = [p, gv] = f_p(gv) = f_p \circ \rho(g)(v). \quad \square$$

8.2 Tensorial Forms on a Principal Bundle

Let $\pi : P \rightarrow M$ be a principal G -bundle, $\rho : G \rightarrow \text{GL}(V)$ be a finite-dimensional representation of G , and $E = P \times_\rho V$ be the associated vector bundle.

Definition 8.9. A V -valued k -form $\varphi \in \Omega^k(P, V)$ is called **tensorial** if it satisfies the following two conditions:

- φ is **horizontal**: $\varphi(X_1, \dots, X_k) = 0$ if at least one of X_i is vertical.
- φ is **G -equivariant of type ρ** : $r_g^* \varphi = \rho(g^{-1}) \cdot \varphi$ for all $g \in G$.

The set of all tensorial k -forms on P of type ρ is denoted by $\Omega_\rho^k(P, V)$.

Example 8.10. The curvature Ω of a connection ω on P is a tensorial 2-form of type Ad .

Below we will show that there is a one-to-one correspondence between $\Omega_\rho^k(P, V)$ and $\Omega^k(M, E)$, where $\Omega^k(M, E)$ is the space of E -valued k -forms on M .

Definition 8.11. Given $x \in M$ and $X_1, \dots, X_k \in T_x M$, choose any point $p \in P_x$ and choose any lift $\tilde{X}_1, \dots, \tilde{X}_k \in T_p P$ of $X_1, \dots, X_k$, i.e., $\tilde{X}_i \in T_p P$ and $\pi_*(\tilde{X}_i) = X_i$. Define $\varphi^\flat$ by

$$\varphi_x^\flat(X_1, \dots, X_k) := f_p(\varphi_p(\tilde{X}_1, \dots, \tilde{X}_k)) \in E_x.$$

where $f_p : V \rightarrow E_x$, $v \mapsto [p, v]$ is the canonical isomorphism defined above.

Conversely, given $\psi \in \Omega^k(M, E)$, $p \in P$ and $Y_1, \dots, Y_k \in T_p P$, define $\psi^\sharp$ by

$$\psi_p^\sharp(Y_1, \dots, Y_k) := f_p^{-1}(\psi_{\pi(p)}(\pi_* Y_1, \dots, \pi_* Y_k)) \in V.$$

Theorem 8.12. *These two maps*

$$\begin{aligned} \Omega_\rho^k(P, V) &\rightarrow \Omega^k(M, E), & \varphi &\mapsto \varphi^\flat, \\ \Omega^k(M, E) &\rightarrow \Omega_\rho^k(P, V), & \psi &\mapsto \psi^\sharp, \end{aligned}$$

are well-defined and inverse to each other, thus we get linear isomorphisms between $\Omega_\rho^k(P, V)$ and $\Omega^k(M, E)$.

Before proving the theorem, let's see some examples and corollaries.

Example 8.13. By the theorem above, the curvature Ω of a connection on a principal G -bundle P can be viewed as an element in $\Omega^2(M, \text{ad}P)$, a 2-form on M with values in the adjoint bundle $\text{ad}P$.

Corollary 8.14. *Take $k = 0$ in Theorem ??, an element in $\Omega_\rho^0(P, V)$ is just a G -equivariant smooth map $f : P \rightarrow V$ of type ρ , i.e.,*

$$f(pg) = (r_g^* f)(p) = \rho(g^{-1})f(p) = g^{-1} \cdot f(p), \quad \forall p \in P, g \in G.$$

Besides, $\Omega^0(M, E)$ is just the space of smooth sections of E . Hence we get a one-to-one correspondence

$$\left\{ \begin{array}{c} G\text{-equivariant maps w.r.t. } \rho \\ f : P \rightarrow V \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{c} \text{sections of the associated bundle} \\ s : M \rightarrow P \times_\rho V \end{array} \right\}$$

Proof of Theorem ??. We first show that φ^b is well-defined. First check that φ^b does not depend on the choice of the lift $\tilde{X}_i$. If $\tilde{X}'_i$ is another lift of X_i , then $\pi_*(\tilde{X}_i - \tilde{X}'_i) = 0$, so $\tilde{X}_i - \tilde{X}'_i$ is vertical. Since φ is horizontal,

$$\varphi_p(\tilde{X}'_1, \dots, \tilde{X}'_k) = \varphi_p(\tilde{X}_1 + \text{vertical}, \dots, \tilde{X}_k + \text{vertical}) = \varphi_p(\tilde{X}_1, \dots, \tilde{X}_k).$$

Next check that φ^b does not depend on the choice of $p \in P_x$. Suppose $p' = pg$ for some $g \in G$, then $r_{g*}\tilde{X}_i$ is a lift of X_i at pg . Since φ is G -equivariant of type ρ , we have

$$\varphi_{pg}(r_{g*}\tilde{X}_1, \dots, r_{g*}\tilde{X}_k) = (r_g^*\varphi_{pg})(\tilde{X}_1, \dots, \tilde{X}_k) = \rho(g^{-1}) \cdot \varphi_p(\tilde{X}_1, \dots, \tilde{X}_k).$$

By Lemma ??,

$$f_{pg}(\varphi_{pg}(r_{g*}\tilde{X}_1, \dots, r_{g*}\tilde{X}_k)) = f_p \circ \rho(g)(\rho(g^{-1}) \cdot \varphi_p(\tilde{X}_1, \dots, \tilde{X}_k)) = f_p(\varphi_p(\tilde{X}_1, \dots, \tilde{X}_k)).$$

Thus φ^b is well-defined.

Let $\psi \in \Omega^k(M, E)$, it's easy to see that $\psi^\sharp$ is horizontal. For G -equivariance, we have

$$\begin{aligned} (r_g^*\psi^\sharp)_p(Y_1, \dots, Y_k) &= \psi^\sharp_{pg}(r_{g*}(Y_1), \dots, r_{g*}(Y_k)) \\ &= f_{pg}^{-1}(\psi_{\pi(pg)}(\pi_*r_{g*}(Y_1), \dots, \pi_*r_{g*}(Y_k))) \\ &= f_{pg}^{-1}(\psi_{\pi(p)}(\pi_*Y_1, \dots, \pi_*Y_k)) \\ &= \rho(g^{-1}) \cdot f_p^{-1}(\psi_{\pi(p)}(\pi_*Y_1, \dots, \pi_*Y_k)) \\ &= \rho(g^{-1}) \cdot \psi^\sharp_p(Y_1, \dots, Y_k). \end{aligned}$$

Hence $\psi^\sharp$ is G -equivariant of type ρ , and thus $\psi^\sharp \in \Omega_\rho^k(P, V)$.

Finally, we check that these two maps are inverse to each other. For any $\varphi \in \Omega_\rho^k(P, V)$, $p \in P$ and $Y_1, \dots, Y_k \in T_pP$,

$$\begin{aligned} (\varphi^b)_p(Y_1, \dots, Y_k) &= f_p^{-1}(\varphi_{\pi(p)}(\pi_*Y_1, \dots, \pi_*Y_k)) \\ &= f_p^{-1}(f_p(\varphi_p(Y_1, \dots, Y_k))) \\ &= \varphi_p(Y_1, \dots, Y_k). \end{aligned}$$

For any $\psi \in \Omega^k(M, E)$, $x \in M$ and $X_1, \dots, X_k \in T_xM$, choose any $p \in P_x$ and any lift $\tilde{X}_1, \dots, \tilde{X}_k \in T_pP$ of $X_1, \dots, X_k$,

$$\begin{aligned} (\psi^\sharp)_x(X_1, \dots, X_k) &= f_p(\psi_p^\sharp(\tilde{X}_1, \dots, \tilde{X}_k)) \\ &= f_p(f_p^{-1}(\psi_x(\pi_*\tilde{X}_1, \dots, \pi_*\tilde{X}_k))) \\ &= \psi_x(X_1, \dots, X_k). \end{aligned} \quad \square$$

8.3 Covariant Derivative

Keep the same notations as above. Notice that given a horizontal distribution H on P , we can decompose the tangent vector $Y_p \in T_pP$ into a horizontal part and a vertical part,

$$Y_p = hY_p + vY_p, \quad hY_p \in H_p, vY_p \in V_pP.$$

In this section, we will use this decomposition to define the covariant derivative of vector-valued forms on P , and thus a covariant derivative on the associated bundle E .

Let's start with the usual exterior derivative $d : \Omega^k(P, V) \rightarrow \Omega^{k+1}(P, V)$.

Proposition 8.15. *If $\varphi \in \Omega^k(P, V)$ is G -equivariant of type ρ , then $d\varphi$ is also G -equivariant of type ρ .*

Proof. For any $g \in G$,

$$r_g^*(d\varphi) = d(r_g^*\varphi) = d(\rho(g^{-1})\varphi) = \rho(g^{-1})d\varphi.$$

The last equality holds since $\rho(g^{-1})$ is a constant linear map on V for a fixed g . $\square$

In general, the exterior derivative does not preserve the horizontality, so it does not preserve the space of tensorial forms.

Definition 8.16. For any $\varphi \in \Omega^k(P, V)$, we define its **horizontal component** $\varphi^h \in \Omega^k(P, V)$ by

$$\varphi_p^h(Y_1, \dots, Y_k) := \varphi_p(hY_1, \dots, hY_k),$$

for any $p \in P$ and $Y_1, \dots, Y_k \in T_pP$.

Proposition 8.17. *If $\varphi \in \Omega^k(P, V)$ is G -equivariant of type ρ , so is φ^h .*

Proof. For $g \in G$, $p \in P$ and $Y_1, \dots, Y_k \in T_pP$,

$$\begin{aligned} (r_g^*\varphi^h)_p(Y_1, \dots, Y_k) &= \varphi_{pg}^h(r_{g*}Y_1, \dots, r_{g*}Y_k) \\ &= \varphi_{pg}(h(r_{g*}Y_1), \dots, h(r_{g*}Y_k)) \\ &= \varphi_{pg}(r_{g*}(hY_1), \dots, r_{g*}(hY_k)) \\ &= (r_g^*\varphi)_p(hY_1, \dots, hY_k) \\ &= \rho(g^{-1}) \cdot \varphi_p(hY_1, \dots, hY_k) \\ &= \rho(g^{-1}) \cdot \varphi_p^h(Y_1, \dots, Y_k). \end{aligned}$$

$\square$

Proposition ?? and ?? together imply that if $\varphi \in \Omega^k(P, V)$ is G -equivariant of type ρ , then $(d\varphi)^h$ is a tensorial form.

Definition 8.18. For any $\varphi \in \Omega^k(P, V)$, we define its **covariant derivative** by

$$D\varphi := (d\varphi)^h.$$

In particular, D maps tensorial k -forms of type ρ to tensorial $(k+1)$ -forms of type ρ , i.e.,

$$D : \Omega_\rho^k(P, V) \rightarrow \Omega_\rho^{k+1}(P, V).$$

Proposition 8.19. *The covariant derivative $D : \Omega_\rho^k(P, V) \rightarrow \Omega_\rho^{k+1}(P, V)$ is an antiderivation of degree +1, i.e., for any $\varphi \in \Omega_\rho^k(P, V)$ and $\psi \in \Omega_\rho^l(P, V)$,*

$$D(\varphi \wedge \psi) = D\varphi \wedge \psi + (-1)^k \varphi \wedge D\psi.$$

Proof. Since φ and ψ are horizontal, we have

$$\begin{aligned} D(\varphi \wedge \psi) &= (d(\varphi \wedge \psi))^h = (d\varphi \wedge \psi + (-1)^k \varphi \wedge d\psi)^h \\ &= (d\varphi)^h \wedge \psi^h + (-1)^k \varphi^h \wedge (d\psi)^h \\ &= D\varphi \wedge \psi + (-1)^k \varphi \wedge D\psi. \end{aligned}$$

$\square$

Example 8.20. By Theorem ??, the curvature Ω of a connection on P is exactly the covariant derivative of ω , since

$$\Omega_p(X_p, Y_p) = d\omega_p(hX_p, hY_p) = (d\omega_p)^h(X_p, Y_p) = (D\omega_p)(X_p, Y_p).$$

By the one-to-one correspondence between $\Omega_\rho^k(P, V)$ and $\Omega^k(M, E)$, we can also define a covariant derivative on $\Omega^k(M, E)$

$$D : \Omega^k(M, E) \rightarrow \Omega^{k+1}(M, E), \quad D\psi := (D\psi^\sharp)^\flat.$$

Example 8.21. Let E be a rank m vector bundle over M , $\text{Fr}(E)$ be its frame bundle. and ω be a connection on $\text{Fr}(E)$ induced by a connection ∇ on E . With the identification $E = \text{Fr}(E) \times_{\rho_{\text{can}}} \mathbb{R}^m$, we claim that the covariant derivative D on $\Omega^k(M, E)$ is exactly ∇ . The proof will be given later.

8.4 A Formula for the Covariant Derivative on Tensorial Forms

We have defined the covariant derivative D on $\Omega^k(P, V)$ by $D\varphi = (d\varphi)^h$. In this section, we will give a more explicit formula for D , but only for $\Omega_\rho^k(P, V)$.

The representation $\rho : G \rightarrow \text{GL}(V)$ induces a Lie algebra representation $\rho_* : \mathfrak{g} \rightarrow \text{End}(V)$, which allows us to define a product of $\tau \in \Omega^k(P, \mathfrak{g})$ and $\varphi \in \Omega^l(P, V)$

Definition 8.22. For $\tau \in \Omega^k(P, \mathfrak{g})$ and $\varphi \in \Omega^l(P, V)$, we define $\tau \cdot \varphi \in \Omega^{k+l}(P, V)$ by

$$(\tau \cdot \varphi)_p(Y_1, \dots, Y_{k+l}) := \frac{1}{k!l!} \sum_{\sigma} \text{sgn}(\sigma) \cdot \rho_*(\tau_p(Y_{\sigma(1)}, \dots, Y_{\sigma(k)})) \varphi_p(Y_{\sigma(k+1)}, \dots, Y_{\sigma(k+l)}).$$

One can check that $\tau \cdot \varphi$ is indeed a $(k+l)$ -form on P with values in V .

Example 8.23. If $V = \mathfrak{g}$ and $\rho = \text{Ad}$, then

$$(\tau \cdot \varphi)_p(Y_1, \dots, Y_{k+l}) = \frac{1}{k!l!} \sum_{\sigma} \text{sgn}(\sigma) \cdot [\tau_p(Y_{\sigma(1)}, \dots, Y_{\sigma(k)}), \varphi_p(Y_{\sigma(k+1)}, \dots, Y_{\sigma(k+l)})].$$

In this case, we write $\tau \cdot \varphi$ instead of $[\tau, \varphi]$.

Theorem 8.24. Let $\pi : P \rightarrow M$ be a principal G -bundle with a connection ω , $\rho : G \rightarrow \text{GL}(V)$ be a finite-dimensional representation of G . If $\varphi \in \Omega_\rho^k(P, V)$, then

$$D\varphi = d\varphi + \omega \cdot \varphi.$$

Before proving the theorem, we first see some examples.

Example 8.25. In case that $V = \mathfrak{g}$ and $\rho = \text{Ad}$, for any $\varphi \in \Omega_{\text{Ad}}^k(P, \mathfrak{g})$,

$$D\varphi = d\varphi + [\omega, \varphi].$$

Notice that this formula only holds for tensorial forms. Since ω is not horizontal, we cannot use this formula to compute $D\omega$, in fact, we know that

$$D\omega = (d\omega)^h = \Omega = d\omega + \frac{1}{2}[\omega, \omega].$$

Since Ω is tensorial, we can apply the formula for Ω . In fact, we have

$$D\Omega = d\Omega + [\omega, \Omega].$$

So the second Bianchi identity can be rewritten as $D\Omega = 0$.

Proof of Theorem ??. To verify the formula, we actually need to show

$$d\varphi(hY_1, \dots, hY_{k+1}) = d\varphi(Y_1, \dots, Y_{k+1}) + (\omega \cdot \varphi)_p(Y_1, \dots, Y_{k+1}) =: \text{I} + \text{II}.$$

Since both sides are multilinear and antisymmetric in $Y_1, \dots, Y_{k+1}$, we can decompose all the Y_i into horizontal and vertical parts and check the equality in several cases. Moreover, since we are verifying the equality, we can assume that the vertical part of $Y_i = \underline{A}_i$ is a fundamental vector field, and the horizontal part of $Y_i = \tilde{B}_i$ is a horizontal lift of some vector field B_i on M . Notice that $\tilde{B}_i$ is right-invariant.

Case 1: All Y_i are horizontal. Then $\text{II} = 0$ since ω vanishes on horizontal vectors, and the formula holds trivially.

Case 2: At least two of the Y_i are vertical. Then

$$\begin{aligned} \text{I} = d\varphi(Y_1, \dots, Y_{k+1}) &= \sum_i (-1)^{i+1} Y_i \varphi(Y_1, \dots, \hat{Y}_i, \dots, Y_{k+1}) \\ &\quad + \sum_{i < j} (-1)^{i+j} \varphi([Y_i, Y_j], Y_1, \dots, \hat{Y}_i, \dots, \hat{Y}_j, \dots, Y_{k+1}). \end{aligned}$$

Since $[\underline{A}_i, \underline{A}_j] = [\underline{A}_i, \underline{A}_j]$ is still vertical, at least one of the arguments is vertical, so $\text{I} = 0$.

On the other hand,

$$\text{II} = \frac{1}{k!} \sum_{\sigma \in S_{k+1}} \text{sgn}(\sigma) \cdot \rho_*(\omega_p(Y_{\sigma(1)})) \varphi_p(Y_{\sigma(2)}, \dots, Y_{\sigma(k+1)}),$$

which is also zero since at least one of the arguments of φ is vertical.

Case 3: Exactly one of the Y_i is vertical, say $Y_1 = \underline{A}$ and $Y_2, \dots, Y_{k+1}$ are horizontal. Then

$$\text{I} = d\varphi(\underline{A}, Y_2, \dots, Y_{k+1}) = \underline{A} \varphi(Y_2, \dots, Y_{k+1}) + \sum_{i=2}^{k+1} (-1)^i \varphi([\underline{A}, Y_i], Y_2, \dots, \hat{Y}_i, \dots, Y_{k+1}).$$

In our assumption, Y_i is a horizontal lift of some vector field B_i on M , so is right-invariant. Hence $[\underline{A}, Y_i] = 0$ for $i = 2, \dots, k+1$. Thus

$$\text{I} = \underline{A} \varphi(Y_2, \dots, Y_{k+1}).$$

On the other hand,

$$\begin{aligned} \text{II} &= \frac{1}{k!} \sum_{\substack{\sigma \in S_{k+1} \\ \sigma(1)=1}} \text{sgn}(\sigma) \cdot \rho_*(\omega(Y_{\sigma(1)})) \varphi(Y_{\sigma(2)}, \dots, Y_{\sigma(k+1)}) \\ &= \rho_*(\omega(\underline{A})) \varphi(Y_2, \dots, Y_{k+1}) = \rho_*(A) \varphi(Y_2, \dots, Y_{k+1}). \end{aligned}$$

If we denote by f the function $\varphi(Y_2, \dots, Y_{k+1})$. For $p \in P$, to calculate $\underline{A}_p f$, let $c(t) = \exp(tA)$ be the integral curve of A in G with $c(0) = e$. Then with $j_p : G \rightarrow P, j_p(g) = pg$, we have

$$\underline{A}_p f = j_{p*}(A)f = j_{p*}(c'(0))f = \left. \frac{d}{dt} \right|_{t=0} (f \circ j_p \circ c).$$

By the right-invariance of the horizontal vector fields $Y_2, \dots, Y_{k+1}$,

$$\begin{aligned}
(f \circ j_p \circ c)(t) &= f(pc(t)) = \varphi_{pc(t)}(Y_{2,pc(t)}, \dots, Y_{k+1,pc(t)}) \\
&= \varphi_{pc(t)}(r_{c(t)*}Y_{2,p}, \dots, r_{c(t)*}Y_{k+1,p}) \\
&= (r_{c(t)}^*\varphi)_p(Y_{2,p}, \dots, Y_{k+1,p}) \\
&= \rho(c(t)^{-1})\varphi_p(Y_{2,p}, \dots, Y_{k+1,p}) \\
&= \rho(\exp(-tA))f(p).
\end{aligned}$$

Thus

$$\underline{A}_p f(p) = \left. \frac{d}{dt} \right|_{t=0} \rho(\exp(-tA))f(p) = -\rho_*(A)f(p).$$

Hence $I + II = 0$. The left-hand side of the formula is also zero because $hY_1 = 0$. $\square$

Exercises

Exercise 8.1 (Forms on P pulled back from M). Since $\pi : P \rightarrow M$ is a surjective submersion, the pull-back $\pi^* : \Omega^*(M, V) \rightarrow \Omega^*(P, V)$ is injective. A differential form $\varphi \in \Omega^*(P, V)$ is said to be **basic** if it is the pull-back $\pi^*\psi$ of some $\psi \in \Omega^*(M, V)$; it is said to be **G -invariant** if $r_g^*\varphi = \varphi$ for all $g \in G$. Show that:

- φ is G -invariant if and only if φ is G -equivariant of type ρ_{trivial} , where ρ_{trivial} is the trivial representation.
- φ is basic if and only if φ is horizontal and G -invariant.

Proof. If φ is G -equivariant of type ρ_{trivial} , then

$$r_g^*\varphi = \rho_{\text{trivial}}(g^{-1}) \cdot \varphi = \varphi, \quad \forall g \in G.$$

This is just the definition of G -invariance.

Since the associated bundle $E = P \times_{\rho_{\text{trivial}}} V$ is just the trivial bundle $M \times V$, and $f_p : V \rightarrow E_{\pi(p)} = V$ is the identity map, by Theorem ??,

$$\begin{aligned}
\Omega^k(M, E) &\cong \Omega^k(M, M \times V) = \Omega^k(M, V) \xrightarrow{\cong} \Omega_{\rho_{\text{trivial}}}^k(P, V) = \{\text{basic forms on } P\} \\
&\quad \psi \mapsto \psi^\sharp.
\end{aligned}$$

where

$$\begin{aligned}
\psi^\sharp(Y_1, \dots, Y_k) &= f_p^{-1}(\psi_{\pi(p)}(\pi_*Y_1, \dots, \pi_*Y_k)) \\
&= \psi_{\pi(p)}(\pi_*Y_1, \dots, \pi_*Y_k) \\
&= (\pi^*\psi)_p(Y_1, \dots, Y_k).
\end{aligned}$$

$\square$

9 Characteristic Classes

Let $\pi : E \rightarrow M$ be a rank r vector bundle, ∇ be a connection on E . Its curvature $R^\nabla := \nabla^2$ is an $\text{End}(E)$ -valued 2-form on M . Under a local trivialization $\{(U_\alpha, \psi_\alpha)\}$ of E , ∇ can be represented by an $r \times r$ matrix of 1-forms ω_α , and R^∇ can be represented by an $r \times r$ matrix of 2-forms Ω_α .

The main goal of this section is the following:

1. An invariant polynomial Φ on $\mathfrak{gl}(r, \mathbb{R})$ determines a global form $\Phi(R^\nabla)$ on M .
2. The global form $\Phi(R^\nabla)$ is closed.
3. The class $[\Phi(R^\nabla)]$ is independent of the choice of ∇ , thus defines an invariant of E .
4. Chern-Weil homomorphism: $\text{Inv}(\mathfrak{gl}(r, \mathbb{R})) \rightarrow H_{\text{dR}}^*(M)$, $\Phi \mapsto [\Phi(R^\nabla)]$.
5. Examples: Chern classes, Pontryagin classes and Euler classes.
6. Generalization to principal G -bundles: $\text{Inv}(\mathfrak{g}) \rightarrow H_{\text{dR}}^*(M)$.

9.1 Chern-Weil Theory

Repeat the settings above: Let $\pi : E \rightarrow M$ be a rank r vector bundle, ∇ a connection on E . The curvature R^∇ is an $\text{End}(E)$ -valued 2-form on M . Under a local trivialization $\{(U_\alpha, \psi_\alpha)\}$ of E , ∇ can be represented by an $r \times r$ matrix of 1-forms ω_α , and R^∇ can be represented by an $r \times r$ matrix of 2-forms Ω_α .

Recall that the curvature form Ω_α is defined by

$$\Omega_\alpha = d\omega_\alpha + \omega_\alpha \wedge \omega_\alpha.$$

It satisfies the second Bianchi identity

$$d\Omega_\alpha = \Omega_\alpha \wedge \omega_\alpha - \omega_\alpha \wedge \Omega_\alpha,$$

and the general Bianchi identity

$$d\Omega_\alpha^k = \Omega_\alpha^k \wedge \omega_\alpha - \omega_\alpha \wedge \Omega_\alpha^k.$$

The transformation formula for ω_α and Ω_α is given by

$$\omega_\beta = g_{\alpha\beta}^{-1} \omega_\alpha g_{\alpha\beta} + g_{\alpha\beta}^{-1} d g_{\alpha\beta}, \quad \Omega_\beta = g_{\alpha\beta}^{-1} \Omega_\alpha g_{\alpha\beta}.$$

where $g_{\alpha\beta}$ is the transition function.

Definition 9.1. Let $X = (x_j^i)$ be an $r \times r$ matrix with indeterminate entries. A polynomial $\Phi(X)$ on $\mathfrak{gl}(r, \mathbb{R}) = \mathbb{R}^{r \times r}$ is a polynomial in the variables x_j^i . A polynomial $\Phi(X)$ on $\mathfrak{gl}(r, \mathbb{R})$ is **Ad-invariant (or simply invariant)** if

$$\Phi(A^{-1}XA) = \Phi(X), \quad \forall A \in \text{GL}(r, \mathbb{R}). \quad (9.1)$$

Remark. If X is a matrix of real numbers, then $\Phi(X)$ is a real number. If X is a matrix of 2-forms, then $\Phi(X)$ is also a differential form.

Fact 8. If (??) holds for all $r \times r$ matrices X of real numbers, then it also holds when X is a matrix of indeterminate. Thus $\text{Tr}(X)$ and $\det(X)$ are invariant polynomials.

Example 9.2. Let $\det(\lambda I + X) = \lambda^r + f_1(X)\lambda^{r-1} + \cdots + f_r(X)$, then $f_k(X)$ is an invariant polynomial of degree k . In fact, $(-1)^k f_k(X)$ is the k -th coefficient of the characteristic polynomial of X .

Example 9.3. Let $\Sigma_k(X) = \text{Tr}(X^k)$ be the trace of X^k , then $\Sigma_k(X)$ is an invariant polynomial of degree k .

Proposition 9.4. Let $\Phi(X)$ be an invariant polynomial on $\mathfrak{gl}(r, \mathbb{R})$. Then $\Phi(\Omega_\alpha)$ and $\Phi(\Omega_\beta)$ coincide on $U_\alpha \cap U_\beta$, thus determine a global form $\Phi(R^\nabla)$ on M .

Proof. Since $\Omega_\beta = g_{\alpha\beta}^{-1} \Omega_\alpha g_{\alpha\beta}$, we have

$$\Phi(\Omega_\beta) = \Phi(g_{\alpha\beta}^{-1} \Omega_\alpha g_{\alpha\beta}) = \Phi(\Omega_\alpha), \quad \text{on } U_\alpha \cap U_\beta. \quad \square$$

Proposition 9.5. The global form $\Phi(R^\nabla)$ is closed.

To prove this proposition, we need the following fact about invariant polynomials.

Fact 9. The algebra of invariant polynomials on $\mathfrak{gl}(r, \mathbb{R})$ is the polynomial algebra generated by $\Sigma_1, \dots, \Sigma_r$ (or equivalently by $f_1, \dots, f_r$), i.e.,

$$\text{Inv}(\mathfrak{gl}(r, \mathbb{R})) = \mathbb{R}[\Sigma_1, \dots, \Sigma_r] = \mathbb{R}[f_1, \dots, f_r].$$

For example, suppose Φ is a homogeneous invariant polynomial of degree 3 on $\mathfrak{gl}(3, \mathbb{R})$, then Φ must be a linear combination of Σ_1^3 , $\Sigma_1 \Sigma_2$ and Σ_3 , i.e.,

$$\Phi(X) = a \Sigma_1(X)^3 + b \Sigma_1(X) \Sigma_2(X) + c \Sigma_3(X), \quad a, b, c \in \mathbb{R}.$$

Proof of Proposition ??. Since Φ is a polynomial in $\Sigma_1, \dots, \Sigma_r$, it suffices to show that $d\text{Tr}(R^\nabla) = 0$ for all k . We have

$$d\text{Tr}(\Omega_\alpha^k) = \text{Tr}(d\Omega_\alpha^k) = \text{Tr}(\Omega_\alpha^k \wedge \omega_\alpha - \omega_\alpha \wedge \Omega_\alpha^k) = 0. \quad \square$$

Proposition 9.6. The class $[\Phi(R^\nabla)]$ is independent of the choice of ∇ . It is a (differential) topological invariant of E .

Proof. For the same reason as above, it suffices to show that $[\text{Tr}(R^\nabla)]$ is independent of the choice of ∇ . Let ∇_0, ∇_1 be two connections on E , then $\nabla_t = (1-t)\nabla_0 + t\nabla_1$ is a family of connections on E , whose connection form is given by

$$\omega_t = (1-t)\omega_0 + t\omega_1,$$

and curvature form is

$$\Omega_t = d\omega_t + \omega_t \wedge \omega_t.$$

So

$$\frac{d\Omega_t}{dt} = \frac{d}{dt}(d\omega_t + \omega_t \wedge \omega_t) = \frac{d}{dt}(d\omega_t) + \frac{d\omega_t}{dt} \wedge \omega_t + \omega_t \wedge \frac{d\omega_t}{dt}$$

Then

$$\begin{aligned}
\frac{d}{dt} \text{Tr}(\Omega_t^k) &= \text{Tr} \left(\frac{d}{dt} \Omega_t^k \right) = \text{Tr} \left(\frac{d\Omega_t}{dt} \wedge \Omega_t^{k-1} + \cdots + \Omega_t^{k-1} \wedge \frac{d\Omega_t}{dt} \right) = k \text{Tr} \left(\frac{d\Omega_t}{dt} \wedge \Omega_t^{k-1} \right) \\
&= k \text{Tr} \left(\left(d \frac{d\omega_t}{dt} + \frac{d\omega_t}{dt} \wedge \omega_t + \omega_t \wedge \frac{d\omega_t}{dt} \right) \wedge \Omega_t^{k-1} \right) \\
&= k \text{Tr} \left(d \left(\frac{d\omega_t}{dt} \right) \wedge \Omega_t^{k-1} - \frac{d\omega_t}{dt} \wedge (\Omega_t^{k-1} \wedge \omega_t - \omega_t \wedge \Omega_t^{k-1}) \right) \\
&= k \text{Tr} \left(d \left(\frac{d\omega_t}{dt} \right) \wedge \Omega_t^{k-1} - \frac{d\omega_t}{dt} \wedge d\Omega_t^{k-1} \right) \\
&= k \text{Tr} \left(d \left(\frac{d\omega_t}{dt} \wedge \Omega_t^{k-1} \right) \right) \\
&= d \, k \text{Tr} \left(\frac{d\omega_t}{dt} \wedge \Omega_t^{k-1} \right).
\end{aligned}$$

Therefore,

$$\begin{aligned}
\text{Tr}(\Omega_1^k) - \text{Tr}(\Omega_0^k) &= \int_0^1 \frac{d}{dt} \text{Tr}(\Omega_t^k) dt = \int_0^1 d \, k \text{Tr} \left(\frac{d\omega_t}{dt} \wedge \Omega_t^{k-1} \right) dt \\
&= d \int_0^1 k \text{Tr} \left(\frac{d\omega_t}{dt} \wedge \Omega_t^{k-1} \right) dt.
\end{aligned} \tag{9.2}$$

Notice that

$$\omega_{t,\beta} = g_{\alpha\beta}^{-1} \omega_{t,\alpha} g_{\alpha\beta} + g_{\alpha\beta}^{-1} d g_{\alpha\beta} \implies \frac{d\omega_{t,\beta}}{dt} = g_{\alpha\beta}^{-1} \frac{d\omega_{t,\alpha}}{dt} g_{\alpha\beta}, \quad \Omega_{t,\beta} = g_{\alpha\beta}^{-1} \Omega_{t,\alpha} g_{\alpha\beta}.$$

So $\text{Tr} \left(\frac{d\omega_t}{dt} \wedge \Omega_t^{k-1} \right)$ defines a global form on M . By (??), we have $[\text{Tr}(\Omega_1^k)] = [\text{Tr}(\Omega_0^k)]$.

For a general invariant polynomial Φ , we can express Φ as a linear combination of monomials $\Sigma_1^{k_1} \cdots \Sigma_r^{k_r}$. Then $\Phi(\Omega_t)$ is a linear combination of $\Sigma_1^{k_1}(\Omega_t) \wedge \cdots \wedge \Sigma_r^{k_r}(\Omega_t)$. Thus $[\Phi(\Omega_1)] = [\Phi(\Omega_t)] = [\Phi(\Omega_0)]$. $\square$

Definition 9.7. By above propositions, we define the **Chern-Weil homomorphism** as

$$\begin{aligned}
c_E : \text{Inv}(\mathfrak{gl}(r, \mathbb{R})) &\rightarrow H_{\text{dR}}^*(M), \\
\Phi &\mapsto [\Phi(R^\nabla)].
\end{aligned}$$

This map sends $\Phi(X)\Psi(X)$ to $[\Phi(R^\nabla) \wedge \Psi(R^\nabla)] = [\Phi(R^\nabla)] \cdot [\Psi(R^\nabla)]$, thus is an algebra homomorphism. This map does not depend on the choice of ∇ .

Definition 9.8. A **characteristic class** of a vector bundle $\pi : E \rightarrow M$ is a map

$$c : \text{Vect}_k(M) \rightarrow H_{\text{dR}}^*(M), \quad E \mapsto c(E),$$

such that if $f : N \rightarrow M$ is a smooth map, then

$$c(f^*E) = f^*c(E).$$

In functional language, a **characteristic class** is a natural transformation from the functor $\text{Vect}_k(-)$ to $H_{\text{dR}}^*(-)$. The naturality condition can be visualized by the following commutative diagram:

$$\begin{array}{ccc} \text{Vect}_k(M) & \xrightarrow{c} & H_{\text{dR}}^*(M) \\ f^* \uparrow & & \uparrow f^* \\ \text{Vect}_k(N) & \xrightarrow{c} & H_{\text{dR}}^*(N) \end{array}$$

Naturality of Chern-Weil homomorphism

Suppose $\pi : E \rightarrow M$ is a rank r vector bundle, ∇ a connection on E . Let $f : N \rightarrow M$ be a smooth map, it induces a pullback bundle $f^*E \rightarrow N$ and a pullback connection $f^*\nabla$ on f^*E

$$(f^*\nabla)(f^*s) := f^*(\nabla s), \quad \forall s \in \Gamma(E).$$

Suppose ω is the connection form of ∇ , then $f^*\omega$ is the connection form of $f^*\nabla$. The curvature form of $f^*\nabla$ is therefore $f^*(d\omega + \omega \wedge \omega) = f^*\Omega$. For any invariant polynomial Φ , we have

$$\Phi(f^*\Omega) = f^*\Phi(\Omega) \implies c_{f^*E}(\Phi) = [\Phi(f^*\Omega)] = f^*[\Phi(\Omega)] = f^*c_E(\Phi).$$

which shows the naturality of Chern-Weil homomorphism.

9.2 Examples of Characteristic Classes

We have introduced a powerful machinery to construct characteristic classes of vector bundles. In this subsection, we will give a few examples of characteristic classes.

9.2.1 Pontryagin Classes

In this subsection, we only consider real vector bundles. Recall that

$$\det(\lambda I + X) = \lambda^r + f_1(X)\lambda^{r-1} + \cdots + f_r(X),$$

in particular, $f_1(X) = \text{Tr}(X)$. If X is a skew-symmetric matrix, then $f_1(X) = 0$. In general we have the following proposition.

Proposition 9.9. *If Ω is skew-symmetric, then $\text{Tr}(\Omega^k) = 0$ for all odd k . Moreover, $\Phi(\Omega) = 0$ for all invariant polynomials Φ of odd degree.*

Proof. Since Ω is skew-symmetric,

$$(\Omega \wedge \cdots \wedge \Omega)^T = (\Omega^T)^T \wedge \cdots \wedge (\Omega^T)^T = (-\Omega) \wedge \cdots \wedge (-\Omega) = (-1)^k \Omega \wedge \cdots \wedge \Omega.$$

So Ω^k is skew-symmetric for odd k , and $\text{Tr}(\Omega^k) = 0$.

For a general invariant polynomial Φ of odd degree, since Φ is a linear combination of monomials $\Sigma_1^{k_1} \cdots \Sigma_r^{k_r}$ with $k_1 + 2k_2 + \cdots + rk_r$ odd, at least one of $k_1, k_3, \dots$ is odd, thus $\Phi(\Omega)$ must be zero. $\square$

Since $[\Phi(R^\nabla)]$ is independent of the choice of ∇ , we can choose a special connection ∇ such that R^∇ is skew-symmetric. To do this, we choose a Riemannian metric on E , then there exists connections on E that are compatible with the metric. Under a local orthonormal frame, the connection form ω is skew-symmetric, so as the curvature form Ω . By the above proposition, $[\Phi(R^\nabla)] = 0$ for all invariant polynomials Φ of odd degree.

Corollary 9.10. *For a real vector bundle E , the ring of characteristic classes of E has two sets of generators:*

$$[\mathrm{Tr}(\Omega^2)], [\mathrm{Tr}(\Omega^4)], [\mathrm{Tr}(\Omega^6)], \dots;$$

and

$$[f_2(\Omega)], [f_4(\Omega)], [f_6(\Omega)], \dots$$

Definition 9.11. The k -th **Pontryagin class** of a real vector bundle E is defined to be

$$p_k(E) := (-1)^k \left[f_{2k} \left(\frac{1}{2\pi} R^\nabla \right) \right] = \left[f_{2k} \left(\frac{\sqrt{-1}}{2\pi} R^\nabla \right) \right] \in H^{4k}(M).$$

Remark. The factor $\sqrt{-1}$ is for the sign convention. The factor $\frac{1}{2\pi}$ is for the integrality of Pontryagin classes. A class $\alpha \in H^q(M)$ is called **integral** if $\int_S \alpha \in \mathbb{Z}$ for all q -dimensional closed oriented submanifolds S of M .

By definition,

$$\begin{aligned} \det \left(\lambda I + \frac{\sqrt{-1}}{2\pi} R^\nabla \right) &= \lambda^r + f_1 \left(\frac{\sqrt{-1}}{2\pi} R^\nabla \right) \lambda^{r-1} + \dots + f_r \left(\frac{\sqrt{-1}}{2\pi} R^\nabla \right) \\ &= \lambda^r + p_1(E) \lambda^{r-2} + \dots + p_{[r/2]}(E) \lambda^{r-2[r/2]}. \end{aligned}$$

Apply $\lambda = 1$,

$$\det \left(I + \frac{\sqrt{-1}}{2\pi} R^\nabla \right) = 1 + p_1(E) + \dots + p_{[r/2]}(E) =: p(E).$$

We call $p(E)$ the **total Pontryagin class** of E .

Proposition 9.12 (Whitney product formula). *For two real vector bundles E and E' , we have*

$$p(E \oplus E') = p(E) \cdot p(E').$$

Proof. Let ∇ and ∇' be connections on E and E' , then $\nabla \oplus \nabla'$ is a connection on $E \oplus E'$. Under a suitable local trivialization, the curvature form of $\nabla \oplus \nabla'$ is given by

$$F_{\nabla \oplus \nabla'} = \begin{pmatrix} R^\nabla & 0 \\ 0 & R^{\nabla'} \end{pmatrix} \implies \Omega_{E \oplus E'} = \begin{pmatrix} \Omega & 0 \\ 0 & \Omega' \end{pmatrix}.$$

So

$$\begin{aligned} p(E \oplus E') &= \det \left(I + \frac{\sqrt{-1}}{2\pi} \Omega_{E \oplus E'} \right) \\ &= \det \begin{pmatrix} I + \frac{\sqrt{-1}}{2\pi} \Omega & \\ & I + \frac{\sqrt{-1}}{2\pi} \Omega' \end{pmatrix} \\ &= \det \left(I + \frac{\sqrt{-1}}{2\pi} \Omega \right) \cdot \det \left(I + \frac{\sqrt{-1}}{2\pi} \Omega' \right) \\ &= p(E) \cdot p(E'). \end{aligned}$$

□

9.2.2 Euler Classes

If in addition E is orientable and we choose an orientation of E . Then we can make the local orthonormal frame $e = [e_1, \dots, e_r]$ of E to be compatible with the orientation, so the transition function between two frames is in $\mathrm{SO}(r)$ and the connection form ω is skew-symmetric. This inspires us to consider the $\mathrm{SO}(r)$ -invariant polynomial on $\mathfrak{so}(r)$.

Fact 10. For r odd, $\mathrm{Inv}(\mathfrak{so}(r))$ is generated by $\Sigma_1, \dots, \Sigma_r$ (or equivalently by $f_1, \dots, f_r$). For r even, $\mathrm{Inv}(\mathfrak{so}(r))$ has one more generator Pf , the **Pfaffian**. That is

$$\begin{aligned}\mathrm{Inv}(\mathfrak{so}(2m+1)) &= \mathbb{R}[\Sigma_1, \dots, \Sigma_{2m+1}] = \mathbb{R}[f_1, \dots, f_{2m+1}], \\ \mathrm{Inv}(\mathfrak{so}(2m)) &= \mathbb{R}[\Sigma_1, \dots, \Sigma_{2m}, \mathrm{Pf}] = \mathbb{R}[f_1, \dots, f_{2m}, \mathrm{Pf}].\end{aligned}$$

The Pfaffian of skew-symmetric matrices

Let's begin with some examples. For a 2×2 skew-symmetric matrix

$$X = \begin{pmatrix} 0 & a \\ -a & 0 \end{pmatrix},$$

we have

$$\det(X) = a^2.$$

For a 4×4 skew-symmetric matrix

$$X = \begin{pmatrix} 0 & a & b & c \\ -a & 0 & d & e \\ -b & -d & 0 & f \\ -c & -e & -f & 0 \end{pmatrix},$$

after a direct calculation,

$$\det(X) = (af - be + cd)^2.$$

One can observe that both determinants are perfect squares. This phenomenon holds for all skew-symmetric matrices:

Fact 11. For any $2m \times 2m$ skew-symmetric matrix $X = (x_j^i)$, $\det(X)$ is a perfect square in $\mathbb{Z}[x_j^i]$. The square root of $\det(X)$ is called the **Pfaffian** of X , denoted by $\mathrm{Pf}(X)$.

Remark. Notice that the square roots of $\det(X)$ are $\pm \mathrm{Pf}(X)$. To determine the sign, we require $\mathrm{Pf}(J_{2m}) = 1$, where

$$J_{2m} = \begin{pmatrix} 0 & 1 & & & \\ -1 & 0 & & & \\ & & \ddots & & \\ & & & 0 & 1 \\ & & & -1 & 0 \end{pmatrix}.$$

Fact 12. Let $A = (a_j^i)$ and $X = (x_j^i)$ be two $2m \times 2m$ matrices with X skew-symmetric, then

$$\mathrm{Pf}(A^T X A) = \det(A) \mathrm{Pf}(X).$$

If $A \in \mathrm{SO}(2m)$, then $\mathrm{Pf}(A^T X A) = \mathrm{Pf}(X)$, so Pf is an invariant polynomial on $\mathfrak{so}(2m)$.

Definition 9.13. For a $2m$ -dimensional oriented real vector bundle $E \rightarrow M$, choose a metric on E and a compatible connection ∇ , the **Euler class** of E is defined to be

$$e(E) := \left[\text{Pf} \left(\frac{1}{2\pi} R^\nabla \right) \right] \in H^{2m}(M).$$

Remark. The Euler class is a characteristic class of oriented real vector bundles. It depends on the orientation: if we reverse the orientation, then $e(E)$ changes to $-e(E)$.

Remark. The definition of Euler class requires a metric and a compatible connection, but the resulting class does not depend on the choice of metric and connection:

- Fixing a metric g on E , any two metric-compatible connections ∇_0 and ∇_1 define the same Euler class.
- Given any two metrics g_0 and g_1 on E , we can choose a family of connections ∇_t such that ∇_0 (resp. ∇_1) is compatible with g_0 (resp. g_1). The whole family ∇_t defines the same Euler class.

Theorem 9.14 (Gauss-Bonnet-Chern theorem). *Let M be a closed oriented Riemannian manifold of dimension $2m$, ∇ be a metric-compatible connection on TM , then*

$$\int_M e(TM) = \int_M \text{Pf} \left(\frac{1}{2\pi} R^\nabla \right) = \chi(M),$$

where $\chi(M)$ is the Euler characteristic of M .

Example 9.15. When M is a closed oriented surface, the above theorem reduces to the classical Gauss-Bonnet theorem: let e_1, e_2 be a local orthonormal frame of TM , θ^1, θ^2 be the dual coframe, then the connection and curvature forms are

$$\omega = \begin{pmatrix} 0 & \omega_2^1 \\ -\omega_2^1 & 0 \end{pmatrix}, \quad \Omega = d\omega + \omega \wedge \omega = \begin{pmatrix} 0 & d\omega_2^1 \\ -d\omega_2^1 & 0 \end{pmatrix},$$

and $d\omega_2^1 = K\theta^1 \wedge \theta^2$, where K is the Gaussian curvature of M . So

$$\text{Pf}(\Omega) = d\omega_2^1 = K\theta^1 \wedge \theta^2.$$

Therefore,

$$\int_M e(TM) = \frac{1}{2\pi} \int_M K\theta^1 \wedge \theta^2 = \chi(M) = 2 - 2g(M).$$

9.2.3 Chern Classes

For a complex vector bundle $E \rightarrow M$ with $\text{rk}_{\mathbb{C}}(E) = r$, we can choose a local trivialization $\{(U_\alpha, \psi_\alpha)\}$ of E such that the transition function $g_{\alpha\beta} \in \text{GL}(r, \mathbb{C})$. Then any complex polynomial Φ on $\mathfrak{gl}(r, \mathbb{C})$ invariant under conjugation by $\text{GL}(r, \mathbb{C})$ defines a characteristic class $[\Phi(R^\nabla)]$ which is independent of the choice of ∇ .

Definition 9.16. The k -th **Chern class** of a complex vector bundle E is defined to be

$$c_k(E) := \left[f_k \left(\frac{\sqrt{-1}}{2\pi} R^\nabla \right) \right] \in H^{2k}(M).$$

The **total Chern class** of E is defined to be

$$c(E) := \det \left(I + \frac{\sqrt{-1}}{2\pi} R^\nabla \right) = 1 + c_1(E) + \cdots + c_r(E).$$

Proposition 9.17. *Chern classes satisfy the naturality property and the Whitney product formula, i.e.,*

$$c(f^*E) = f^*c(E), \quad c(E \oplus E') = c(E) \cdot c(E').$$

Remark. If we choose a Hermitian metric on E and a compatible connection ∇ , then the curvature form R^∇ is skew-Hermitian, so $\frac{\sqrt{-1}}{2\pi} R^\nabla$ is Hermitian. Therefore, $\sum_k \left(\frac{\sqrt{-1}}{2\pi} R^\nabla \right)^k = \text{Tr} \left(\left(\frac{\sqrt{-1}}{2\pi} R^\nabla \right)^k \right)$ is a real form, and $c_k(E)$ is a real cohomology class. In fact, $c_k(E)$ is an integral cohomology class, i.e., $c_k(E) \in H^{2k}(M; \mathbb{Z})$.

Proposition 9.18. *Let E be a complex vector bundle, $\text{rk}_{\mathbb{C}}(E) = r$, $E_{\mathbb{R}}$ be the underlying real vector bundle of E . $E_{\mathbb{R}}$ is oriented and if $e = [e_1, \dots, e_r]$ is a local frame of E , then $e_{\mathbb{R}} = [e_1, \sqrt{-1}e_1, \dots, e_r, \sqrt{-1}e_r]$ is a local oriented frame of $E_{\mathbb{R}}$. We set the orientation determined by $e_{\mathbb{R}}$ to be the orientation of $E_{\mathbb{R}}$, then the top Chern class $c_r(E)$ is equal to the Euler class $e(E_{\mathbb{R}})$.*

Proof. Choose a Hermitian metric $\langle \cdot, \cdot \rangle$ on E and a compatible connection ∇ , then the curvature R^∇ is skew-Hermitian. One can define a Riemannian metric on $E_{\mathbb{R}}$ by

$$\langle u, v \rangle_{\mathbb{R}} := \text{Re} \langle u, v \rangle \quad \forall u, v \in E_{\mathbb{R}}.$$

Then

$$X \langle u, v \rangle_{\mathbb{R}} = \text{Re} \left(X \langle u, v \rangle \right) = \text{Re} \left(\langle \nabla_X u, v \rangle + \langle u, \nabla_X v \rangle \right) = \langle \nabla_X u, v \rangle_{\mathbb{R}} + \langle u, \nabla_X v \rangle_{\mathbb{R}}.$$

So ∇ is compatible with $\langle \cdot, \cdot \rangle_{\mathbb{R}}$, and the curvature $(R^\nabla)_{\mathbb{R}}$ is skew-symmetric.

Under a local unitary frame $e = [e_1, \dots, e_r]$ of E , the curvature form Ω is skew-Hermitian, and so $\sqrt{-1}\Omega$ is Hermitian. The frame $e_{\mathbb{R}} = [e_1, \sqrt{-1}e_1, \dots, e_r, \sqrt{-1}e_r]$ is a local orthonormal frame of $E_{\mathbb{R}}$, and the curvature form under $e_{\mathbb{R}}$ is skew-symmetric.

If $\Omega = (z_j^i) = (x_j^i + \sqrt{-1}y_j^i)$ with $x_j^i, y_j^i \in \mathbb{R}$, then $\Omega = A + \sqrt{-1}B$ with $A = (x_j^i)$ and $B = (y_j^i)$. Then the curvature form under $e_{\mathbb{R}}$ is given by

$$\Omega_{\mathbb{R}} = \begin{pmatrix} x_1^1 & -y_1^1 & \cdots & x_r^1 & -y_r^1 \\ y_1^1 & x_1^1 & \cdots & y_r^1 & x_r^1 \\ \vdots & \vdots & & \vdots & \vdots \\ x_1^r & -y_1^r & \cdots & x_r^r & -y_r^r \\ y_1^r & x_1^r & \cdots & y_r^r & x_r^r \end{pmatrix} \xrightarrow{\sim} \begin{pmatrix} x_1^1 & \cdots & x_r^1 & -y_1^1 & \cdots & -y_r^1 \\ \vdots & & \vdots & \vdots & & \vdots \\ x_1^r & \cdots & x_r^r & -y_1^r & \cdots & -y_r^r \\ y_1^1 & \cdots & y_r^1 & x_1^1 & \cdots & x_r^1 \\ \vdots & & \vdots & \vdots & & \vdots \\ y_1^r & \cdots & y_r^r & x_1^r & \cdots & x_r^r \end{pmatrix} = \begin{pmatrix} A & -B \\ B & A \end{pmatrix}$$

So

$$\begin{aligned}\det\left(\frac{1}{2\pi}\Omega_{\mathbb{R}}\right) &= \det\left(\frac{1}{2\pi}\begin{pmatrix} A & -B \\ B & A \end{pmatrix}\right) = \frac{1}{(2\pi)^{2r}} |\det(A + \sqrt{-1}B)|^2 \\ &= \left|\det\left(\frac{1}{2\pi}\Omega\right)\right|^2 = \left|\det\left(\frac{\sqrt{-1}}{2\pi}\Omega\right)\right|^2 = \left(\det\left(\frac{\sqrt{-1}}{2\pi}\Omega\right)\right)^2.\end{aligned}$$

The last equality holds since $\det\left(\frac{\sqrt{-1}}{2\pi}\Omega\right)$ is a real form. Therefore,

$$\text{Pf}\left(\frac{1}{2\pi}(R^{\nabla})_{\mathbb{R}}\right) = \pm \det\left(\frac{\sqrt{-1}}{2\pi}R^{\nabla}\right).$$

To determine the sign, set $\Omega = -\sqrt{-1}I_r$, then

$$\Omega_{\mathbb{R}} = \begin{pmatrix} 0 & 1 & & \\ -1 & 0 & & \\ & & \ddots & \\ & & & 0 & 1 \\ & & & -1 & 0 \end{pmatrix} = J_{2r}$$

So

$$\det(\sqrt{-1}\Omega) = \det(I_r) = 1 = \text{Pf}(J_{2r}) = \text{Pf}(\Omega_{\mathbb{R}}).$$

The sign under this setting is +1, thus

$$\text{Pf}\left(\frac{1}{2\pi}(R^{\nabla})_{\mathbb{R}}\right) = \det\left(\frac{\sqrt{-1}}{2\pi}R^{\nabla}\right) \implies e(E_{\mathbb{R}}) = c_r(E). \quad \square$$

9.3 Characteristic classes of Principal Bundles

Let M be a smooth manifold. There are **one-to-one correspondences** between vector bundles (with additional structures) and principal bundles on M .

Vector Bundles (with additional structure)	Principal Bundles
$\mathbb{R}^r$ -bundles	$\text{GL}(r, \mathbb{R})$ -bundles
$\mathbb{C}^r$ -bundles	$\text{GL}(r, \mathbb{C})$ -bundles
Oriented $\mathbb{R}^r$ -bundles (i.e., with trivial determinant)	$\text{SL}(r, \mathbb{R})$ -bundles
$\mathbb{C}^r$ -bundles with trivial determinant line bundle	$\text{SL}(r, \mathbb{C})$ -bundles
$\mathbb{R}^r$ -bundles with Riemannian metric	$\text{O}(r)$ -bundles
$\mathbb{C}^r$ -bundles with Hermitian metric	$\text{U}(r)$ -bundles
Oriented $\mathbb{R}^r$ -bundles with Riemannian metric	$\text{SO}(r)$ -bundles
$\mathbb{C}^r$ -bundles with Hermitian metric and trivial determinant	$\text{SU}(r)$ -bundles

So the characteristic classes of vector bundles can also define for the corresponding principal bundles. In fact, there is a more general Chern-Weil theory for principal G -bundles.

Let $\pi : P \rightarrow M$ be a principal G -bundle, $\mathfrak{g}$ be the Lie algebra of G . ω is a connection form on P , Ω is the curvature form. Recall that

- $\omega \in \Omega^1(P, \mathfrak{g})$, $r_g^* \omega = \text{Ad}_{g^{-1}} \omega$, and $\omega(\underline{A}) = A$ for all $A \in \mathfrak{g}$;
- $\Omega \in \Omega^2(P, \mathfrak{g})$, $r_g^* \Omega = \text{Ad}_{g^{-1}} \Omega$, and $\Omega(X, Y) = 0$ if X or Y is vertical.
- There is a one-to-one correspondence between $\Omega_\rho^k(P, V)$ and $\Omega^k(M, P \times_\rho V)$:

$$\left\{ \begin{array}{l} k\text{-forms on } P \text{ valued in } V \text{ that are} \\ G\text{-equivariant and horizontal} \end{array} \right\} \xleftrightarrow{1-1} \left\{ k\text{-forms on } M \text{ valued in } P \times_\rho V \right\}$$

- There is a one-to-one correspondence between $\Omega_{\text{basic}}^k(P, V)$ and $\Omega^k(M, V)$:

$$\left\{ \begin{array}{l} k\text{-forms on } P \text{ valued in } V \text{ that are} \\ G\text{-invariant and horizontal} \end{array} \right\} \xleftrightarrow{1-1} \left\{ k\text{-forms on } M \text{ valued in } V \right\}$$

$$\pi^* \phi \longleftrightarrow \phi.$$

- The covariant derivative $D : \Omega^k(P, V) \rightarrow \Omega^{k+1}(P, V)$ is defined by $D\varphi = (d\varphi)^h$. In particular, it maps $\Omega_\rho^k(P, V)$ to $\Omega_\rho^{k+1}(P, V)$;
- The Bianchi identity: $D\Omega = d\Omega + [\omega, \Omega] = 0$;
- If $\varphi \in \Omega_\rho^k(P, V)$, then

$$D\varphi = d\varphi + \omega \cdot \varphi.$$

In particular, if $\varphi \in \Omega_{\text{basic}}^k(P, V)$, then $D\varphi = d\varphi$.

Definition 9.19 (Invariant polynomials). Let G be a Lie group, $\mathfrak{g}$ be its Lie algebra. A polynomial of degree k on $\mathfrak{g}$ is an element of $\text{Sym}^k(\mathfrak{g}^*)$. We call a polynomial $\Phi \in \text{Sym}^k(\mathfrak{g}^*)$ **Ad(G)-invariant** if

$$\Phi(\text{Ad}_g X) = \Phi(X) \quad \forall g \in G, X \in \mathfrak{g}.$$

The set of all $\text{Ad}(G)$ -invariant polynomials on $\mathfrak{g}$ is denoted by $\text{Inv}(\mathfrak{g})$.

Theorem 9.20. For any invariant polynomial $\Phi \in \text{Inv}(\mathfrak{g})$, we have

- $\Phi(\Omega)$ is a basic form, thus defines a form Λ on M such that $\pi^* \Lambda = \Phi(\Omega)$;
- $\Phi(\Omega)$ is closed, so as Λ ;
- the cohomology class $[\Lambda] \in H^{2k}(M)$ is independent of the choice of connection ω on P .

The cohomology class $[\Lambda]$ is called the **characteristic class** of P associated to Φ . The map

$$c : \text{Inv}(\mathfrak{g}) \rightarrow H^*(M)$$

$$\Phi \mapsto [\Lambda],$$

is called the **Chern-Weil homomorphism**.

Proof. Step 1: $\Phi(\Omega)$ is a basic form. Since Ω is a tensorial form of type Ad , we have

$$r_g^* \Phi(\Omega) = \Phi(r_g^* \Omega) = \Phi(\text{Ad}_{g^{-1}} \Omega) = \Phi(\Omega),$$

so $\Phi(\Omega)$ is G -invariant. Clearly, $\Phi(\Omega)$ is horizontal. Thus $\Phi(\Omega)$ is a basic form and $\Phi(\Omega) = \pi^* \Lambda$ for some form Λ on M .

Step 2: $\Phi(\Omega)$ is closed. Since $\Phi(\Omega)$ is basic,

$$d\Phi(\Omega) = D\Phi(\Omega) = \Phi(D\Omega) = 0.$$

For the closedness of Λ ,

$$\pi^*(d\Lambda) = d(\pi^* \Lambda) = d\Phi(\Omega) = 0.$$

Thus $d\Lambda = 0$ by the injectivity of π^* .

Step 3: Independence of ω . Let ω_0 and ω_1 be two connections on P , Ω_0 and Ω_1 be the curvature forms, Λ_0 and Λ_1 are forms on M such that $\pi^* \Lambda_i = \Phi(\Omega_i)$, $i = 0, 1$.

Consider the family of connections $\omega_t = (1 - t)\omega_0 + t\omega_1$, and the corresponding curvature forms Ω_t .

□

10 Transgressions and the Chern-Simons Form

Notation: ω, Ω are forms on the bundle P , while A, F are forms on M .

10.1 Gauss-Bonnet Theorem

Our story begins with the Gauss-Bonnet theorem, which is a great theorem connecting local geometric quantity with global topologic quantity.

Theorem 10.1 (Gauss-Bonnet Theorem). *Let M^2 be a closed oriented surface with Riemannian metric, K be its Gauss curvature. Then*

$$\frac{1}{2\pi} \int_M K d\text{Vol}_M = \chi(M),$$

where $\chi(M)$ is the Euler characteristic of M . If we denote the curvature 2-form of the Levi-Civita connection by F , then $F = K d\text{Vol}_M$, thus we can rewrite the Gauss-Bonnet theorem as

$$\frac{1}{2\pi} \int_M F = \chi(M).$$

To prove this theorem, we need to use the vector field V on M .

Definition 10.2. Let V be a smooth vector field on M with isolated zeros $\{x_i\}$. On $M - \{x_i\}$, we can normalize V to be a unit vector field $\tilde{V} = \frac{V}{|V|}$. On each zero x_i , choose a small neighborhood $B_r(x_i)$, then $\tilde{V}|_{\partial B_r(x_i)}$ can be viewed as a map from S^1 to S^1 . We define the index of V at x_i as the degree of this map, i.e.

$$\text{Ind}_{x_i}(V) = \deg(\tilde{V}|_{\partial B_r(x_i)} : S^1 \rightarrow S^1).$$

The index of V is defined to be the sum of the indices at all zeros,

$$\text{Ind}(V) = \sum_i \text{Ind}_{x_i}(V).$$

The following picture shows some zeros with different indices.

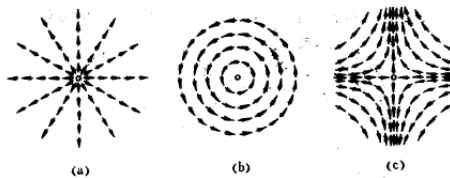


图 89

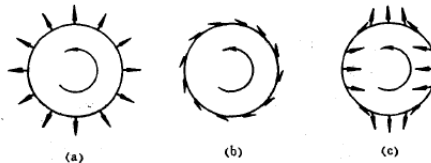


图 90

Remark. The index of V at x_i is independent of the choice of $B_r(x_i)$, thus is well-defined. Since $\text{Ind}(V)$ is an integer, it is invariant under homotopy of V .

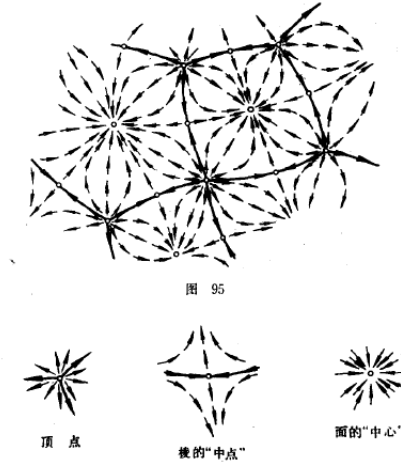
Theorem 10.3. (*Poincaré-Hopf Index Theorem*) Let M be a closed oriented surface, V be a smooth vector field on M with isolated zeros. Then

$$\text{Ind}(V) = \chi(M).$$

Sketch of the proof.

Step 1: Independence of V . Omit.

Step 2: Construction of a special V . Given a triangulation of M , we can construct a vector field V such that $\text{Ind}(V) = \chi(M)$, thus we have $\text{Ind}(V) = \chi(M)$ for any V . The following picture shows how to construct such a vector field V : $\square$



Proof of Gauss-Bonnet Theorem. Let V be a smooth vector field on M with isolated zeros $\{x_i\}$. We construct another smooth vector field W with the same zeros such that $\{\tilde{V}, \tilde{W}\} := \{\frac{V}{|V|}, \frac{W}{|W|}\}$ is an orthonormal frame on $M - \{x_i\}$ compatible with the orientation of M .

Let A be the connection 1-form of the Levi-Civita connection with respect to the frame $\{V, W\}$, i.e.,

$$A(X) := \langle \nabla_X \tilde{V}, \tilde{W} \rangle, \quad \forall X \in T(M - \{x_i\}).$$

then the curvature 2-form $F = dA$. Let $B_r(x_i)$ be a small neighborhood of x_i containing no other zeros, then

$$\begin{aligned} \frac{1}{2\pi} \int_M F &= \lim_{r \rightarrow 0} \frac{1}{2\pi} \int_{M - \cup_i B_r(x_i)} F \\ &= \lim_{r \rightarrow 0} \frac{1}{2\pi} \int_{M - \cup_i B_r(x_i)} dA \\ &= \lim_{r \rightarrow 0} \sum_i \frac{1}{2\pi} \int_{\partial(M - B_r(x_i))} A \\ &= \lim_{r \rightarrow 0} \sum_i \frac{1}{2\pi} \int_{\partial B_r(x_i)} A \end{aligned}$$

Notice that the left hand side is independent of the choice of metric and connection (by characteristic class theory), thus we can choose any metric and connection we like. In particular, we can choose the Euclidean metric and the trivial connection in each neighborhood $B_r(x_i)$. Under a local coordinate system $(B_r(x_i); x, y)$, we can write

$$\tilde{V} = \cos \theta \frac{\partial}{\partial x} + \sin \theta \frac{\partial}{\partial y}, \quad \tilde{W} = -\sin \theta \frac{\partial}{\partial x} + \cos \theta \frac{\partial}{\partial y},$$

where $\theta = \theta(x, y)$ is a smooth function on $B_r(x_i)$. Then

$$\begin{aligned} A\left(\frac{\partial}{\partial x}\right) &= \langle \nabla_{\frac{\partial}{\partial x}} \tilde{V}, \tilde{W} \rangle = \langle -\theta_x \sin \theta \frac{\partial}{\partial x} + \theta_x \cos \theta \frac{\partial}{\partial y}, -\sin \theta \frac{\partial}{\partial x} + \cos \theta \frac{\partial}{\partial y} \rangle = \theta_x, \\ A\left(\frac{\partial}{\partial y}\right) &= \langle \nabla_{\frac{\partial}{\partial y}} \tilde{V}, \tilde{W} \rangle = \langle -\theta_y \sin \theta \frac{\partial}{\partial x} + \theta_y \cos \theta \frac{\partial}{\partial y}, -\sin \theta \frac{\partial}{\partial x} + \cos \theta \frac{\partial}{\partial y} \rangle = \theta_y. \end{aligned}$$

Thus we have $A = \theta_x dx + \theta_y dy = d\theta$, so

$$\lim_{r \rightarrow 0} \frac{1}{2\pi} \int_{\partial B_r(x_i)} A = \lim_{r \rightarrow 0} \frac{1}{2\pi} \int_{\partial B_r(x_i)} d\theta = \text{Ind}_{x_i}(V).$$

Therefore,

$$\frac{1}{2\pi} \int_M F = \sum_i \text{Ind}_{x_i}(V) = \text{Ind}(V) = \chi(M). \quad \square$$

Remark. A is not a globally defined 1-form on M , but its exterior derivative dA can be extended to a globally defined 2-form on M .

Gauss-Bonnet-Chern Theorem

The Gauss-Bonnet-Chern theorem is a generalization of the Gauss-Bonnet theorem to higher dimensions.

Theorem 10.4 (Gauss-Bonnet-Chern Theorem). *Let M^{2n} be a closed oriented Riemannian manifold, F be the curvature 2-form of the Levi-Civita connection. Then*

$$\int_M \text{Pf}\left(\frac{F}{2\pi}\right) = \chi(M),$$

where Pf is the Pfaffian of a skew-symmetric matrix.

To prove this theorem, we need to use the following generalization of the Poincaré-Hopf index theorem.

Theorem 10.5 (General Poincaré-Hopf Index Theorem). *Let M be a closed oriented manifold, V be a smooth vector field on M with isolated zeros. Then*

$$\text{Ind}(V) = \chi(M).$$

Sketch of the proof.

Step 1: Independence of V . Omit.

Step 2: Construction of a special V . Choose V to be the gradient vector field of a Morse function f on M , then the zeros of V are exactly the critical points of f , and the index of V at a critical point x_i is $(-1)^{\lambda_i}$, where λ_i is the Morse index of x_i . So

$$\text{Ind}(V) = \sum_i (-1)^{\lambda_i} = \sum_{\lambda} (-1)^{\lambda} \#\{\text{critical points of index } \lambda\} = \chi(M). \quad \square$$

Sketch of the proof of G.B.C. Theorem. Consider the unit sphere bundle $\pi_S : SM \rightarrow M$, Chern constructed a global angular form Ψ on SM such that:

- Ψ is the "transgression" of $\text{Pf}\left(\frac{F}{2\pi}\right)$, i.e.,

$$d\Psi = \pi_S^* \text{Pf}\left(\frac{F}{2\pi}\right). \quad (10.1)$$

- For any $x \in M$,

$$\int_{S_x^{2n-1}} \iota_x^* \Psi = 1, \quad (10.2)$$

where $\iota_x : S_x^{2n-1} \hookrightarrow SM$ is the inclusion of the fiber at x .

Now let V be a normal vector field on M with isolated singular points $\{x_i\}$. V can be viewed as a section of SM over $M - \{x_i\}$. Let $B_r(x_i)$ be a small neighborhood of x_i containing no other singular points, then

$$\begin{aligned} \int_M \text{Pf}\left(\frac{F}{2\pi}\right) &= \lim_{r \rightarrow 0} \int_{M - \cup_i B_r(x_i)} \text{Pf}\left(\frac{F}{2\pi}\right) \\ &= \lim_{r \rightarrow 0} \int_{M - \cup_i B_r(x_i)} V^* \pi_S^* \text{Pf}\left(\frac{F}{2\pi}\right) \\ &= \lim_{r \rightarrow 0} \int_{M - \cup_i B_r(x_i)} V^* d\Psi \\ &= \lim_{r \rightarrow 0} \sum_i \int_{\partial(M - B_r(x_i))} V^* \Psi \\ &= \lim_{r \rightarrow 0} \sum_i \int_{\partial B_r(x_i)} V^* \Psi \end{aligned}$$

Notice that the last integral actually computes the degree of $V|_{\partial B_r(x_i)}$, where

$$V|_{\partial B_r(x_i)} : \partial B_r(x_i) \rightarrow SM|_{B_r(x_i)} \cong B_r(x_i) \times S^{2n-1} \simeq S^{2n-1},$$

Therefore,

$$\int_M \text{Pf}\left(\frac{F}{2\pi}\right) = \sum_i \deg(V|_{\partial B_r(x_i)}) = \sum_i \text{Ind}_{x_i}(V) = \text{Ind}(V) = \chi(M). \quad \square$$

This proof highly depends on the magical construction of the global angular form Ψ , which is a special case of the transgression form we'll talk about later.

Remark. Back in the 2-dimensional case, the unit sphere bundle SM is just the oriented orthonormal frame bundle $SO(TM)$, both of which are S^1 -bundles over M . Let ω be the Ehresmann connection of $SO(TM)$ induced by the Levi-Civita connection, Ω be the curvature of ω . Since $SO(2)$ is abelian, we have

$$\Omega = d\omega + \frac{1}{2}[\omega, \omega] = d\omega, \quad \text{Pf}\left(\frac{\Omega}{2\pi}\right) = d \text{Pf}\left(\frac{\omega}{2\pi}\right) =: d\Psi.$$

The relation between $\Omega \in \Omega^2(SO(TM), \mathfrak{so}(2))$ and $F \in \Omega^2(M)$ is

$$\text{Pf}\left(\frac{\Omega}{2\pi}\right) = \pi_S^*\left(\frac{F}{2\pi}\right).$$

So

$$d\Psi = \pi_S^*\left(\frac{F}{2\pi}\right). \quad (10.3)$$

Notice that the inclusion $\iota_x : S_x^1 \hookrightarrow SO(TM)$ pulls back ω to the Maurer-Cartan form ω_{MC} on S^1 , where

$$\begin{aligned} \omega_{MC} &= g^{-1}dg = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} d \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \\ &= \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} -\sin \theta d\theta & -\cos \theta d\theta \\ \cos \theta d\theta & -\sin \theta d\theta \end{pmatrix} = \begin{pmatrix} 0 & d\theta \\ -d\theta & 0 \end{pmatrix} \end{aligned}$$

Thus

$$\iota_x^* \Psi = \iota_x^* \text{Pf}\left(\frac{\omega}{2\pi}\right) = \text{Pf}\left(\frac{\iota_x^* \omega}{2\pi}\right) = \text{Pf}\left(\frac{\omega_{MC}}{2\pi}\right) = \frac{d\theta}{2\pi}, \quad (10.4)$$

is the angular form on S_x^1 with integral 1. Formulas (??) and (??) are exactly the properties of the global angular form Ψ in the proof of the G.B.C. theorem. The form A is just the pullback of Ψ via the section $V : M - \{x_i\} \rightarrow SO(TM)$, i.e., $A = V^* \Psi$.

10.2 Transgressions

Now let's focus on a G -principal bundle $\pi : P \rightarrow M$ with connection A and curvature F . The Chern-Weil theory tells us any invariant polynomial Φ on $\text{Inv}^k(\mathfrak{g})$ defines a closed $2k$ -form $\Phi(F)$ on M and its cohomology class $[\Phi(F)]$ is independent of the choice of connection.

Consider the pullback bundle π^*P over P ,

$$\begin{array}{ccc} \pi^*P & & P \\ \downarrow & & \downarrow \pi \\ P & \xrightarrow{\pi} & M \end{array}$$

π^*P can be identified with the fiber product $P \times_M P$, thus has a tautological section $\Delta : P \rightarrow \pi^*P$, $\Delta(p) = (p, p)$. This means π^*P is a trivial bundle, so all characteristic classes of π^*P vanish. In particular, $[\Phi(\pi^*F)] = 0$ in $H^{2k}(P)$. Thus there exists a $(2k-1)$ -form $\Psi(A, F)$ on P such that

$$d\Psi(A, F) = \Phi(\pi^*F) = \pi^*\Phi(F).$$

The form $\Psi(A, F)$ is called the transgression of $\Phi(F)$.

Remark. Let $SO(TM)$ be the orthonormal frame bundle of a Riemannian manifold M^{2n} , then the transgression of $\text{Pf}\left(\frac{F}{2\pi}\right) \in \Omega^{2n}(M)$ is a $(2n-1)$ -form Ψ on $SO(TM)$ such that $d\Psi = \pi^*\text{Pf}\left(\frac{F}{2\pi}\right)$. The unit sphere bundle SM is the associated bundle of $SO(TM)$ with $SO(2n)$ naturally acting on S^{2n-1} , thus we can induce a form $\tilde{\Psi}$ on SM from Ψ . It turns out that $\tilde{\Psi}$ is exactly the global angular form Chern constructed in the proof.

There is a more explicit way to construct the transgression $\Psi(A, F)$. To introduce it, we need to review the Chern-Weil theory.

Given two connection ∇_0 and ∇_1 on vector bundle E , the independence of characteristic classes on the choice of connection is proved by the following formula:

$$\text{Tr}(\Omega_1^k) - \text{Tr}(\Omega_0^k) = d \int_0^1 k \text{Tr} \left(\frac{d\omega_t}{dt} \wedge \Omega_t^{k-1} \right) dt.$$

Here $\omega_t = (1-t)\omega_0 + t\omega_1$ is the connection form of $\nabla_t = (1-t)\nabla_0 + t\nabla_1$, Ω_t is the curvature of ∇_t . If we fix ∇_0 to be flat, i.e., $\Omega_0 = 0$, then the right hand side exactly gives a transgression of $\text{Tr}(\Omega_1^k)$.

10.3 Chern-Simons Form

Historically, Simons wanted to study the signature of a 4-manifold X^4 , which is related to the first Pontryagin class p_1 because of **the Hirzebruch signature theorem**:

$$\sigma(X) = \frac{1}{3} \int_X p_1(TX).$$

Notice that

$$p_1(TX) = f_2\left(\frac{F}{2\pi}\right) = \frac{1}{2} \left[\left(\text{Tr} \left(\frac{F}{2\pi} \right) \right)^2 - \text{Tr} \left(\frac{F^2}{4\pi^2} \right) \right] = -\frac{1}{8\pi^2} \text{Tr}(F^2).$$

Simons constructed a transgression of $p_1(TX)$, which is now called the Chern-Simons form.

Let M be a closed oriented 3-manifold, there is a fact that its oriented orthonormal frame bundle $SO(TM)$ is always trivial. So we can choose a global section $s : M \rightarrow SO(TM)$, and fix a trivial connection ∇_0 on $SO(TM)$ such that its connection form is zero with respect to s . For any connection ∇_1 , let A be its connection form with respect to s . Now the connection $\nabla_t = (1-t)\nabla_0 + t\nabla_1$ has connection form

$$A_t = tA,$$

and curvature

$$F_t = dA_t + \frac{1}{2}[A_t, A_t] = t dA + t^2 A \wedge A.$$

So the Chern-Simons form is

$$\begin{aligned} -\frac{1}{8\pi^2} \int_0^1 k \operatorname{Tr} \left(\frac{dA_t}{dt} \wedge F_t^{k-1} \right) dt &= -\frac{1}{8\pi^2} \int_0^1 2 \operatorname{Tr} (A \wedge (t dA + t^2 A \wedge A)) dt \\ &= -\frac{1}{8\pi^2} \operatorname{Tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right). \end{aligned} \quad (10.5)$$

Remark. The Chern-Simons form is a 3-form on the oriented 3-manifold M . It is closed for the dimension reason.

Definition 10.6 (Chern-Simons functional). Let M be a closed oriented 3-manifold, $SO(TM)$ be its oriented orthonormal frame bundle, ∇ be a connection on $SO(TM)$. For any global trivialization $s : M \rightarrow SO(TM)$, let A be the connection form of ∇ with respect to s , then the Chern-Simons functional of ∇ is defined to be

$$CS(A) := \frac{1}{8\pi^2} \int_M \operatorname{Tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right).$$

Remark. Notice that different choices of global trivialization s, s' differ by a gauge transformation $g : M \rightarrow SO(3)$. One can show that although $CS(A) \neq CS(A')$ in general, but $CS(A) - CS(A')$ is always an integer. Thus $CS(A)$ is well-defined as an element in $\mathbb{R}/\mathbb{Z}$.

Critical points of the Chern-Simons functional

If the connection ∇ is a critical point of the Chern-Simons functional. Fix a global trivialization $s : M \rightarrow SO(TM)$ and let A be the connection form of ∇ w.r.t. s . Then for any variation $a \in \Omega^1(M, \mathfrak{so}(3))$, the perturbation $\nabla + ta$ has connection form $A + ta$, so

$$\begin{aligned} &\left. \frac{d}{dt} \right|_{t=0} CS(A + ta) \\ &= \left. \frac{d}{dt} \right|_{t=0} \frac{1}{8\pi^2} \int_M \operatorname{Tr} \left((A + ta) \wedge d(A + ta) + \frac{2}{3} (A + ta) \wedge (A + ta) \wedge (A + ta) \right) \\ &= \frac{1}{8\pi^2} \int_M \operatorname{Tr} \left(a \wedge dA + A \wedge da + \frac{2}{3} (a \wedge A \wedge A + A \wedge a \wedge A + A \wedge A \wedge a) \right) \\ &= \frac{1}{8\pi^2} \int_M \operatorname{Tr} (a \wedge dA + d(A \wedge a) + dA \wedge a + 2a \wedge A \wedge A) \\ &= \frac{1}{4\pi^2} \int_M \operatorname{Tr} (a \wedge (dA + A \wedge A)) + \frac{1}{8\pi^2} \int_M d \operatorname{Tr} (A \wedge a) \\ &= \frac{1}{4\pi^2} \int_M \operatorname{Tr} (a \wedge F). \end{aligned}$$

Since a is arbitrary, we have $F = 0$.

Theorem 10.7. *The critical points of the Chern-Simons functional are exactly the flat connections on $SO(TM)$.*

10.4 Gauge Transformation

Definition 10.8. Let $\pi : P \rightarrow M$ be a G -principal bundle, the gauge transformation of P is a G -bundle automorphism $\varphi : P \rightarrow P$, i.e., φ is a diffeomorphism such that $\varphi(pg) = \varphi(p)g$ and $\pi \circ \varphi = \pi$.

$$\begin{array}{ccc} P & \xrightarrow{\varphi} & P \\ & \searrow \pi \quad \swarrow \pi & \\ & M & \end{array}$$

All gauge transformations of P form the gauge group of P , denoted by $\mathcal{G}(P)$.

Example 10.9. For trivial bundle $P = M \times G$, each smooth function $f : M \rightarrow G$ defines a gauge transformation $\varphi(x, g) = (x, f(x)g)$. In fact, all gauge transformations are of this form, so $\mathcal{G}(M \times G) \cong C^\infty(M, G)$, where multiplication in $C^\infty(M, G)$ is defined pointwise.

Let $\varphi : P \rightarrow P$ be a gauge transformation, then for any $p \in P$, there exists a unique $g_p \in G$ such that $\varphi(p) = pg_p$. Thus we can define a smooth function $f : P \rightarrow G$ by $f(p) = g_p$.

Proposition 10.10. $f : P \rightarrow G$ defined above satisfies $f(pg) = g^{-1}f(p)g$, $\forall p \in P, g \in G$.

Proof. Since φ is G -equivariant,

$$pgf(pg) = \varphi(pg) = \varphi(p)g = pf(p)g \implies f(pg) = g^{-1}f(p)g. \quad \square$$

The above proposition implies that f can be viewed as a section of the associated bundle $\text{Ad}P := P \times_{\text{cong}} G$ with G acting on itself by conjugation.

Theorem 10.11. *There is a one-to-one correspondence between the gauge group $\mathcal{G}(P)$ and the space of sections $C^\infty(M, \text{Ad}P)$. If we define the multiplication in $C^\infty(M, \text{Ad}P)$ pointwise, then this correspondence is a group isomorphism.*

Proof.

$$\left\{ \begin{array}{c} \text{sections of } \text{Ad}P \\ s : M \rightarrow \text{Ad}P \end{array} \right\} \xrightarrow{1-1} \left\{ \begin{array}{c} G\text{-equivariant maps} \\ f : P \rightarrow G \end{array} \right\} \xrightarrow{1-1} \left\{ \begin{array}{c} \text{gauge transformations} \\ \varphi(p) = pf(p) \end{array} \right\}. \quad \square$$

Example 10.12. For the trivial bundle $P = M \times G$, we have $\text{Ad}P = M \times G$, so $C^\infty(M, \text{Ad}P) \cong C^\infty(M, G)$.

Example 10.13. For abelian G , the conjugation action is trivial, so $\text{Ad}P = P \times_{\text{cong}} G \cong M \times G$, thus $\mathcal{G}(P) \cong C^\infty(M, G)$.

Gauge transformation acts on connections

Let $\varphi : P \rightarrow P$ be a gauge transformation, then for any connection ω on P , we can pull back ω via φ to get $\varphi^*\omega$.

Theorem 10.14. $\varphi^*\omega$ is still a connection on P .

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10.5 Holonomy and flat connections

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Appendix

Vector valued-differential forms

Through out this section, let M be a smooth manifold and V be a vector space. $\Omega^k(M)$ denotes the space of smooth k -forms on M .

V -valued forms

Definition 10.15. Define the space of V -valued k -forms on M as

$$\Omega^k(M, V) := C^\infty(M, (\wedge^k T^*M) \otimes V).$$

Fix a basis $\{v_i\}$ of V , then any $\alpha \in \Omega^k(M, V)$ can be written as

$$\alpha = \alpha^i \otimes v_i (= \alpha^i v_i), \quad \alpha^i \in \Omega^k(M).$$

Scalar multiplication: We can define a scalar multiplication

$$\begin{aligned} \Omega^a(M) \times \Omega^k(M, V) &\rightarrow \Omega^{a+k}(M, V) \\ (\lambda, \alpha \otimes v) &\mapsto (\lambda \wedge \alpha) \otimes v. \end{aligned}$$

If $\alpha = \alpha^i v_i$ then $\lambda \cdot \alpha := (\lambda \wedge \alpha^i) \otimes v_i$. More generally, we can define

$$\begin{aligned} \Omega^a(M, \text{End}(V)) \times \Omega^k(M, V) &\rightarrow \Omega^{a+k}(M, V) \\ (\eta \otimes A, \alpha \otimes v) &\mapsto (\eta \wedge \alpha) \otimes Av. \end{aligned}$$

Exterior derivative: For $\alpha \in \Omega^k(M, V)$, define

$$d\alpha := (d\alpha^i) \otimes v_i \in \Omega^{k+1}(M, V).$$

Still, this definition is well-defined.

Proposition 10.16. *The exterior derivative d satisfies the following properties:*

- $d^2 = 0$.
- $d(\lambda \cdot \alpha) = d\lambda \cdot \alpha + (-1)^a \lambda \cdot d\alpha$, $\forall \lambda \in \Omega^a(M), \alpha \in \Omega^k(M, V)$,

Products of vector-valued forms

Let V, W, Z be vector spaces and $\mu : V \times W \rightarrow Z$ be a bilinear map.

Definition 10.17. We can define a product:

$$\begin{aligned} \Omega^k(M, V) \times \Omega^l(M, W) &\rightarrow \Omega^{k+l}(M, Z) \\ (\alpha, \beta) &\mapsto \alpha \cdot \beta \end{aligned}$$

where

$$\begin{aligned} (\alpha \cdot \beta)_p(X_1, \dots, X_{k+l}) &:= \frac{1}{k!l!} \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) \mu \left(\alpha_p(X_{\sigma(1)}, \dots, X_{\sigma(k)}), \beta_p(X_{\sigma(k+1)}, \dots, X_{\sigma(k+l)}) \right) \\ &= \sum_{(k,l)\text{-shuffles } \sigma} \text{sgn}(\sigma) \mu \left(\alpha_p(X_{\sigma(1)}, \dots, X_{\sigma(k)}), \beta_p(X_{\sigma(k+1)}, \dots, X_{\sigma(k+l)}) \right).^{11} \end{aligned}$$

¹¹We say $\sigma \in S_{k+l}$ is a (k, l) -shuffle if $\sigma(1) < \dots < \sigma(k)$ and $\sigma(k+1) < \dots < \sigma(k+l)$.

Proposition 10.18. *This product satisfies the following properties:*

- If $\alpha = \alpha^i v_i$ and $\beta = \beta^j w_j$, then $\alpha \cdot \beta = (\alpha^i \wedge \beta^j) \otimes \mu(v_i, w_j)$.
- If $\lambda \in \Omega^a(M)$, then $\lambda \cdot (\alpha \cdot \beta) = (\lambda \cdot \alpha) \cdot \beta = (-1)^{ka} \alpha \cdot (\lambda \cdot \beta)$.
- $d(\alpha \cdot \beta) = d\alpha \cdot \beta + (-1)^k \alpha \cdot d\beta$.

Notice that in general, $\alpha \cdot \beta$ is not associative, commutative, or anticommutative.

Example 10.19. Let $\mu : \mathbb{R}^{m \times n} \times \mathbb{R}^{n \times p} \rightarrow \mathbb{R}^{m \times p}$ be the matrix multiplication. $\alpha = \alpha^{ij} e_{ij} \in \Omega^k(M, \mathbb{R}^{m \times n})$ and $\beta = \beta^{ij} e_{ij} \in \Omega^l(M, \mathbb{R}^{n \times p})$, then

$$\alpha \cdot \beta = (\alpha^{ij} \wedge \beta^{jk}) e_{ik} =: \alpha \wedge \beta$$

is the usual matrix-valued wedge product.

g-valued forms

In the special case when $V = \mathfrak{g}$ is a Lie algebra or more precisely the matrix algebra $\mathfrak{gl}(n, \mathbb{R})$, we take $\mu : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ to be the Lie bracket, and we write $[\alpha, \beta]$ instead of $\alpha \cdot \beta$.

$$\begin{aligned} [\alpha, \beta]_p(X_1, \dots, X_{k+l}) &:= \frac{1}{k!l!} \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) [\alpha_p(X_{\sigma(1)}, \dots, X_{\sigma(k)}), \beta_p(X_{\sigma(k+1)}, \dots, X_{\sigma(k+l)})] \\ &= \sum_{(k,l)\text{-shuffles } \sigma} \text{sgn}(\sigma) [\alpha_p(X_{\sigma(1)}, \dots, X_{\sigma(k)}), \beta_p(X_{\sigma(k+1)}, \dots, X_{\sigma(k+l)})]. \end{aligned}$$

For example, if $\alpha, \beta \in \Omega^1(M, \mathfrak{g})$, then $[\alpha, \beta]_p(X, Y) = [\alpha_p(X), \beta_p(Y)] - [\alpha_p(Y), \beta_p(X)]$.

Example 10.20. If $\mathfrak{g} = \mathfrak{gl}(n, \mathbb{R})$, $[A, B] = AB - BA$, then

$$[\alpha, \beta] = \alpha \wedge \beta - (-1)^{|\alpha||\beta|} \beta \wedge \alpha.$$

Hence

$$[\alpha, \alpha] = \begin{cases} 2\alpha \wedge \alpha, & \text{if } |\alpha| \text{ is odd,} \\ 0, & \text{if } |\alpha| \text{ is even.} \end{cases}$$

Proposition 10.21. For $\alpha = \alpha^i A_i, \beta = \beta^j A_j$, we have

$$[\alpha, \beta] = (\alpha^i \wedge \beta^j) \otimes [A_i, A_j].$$

Corollary 10.22. If $\alpha \in \Omega^k(M, \mathfrak{g})$ and $\beta \in \Omega^l(M, \mathfrak{g})$, then

$$[\alpha, \beta] = (-1)^{kl} [\beta, \alpha].$$

Example 10.23. If $\alpha, \beta \in \Omega^1(M, \mathfrak{g})$, then $[\alpha, \beta] = [\beta, \alpha]$, thus $[\alpha, \alpha]$ is not necessarily zero.

Proposition 10.24. For $\alpha \in \Omega^k(M, \mathfrak{g})$, $\beta \in \Omega^l(M, \mathfrak{g})$ and $\gamma \in \Omega^m(M, \mathfrak{g})$, we have

$$[\alpha, [\beta, \gamma]] + (-1)^{k(l+m)} [\beta, [\gamma, \alpha]] + (-1)^{m(k+l)} [\gamma, [\alpha, \beta]] = 0.$$

Example 10.25. If $\alpha \in \Omega^1(M, \mathfrak{g})$, then $[\alpha, [\alpha, \alpha]] = 0$.

Proposition 10.26. $d[\alpha, \beta] = [d\alpha, \beta] + (-1)^{|\alpha|} [\alpha, d\beta]$.

Vector Bundle valued forms

Let $E \rightarrow M$ be a vector bundle. We can define

$$\Omega^k(M, E) := C^\infty(M, (\wedge^k T^*M) \otimes E).$$

Let $e_1, \dots, e_n$ be a local frame of E over an open subset $U \subset M$, then any $\alpha \in \Omega^k(M, E)$ can be written as

$$\alpha = \alpha^i e_i (= \alpha^i \otimes e_i), \quad \alpha^i \in \Omega^k(U).$$

Example 10.27. Let ∇ be a connection on E , then ∇^2 is a $\text{End}(E)$ -valued 2-form on M .

Maurer-Cartan forms

Let G be a Lie group and $\mathfrak{g}$ be its Lie algebra.

Definition 10.28. The **Maurer-Cartan form** θ on G is the unique $\mathfrak{g}$ -valued 1-form $\theta \in \Omega^1(G, \mathfrak{g})$ such that

- $\theta_e(X_e) = X_e, \forall X_e \in T_e G = \mathfrak{g}$.
- θ is left-invariant, i.e. $(l_g)^* \theta = \theta, \forall g \in G$. That is

$$\theta_g(X_g) = (l_{g^{-1}}^* \theta_e)(X_g) = \theta_e(l_{g^{-1}*} X_g) = l_{g^{-1}*} X_g, \quad \forall X_g \in T_g G.$$

Proposition 10.29. *The Maurer-Cartan form θ satisfies the **Maurer-Cartan equation***

$$d\theta + \frac{1}{2}[\theta, \theta] = 0.$$

Proof. It suffices to check this equation on left-invariant vector fields. For any left-invariant vector field X , $\theta(X)$ is a constant function on G , thus

$$\begin{aligned} (d\theta)(X, Y) &= X(\theta(Y)) - Y(\theta(X)) - \theta([X, Y]) = -[X_e, Y_e] \text{ (constant } \mathfrak{g}\text{-valued function)} \\ \frac{1}{2}[\theta, \theta](X, Y) &= \frac{1}{2}([\theta(X), \theta(Y)] - [\theta(Y), \theta(X)]) = [X_e, Y_e]. \text{ (constant } \mathfrak{g}\text{-valued function)} \end{aligned}$$

Thus

$$(d\theta + \frac{1}{2}[\theta, \theta])(X, Y) = 0. \quad \square$$

Proposition 10.30. *Under right translation, the Maurer-Cartan form θ transforms as*

$$(r_g)^* \theta = (\text{Ad}_{g^{-1}}) \theta, \quad \forall g \in G.$$

Proof. It suffices to check this equation on left-invariant vector fields. For any left-invariant vector field X , both $(r_g^* \theta)(X)$ and $\text{Ad}_{g^{-1}}(\theta(X))$ are constant $\mathfrak{g}$ -valued functions on G , and

$$(r_g^* \theta_g)(X_e) = \theta_g(r_{g*} X_e) = l_{g^{-1}*}(r_{g*} X_e) = \text{Ad}_{g^{-1}}(X_e) = \text{Ad}_{g^{-1}}(\theta_e(X_e)). \quad \square$$